



Superconducting quantum computers: who is leading the future?

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Abstract

Superconducting quantum computing has emerged as a leading platform in the pursuit of practical quantum computers, driven by rapid advances from industry, academia, and government initiatives. This review examines the state of superconducting quantum technology, with emphasis on qubit design, processor architecture, scalability, and supporting quantum software. We compare the hardware strategies and performance milestones of key players—including IBM Quantum, Google Quantum AI, Rigetti Computing, Intel Quantum, QuTech, and Oxford Quantum Circuits—highlighting innovations in qubit coherence, control, and system integration. Landmark demonstrations such as quantum supremacy experiments are discussed alongside progress toward real-world applications in the noisy intermediate-scale quantum (NISQ) era. Beyond hardware, attention is given to the broader software and service ecosystem, including quantum programming frameworks, operating environments, and cloud-accessible platforms such as Amazon Braket, Azure Quantum, and OriginQ Cloud, which enable remote access and algorithm development. Persistent challenges in superconducting quantum computing—such as error correction, system stability, and large-scale integration—are assessed in light of emerging approaches aimed at fault-tolerant quantum computing. As the field moves from the NISQ era toward fault-tolerant quantum computing, we capture the defining hardware achievements and characteristics of current superconducting processors, while examining the ongoing efforts and challenges in overcoming NISQ-era limitations. These developments offer critical insights into the path toward scalable quantum systems and their transformative impact on future technologies, while also underscoring the strategic and societal considerations that require balancing innovation with responsible oversight and thoughtful governance.

Keywords: Superconducting quantum processors; IBM Quantum; Google Quantum AI; Rigetti Computing; Zuchongzhi Quantum computers; Oxford Quantum Circuits; Intel Quantum; QuTech; Quantum Inspire; Alibaba Cloud; Microsoft's Azure Quantum; QuantWare. OriginQ Cloud; Alice and Bob; IQM; Amazon Braket

1 Introduction

Quantum computing has emerged as one of the most exciting and transformative technologies of the 21st century [1], with the potential to solve problems that are beyond the reach of even the most powerful classical computers [2]. From designing new materials

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and accelerating drug discovery [3] to optimizing global supply chains [4] and advancing financial modeling [5], quantum computers could revolutionize industries and scientific research [2]. At the heart of this revolution is the pursuit of fully fault-tolerant quantum computers—machines capable of performing complex computations reliably, even in the face of the delicate and error-prone nature of quantum states [6]. Achieving fault tolerance is more than just a technical milestone; it's the key to unlocking practical quantum applications at scale.

Among the various quantum computing architectures [1, 7–11], superconducting quantum processors have emerged as front-runners [2, 12]. Leveraging decades of advances in semiconductor fabrication, superconducting qubits [13–25] offer a promising combination of scalability, compatibility with existing infrastructure, and consistent performance improvements. However, the road to fully fault-tolerant quantum systems is far from straightforward. Key challenges such as mitigating noise, extending coherence times, and implementing robust quantum error correction remain formidable obstacles [26]. Despite impressive progress, these technical barriers must be overcome before quantum computing can truly deliver on its transformative potential.

The combination of public and private investment has created an environment where innovation thrives, accelerating the pace of development and bringing practical quantum systems closer to reality. But the question remains: how soon can superconducting quantum processors move beyond the NISQ era [2] and start delivering real-world impact? Addressing this urgency requires a closer look at the current state of superconducting quantum computing and the collaborative efforts shaping its future. Understanding the innovations, strategies, and challenges faced by key players in the field is essential for grasping the trajectory of this technology and its path toward scalable, fault-tolerant systems. This article aims to explore that landscape, focusing on the quantum processors developed by industry leaders, academic researchers, and government agencies. By analyzing their contributions and the technical hurdles they face [26], this paper provides a comprehensive overview of the progress toward fully fault-tolerant quantum computers. In doing so, it not only highlights the technological advancements but also the collaborative and competitive dynamics defining this critical moment in quantum computing's evolution.

The remainder of this paper is organized as follows: Sect. 2 discusses the pivotal role of industry leaders in advancing superconducting quantum computing. Section 3 reviews key superconducting qubit architectures, including transmon qubits, fluxonium qubits, heavy fluxonium qubits, $0-\pi$ qubits, unimon qubits, and other emerging designs. Section 4 summarizes major milestones in demonstrating quantum computational supremacy with superconducting processors. The contributions of various stakeholders are examined in Sects. 5 to 20, while Sect. 21 takes a deep dive into the challenges and bottlenecks hindering scalable quantum systems [26, 27]. Finally, Sect. 22 explores the path toward real-world applications, and Sect. 23 wraps up with a discussion on the future of superconducting quantum computing.

2 Accelerating the landscape of superconducting quantum computing

The rapid pace of innovation in quantum computing reflects a collective ambition to unlock unprecedented computational power [2]. Among the many competing approaches [7, 9, 10], superconducting qubits have emerged as a leading contender, thanks to their relatively advanced development, compatibility with existing semiconductor fabrication techniques, and potential for scalable architectures. Yet the journey from NISQ computers to

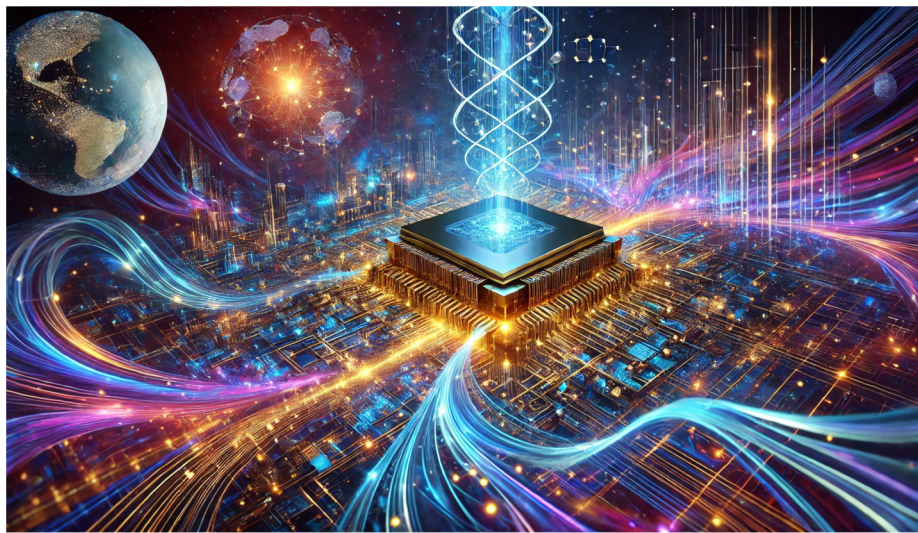


Figure 1 A conceptual digital artwork of a superconducting quantum computer, featuring a glowing processor, dynamic qubit networks, and global data streams. The design symbolizes scalability, innovation, and the worldwide pursuit of quantum computing breakthroughs

practical quantum computers remains filled with formidable challenges [26, 27]. Achieving longer coherence times, reducing error rates, and implementing robust quantum error correction (QEC) are just a few of the critical hurdles the field must overcome.

What makes superconducting quantum computing particularly compelling is the collaborative ecosystem driving its progress. Industry leaders [28–30], academic researchers [31], and governmental agencies [32] are not just competing — they’re actively contributing to a shared foundation of knowledge. Through open-source tools, published benchmarks, and joint research initiatives, these stakeholders are collectively pushing the boundaries of what quantum technology can achieve [2]. This convergence of expertise and resources is vital for addressing the complexity of building fault-tolerant quantum systems.

Crucially, the focus has shifted from isolated achievements —like increasing qubit count— to systemic progress. Building practical quantum computers requires advances across multiple dimensions: materials science to improve coherence times, cryogenic engineering to maintain stable operating environments, and hybrid classical-quantum interfaces to manage and interpret quantum information. These interconnected challenges [26, 27] underscore the need for interdisciplinary solutions, blending insights from physics, engineering, and computer science.

As the field matures, companies pioneering superconducting quantum processors play a dual role: advancing their proprietary technologies while contributing to the broader scientific and industrial ecosystem (see Fig. 1). This shared momentum accelerates the timeline for real-world applications, with transformative potential across industries like drug discovery, logistics optimization, and secure communications. These efforts are not just shaping the future of computing — they’re laying the foundation for solving some of the world’s most complex problems.

But the transition from experimental systems to commercial solutions requires more than technical breakthroughs. It demands strategic partnerships and industry-wide col-

laboration to align innovation with practical needs. In what follows, we explore how leading companies are navigating this critical phase, bringing superconducting quantum computing closer to commercialization and demonstrating its potential across diverse fields, from pharmaceuticals and finance to global supply chains and beyond.

3 Superconducting qubit architectures

Superconducting qubits [13–17, 19–25, 27] have emerged as strong candidates for building scalable quantum computers due to their high controllability and compatibility with existing fabrication techniques. However, achieving the performance necessary for practical quantum computing remains a challenge [26, 27]. Various types of qubits have been developed to address different aspects of coherence, noise resilience, and gate fidelity.

3.1 The transmon qubits

Most superconducting multi-qubit processors today are based on *transmon qubits* [12, 32–34], which are reliably fabricated [34] and feature coherence times extending to several hundred microseconds [35, 36]. This has enabled exceptionally high gate fidelities, with single-qubit gates achieving 99.98 – 99.99% and two-qubit gates reaching 99.8 – 99.9% [37–40]. The transmon qubit is an advancement of the charge qubit [41], incorporating a shunt capacitor in parallel with a Josephson junction. This modification significantly reduces the qubit's sensitivity to charge noise by suppressing fluctuations in its transition frequency. Despite that, the large shunt capacitance also results in relatively low anharmonicity, around 200–300 MHz, which is just 5% of the typical qubit frequency [21, 42]. This limitation constrains the speed at which quantum gates can be implemented with transmons, as leakage to non-computational states must be avoided [37, 43]. Additionally, the low anharmonicity affects the readout speed of transmon qubits, with strong readout signals potentially causing the qubit to transition to undesired states outside the cosine potential [44]. To improve both the speed of operations and gate fidelities, especially given the finite coherence time, a higher anharmonicity is desired.

3.2 The fluxonium qubits

Recent advancements in *fluxonium qubits* have positioned them as a promising alternative to transmons, this is due to their high anharmonicity and extended relaxation and coherence times [45–47]. These improvements have led to remarkable gate fidelities, including over 99.99% for single-qubit gates [48] and 99.7% for two-qubit gates [46]. Fluxonium qubits are characterized by a small Josephson junction shunted by a superinductor, which can be realized using a variety of materials such as large Josephson junctions [45, 47, 49], nanowires with high kinetic inductance [50], granular aluminum wire [51], or through a geometric superinductor [52]. In fluxonium qubits, the superinductor plays a crucial role in reducing the dephasing and relaxation rates caused by flux noise. Moreover, the structure of the fluxonium qubit provides strong protection against dephasing from low-frequency charge noise.

3.3 Heavy fluxonium qubits

By incorporating a large shunt capacitor, fluxonium qubits can be modified into what is known as heavy fluxonium [47, 53], where the transition matrix element between the ground and first excited states is suppressed, leading to significantly enhanced relaxation

times, potentially reaching the millisecond range [53]. Despite that, due to their low operating frequency and small transition matrix elements, particularly near the half-flux quantum point, special techniques are required for controlling, reading out, and resetting these high-coherence fluxonium qubits [47]. Additionally, these qubits do not offer complete protection against both relaxation and dephasing due to flux noise at any single operational point. Parasitic capacitances in the superinductor can further complicate the reliable fabrication of fluxonium qubits and introduce unwanted parasitic modes.

3.4 Fluxonium alternatives

Reducing the total inductance in the junction array of fluxonium qubits enables the creation of alternative qubits like the *plasmonium qubit* [54], which operates at zero flux, and the *quarton qubit* [55], which functions at the half-flux-quantum point. These variants offer compact designs, high anharmonicity, and better protection against charge noise than transmons, aligning with current coherence time expectations. Alternatively, increasing the superinductance transforms the fluxonium into a *quasicharge qubit* [56], whose charge-basis eigenstates resemble those of early charge qubits but still maintain protection from charge noise.

3.5 The $0-\pi$ qubit, and other designs

Other types of qubits that provide some level of protection against relaxation and dephasing include the $0-\pi$ qubit [57], *bifluxon* [58], and *two-Cooper-pair tunneling* qubit [59]. The $0-\pi$ qubit, in particular, benefits from topological protection against charge and flux noise, though this feature complicates its operation and makes its circuit design more vulnerable to parasitic capacitance. Despite notable progress in the development of fluxonium and protected qubits, they have yet to demonstrate a clear and widespread advantage over transmons. As such, research into better superconducting qubit technologies remains ongoing.

3.6 The unimon qubits

A new type of superconducting qubit, known as the *unimon* [60], which combines several desirable features, such as increased anharmonicity, complete insensitivity to dc charge noise, reduced sensitivity to flux noise, and a relatively simple design that includes just a single Josephson junction within a resonator. The unimon qubits aiming to address the challenges of noise resilience, gate fidelity, and qubit design complexity [60]. Unlike conventional designs such as transmons or fluxonium qubits, the unimon integrates a single Josephson junction within a grounded coplanar-waveguide (CPW) resonator, eliminating the need for charge islands or superinductors. This structure combines simplicity with robustness, offering protection against low-frequency charge noise and partial resilience to flux noise. The unimon operates in a unique parameter regime where the Josephson energy and inductive energy achieve near cancellation, resulting in high anharmonicity without compromising coherence properties.

Experimental studies on five unimon qubits demonstrated their frequency tunability and anharmonicity across different configurations. At the flux-insensitive sweet spot, qubit Q_A achieved an anharmonicity of 744 MHz, significantly surpassing the 200 – 300 MHz range typical of transmon qubits. These high anharmonicities facilitate faster

quantum gate operations while minimizing leakage to non-computational states. Single-qubit gates achieved fidelity up to 99.9% with gate durations as short as 13.3 ns, demonstrating stability over extended periods [60].

The unimon's architecture (as detailed in Sect. 10.4) positions it as a promising candidate for advancing superconducting qubits. Its geometric inductance offers enhanced reproducibility compared to junction-array-based superinductors, making it suitable for scalable quantum processors. Design optimizations and material advancements [27] are expected to further reduce dielectric losses, potentially pushing gate fidelities beyond 99.99%, a threshold critical for efficient quantum error correction. Additionally, the unimon's high anharmonicity supports the development of faster, high-fidelity two-qubit gates and enables exploration of coupling schemes to suppress residual interactions. However, scaling up unimon-based systems requires addressing challenges such as flux noise sensitivity, on-chip crosstalk, and multi-qubit gate implementation.

This paper also explores several promising alternatives. For instance, Cat-qubit technology is examined in Sect. 18.2, while topological qubits are presented in Sect. 19.1. Additionally, the three-dimensional design known as Coaxmon qubits is discussed in Sect. 9.2. As advancements continue, these diverse superconducting qubit technologies have the potential to push the boundaries of quantum computation, particularly in the NISQ era and beyond.

4 Quantum supremacy with superconducting quantum processors

"Quantum supremacy," the term proposed by John Preskill in 2012 [61], refers to the milestone where a quantum computer surpasses classical computing systems by solving problems previously considered intractable [2]. Achieving this milestone [2] has been a focal point of scientific inquiry, driving efforts to design quantum computers capable of performing such extraordinary tasks [19, 61]. Significant advancements have been made toward demonstrating quantum advantage, utilizing both cryogenic and photonic-based platforms [1]. Table 1 provides a concise overview of experiments that have achieved quantum supremacy using superconducting quantum processors. For a more extensive review of the broader pursuit of quantum supremacy, including experiments utilizing platforms beyond superconducting processors—such as photonic quantum computers [1, 11]—and foundational proof-of-principle demonstrations across diverse quantum computing domains, readers are referred to the comprehensive review in [2].

5 IBM quantum

5.1 Overview

IBM Quantum has established itself as a prominent force in the quantum computing arena, driving advancements in both hardware and software capabilities [28]. Since 2016, it has offered access to its quantum hardware through IBM Cloud[†]. The systems feature advanced superconducting quantum processors [28] utilizing transmon qubits [24]. Over the years, IBM has achieved remarkable progress [28] in increasing qubit counts, enhancing qubit coherence times, and developing error mitigation techniques essential for improving the reliability of quantum computations on NISQ devices [63, 67]. These innovations have solidified IBM Quantum's position as a leader in quantum computing research, with significant applications in areas such as computational chemistry, optimization, cryptography, and machine learning [63, 67–70].

Table 1 Quantum computational supremacy with superconducting quantum processors [2]

Developer	Quantum processor	Computations	Qubits	Year	Description
Google Quantum AI	Sycamore 54	Sycamore, which demonstrated a landmark achievement by performing a computation (RCS) in just 200 seconds—a task estimated to take Summit, the most powerful supercomputer at the time of the experiment, approximately 10,000 years to complete [12].	54 ^a	2019	Sect. 6.3.1
China ^b	Zuchongzhi 2.0	The sampling task accomplished by Zuchongzhi, a quantum computing system comprising 56 superconducting qubits, within approximately 1.2 hours would necessitate the most potent supercomputer (Summit) to dedicate at least 8 years for completion [62].	56	2022	Sect. 8.2
China ^b	Zuchongzhi 2.1	An upgraded version, Zuchongzhi 2.1, a 66-qubit programmable superconducting quantum computing system, which significantly enhanced quantum advantage over its predecessor [32].	66	2022	Sect. 8.3
IBM Quantum	Eagle (<i>ibm_kyiv</i>)	Kim <i>et al.</i> [63] present a groundbreaking demonstration of quantum computation that surpasses the capabilities of classical simulation, determining the magnetization of a quantum material model using 127 qubits, to execute circuits containing up to 60 layers of 2-qubit gates, amounting to a total of 2880 CX gates. These circuit depths are far beyond the reach of brute-force classical simulation methods [63]. The success of the experiment was enabled by advances in coherence times, improved calibration techniques, and effective noise management in large-scale superconducting processors [64].	127	2024	Sect. 5.3
Google Quantum AI	Willow	The Willow chip achieved a groundbreaking milestone [65] by solving a computational problem (RCS [12]) in less than five minutes, a task that would require the most advanced supercomputers of today an unfathomable 10^{25} years to complete [66]. This timescale, equivalent to 10 septillion years. To put it in perspective, that is 10,000,000,000,000,000,000,000 years, far surpassing known timescales in physics and the current age of the universe, and challenges our understanding of computational limits [66].	72,105	2024	Sect. 6.4

^aDuring the supremacy experiment [12], one qubit was non-functional, resulting in operation with 86 couplers and 53 qubits.

^bThe Center for Excellence in Quantum Information and Quantum Physics (CEIQIP) under the Chinese Academy of Sciences (CAS).

In 2023, IBM Quantum unveiled *Condor*, a quantum processor featuring 1121 superconducting qubits arranged in a honeycomb configuration [71]. This development follows the success of earlier groundbreaking processors like *Eagle*, a 127-qubit chip launched in 2021 [72, 73], and *Osprey*, a 433-qubit processor announced in November 2022 [74, 75]. Alongside *Condor*, IBM introduced another quantum chip called *Heron*, which boasts 133 qubits and has achieved a record-low error rate that is three times lower than that of its predecessor. This shift represents a departure from IBM's previous strategy of annually doubling qubit counts, emphasizing instead the importance of enhanced error resistance over mere increases in qubit scalability [67, 71].

5.2 IBM quantum's quantum computers

IBM Quantum stands at the forefront of quantum computing, providing access to cutting-edge quantum hardware tailored to the needs of researchers, developers and industry professionals. The organization has developed multiple generations of quantum computers [28], each pushing the boundaries of quantum computing capabilities. Currently, IBM Quantum operates seven utility-scale systems, with more in the pipeline [76].

These systems are renowned for their exceptional reliability, achieving over 95% up-time collectively. They also exhibit outstanding stability, with 2-qubit gate errors showing minimal fluctuations, not exceeding 0.001 over several months (median 2-qubit gate errors recorded across all accessible *Eagle* processors from July 20 to September 20, 2023) [73, 77]. This combination of reliability and precision positions IBM Quantum as a leader in the rapidly evolving field of quantum computing.

5.2.1 Machines' performance

This section provides an analysis of the performance metrics for the latest IBM Quantum machines. Table 2 summarizes key aspects of hardware performance, qubit characteristics, and specifications of these superconducting quantum computers. Performance metrics include relaxation time (T_1) and coherence time (T_2), both measured in microseconds (μs), as well as qubit frequencies expressed in gigahertz (GHz) and the qubit energy level nonlinearity (anharmonicity) also measured in gigahertz (GHz). Furthermore, the table details readout assignment error, the duration of the readout process for a qubit (readout length) measured in nanoseconds (ns), and the probabilities of measuring $|0\rangle$ after preparing $|1\rangle$ ($P(0|1)$) and measuring $|1\rangle$ after preparing $|0\rangle$ ($P(1|0)$). It also includes the error rates associated with both single-qubit and 2-qubit gates, providing a thorough overview of the operational fidelity of these quantum systems.

5.2.2 *Condor*—IBM's 1121-qubit quantum processor

IBM Quantum has introduced the IBM *Condor*, an innovative quantum processor featuring 1121 superconducting qubits and leveraging IBM's state-of-the-art cross-resonance gate technology [71]. *Condor* sets a new standard in chip design, achieving an impressive 50% increase in qubit density, along with advancements in qubit fabrication techniques and laminate dimensions. Additionally, it incorporates over a mile of high-density cryogenic flex I/O wiring, all housed within a single dilution refrigerator.

With performance metrics that rival those of IBM Quantum's previous 433-qubit processor, *Osprey* (see Table 3), *Condor* represents a significant leap forward in the field of quantum computing. This development addresses key scalability challenges and offers critical insights that will inform the design of future quantum hardware.

Table 2 Performance metrics for IBM Quantum’s state-of-the-art quantum computers. This table presents the median values of key system configuration parameters, including both qubits performance and connectivity, for each quantum computer. Data for the following quantum systems accessed on October 9, 2024: *ibm_fez*, *ibm_toronto*, *ibm_kyiv*, and *ibm_sherbrooke*. While data for the remaining systems—*ibm_brussels*, *ibm_brisbane*, *ibm_rensselaer*, *ibm_kawasaki*, *ibm_quebec*, *ibm_nazca*, and *ibm_strasbourg*—accessed on October 10, 2024. Please note that IBM Quantum systems are regularly updated and may be modified, upgraded, or replaced; consequently, performance metrics are subject to change over time.

[illegible]

Table 2 (Continued)

Machines	(1) ibm_fez	(2) ibm_toronto	(3) ibm_kyiv	(4) ibm_sherbrooke	(5) ibm_brussels	(6) ibm_brisbane	(7) ibm_rensseleer	(8) ibm_kawasaki	(9) ibm_quebec	(10) ibm_nazca	(11) ibm_strasbourg
2Q Error (best) ^d	1.68×10^{-3}	1.10×10^{-3}	3.31×10^{-3}	3.10×10^{-3}	2.53×10^{-3}	3.62×10^{-3}	3.49×10^{-3}	2.93×10^{-3}	2.71×10^{-3}	4.92×10^{-3}	2.86×10^{-3}
2Q Error (layered) ^e	4.16×10^{-3}	6.75×10^{-3}	1.35×10^{-2}	1.63×10^{-2}	1.79×10^{-2}	1.81×10^{-2}	2.32×10^{-2}	2.58×10^{-2}	2.59×10^{-2}	2.98×10^{-2}	2.69×10^{-2}
CZ Error [*]	3.116×10^{-3}	3.297×10^{-3}	—	—	—	—	—	—	—	—	—
ECR Error [*]	—	—	1.135×10^{-2}	8.300×10^{-3}	7.269×10^{-3}	7.494×10^{-3}	8.675×10^{-3}	7.575×10^{-3}	7.279×10^{-3}	1.109×10^{-3}	8.377×10^{-3}
Gate time [*] (ns)	68	68	561.778	533.333	660	660	664.889	586.667	593.778	660	660

^aQuantum volume

^bHardware-aware circuit layer operations per second.

^cSupports OpenQASM 3 and dynamic circuits.

^dThe lowest two qubit error rate from all edges measured by isolated randomized benchmarking.

^eThe average two qubit error rate per layered gate (EPLG) for a 100-qubit chain.

The * symbol denotes median values for the reported performance metrics.

Table 3 Performance metrics and qubit characteristics of IBM Quantum's 433-qubit quantum computer, named *ibm_seattle*. This system is built based on *Osprey* r1 processor. The fundamental gates utilized by this machine include ECR, RZ, ID, SX, and X. It has an average ECR error of 109.266×10^{-3} , an average SX error of 45.722×10^{-4} , a mean readout error of 11.202×10^{-2} , an average T_1 time of 91.678 μ s, and an average T_2 time of 61.865 μ s. Accessed June 29, 2023

Properties	Min.	Max.	Mean	Median	1 st Quartile	3 rd Quartile	StdError	Stdev.	Unit
T_1	1.3187	237.918	91.678	88.349	66.4736	110.752	1.90127	38.6851	(μ s)
T_2	4.7695	181.689	61.865	58.732	33.4082	84.3429	1.69673	34.5234	(μ s)
Qubit anharmonicity	-0.4027	0.0	-0.279	-0.300	-0.3019	-0.2974	0.00361	0.07501	(GHz)
Qubit frequency	0.0	5.52679	4.9678	5.1818	5.09486	5.28126	0.05157	1.07283	(GHz)
$P(1\rangle 0\rangle)$	0.36	71.64	7.18709	4.18	1.72	9.635	0.40597	8.26025	$\times 10^{-2}$
$P(0\rangle 1\rangle)$	0.40	51.580	7.06688	4.380	2.085	9.655	0.35232	7.16863	$\times 10^{-2}$
Readout error	0.72	100	11.202	4.909	1.970	10.93	0.976	20.319	$\times 10^{-2}$
Readout length	2000	2000	2000	2000	2000	2000	0.0	0.0	(ns)
SX error	0.179	1000	45.722	0.625	0.4599	1.290	9.843	204.837	$\times 10^{-3}$
ID error	0.179	1000	45.722	0.625	0.4599	1.290	9.843	204.837	$\times 10^{-3}$
Pauli-X error	0.179	1000	45.722	0.625	0.4599	1.290	9.843	204.837	$\times 10^{-3}$
ECR error	7.1985	1000	109.266	21.548	15.511	39.8778	11.7846	264.563	$\times 10^{-3}$
ECR gate time	135	1404	635	660	660	660	6.18734	138.905	(ns)

5.3 Evidence of utility before fault tolerance

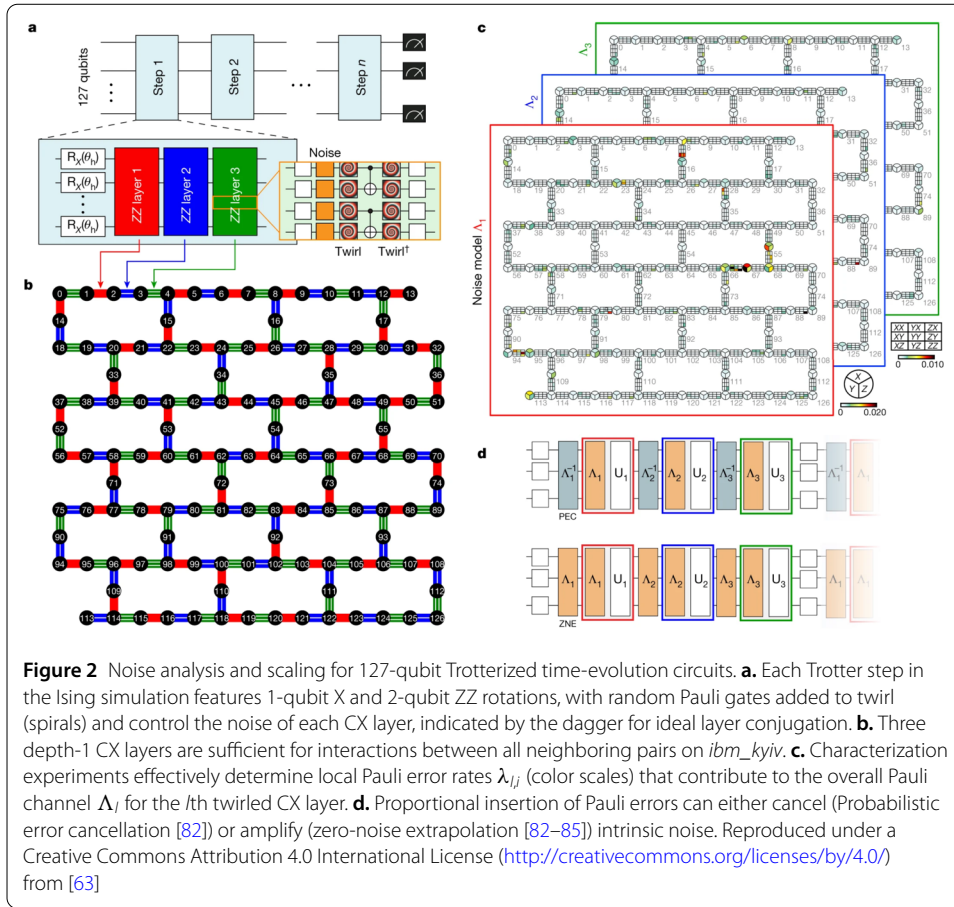
5.3.1 Achieving quantum advantage

Kim *et al.* [63] demonstrate a remarkable quantum-computational achievement that exceeds the limits of classical simulation, unveiling a significant step toward near-term practical quantum computing [2, 26]. This experiment showcased that quantum computers can perform circuits that exceed the capabilities of brute-force classical simulations. The primary advantage of their system lies in its scale rather than its speed, as classical computers lack the sufficient memory to represent the vast number of possibilities generated by these 127 qubits.

Achieving quantum advantage can be pursued through two approaches: first, by showcasing the capability of current devices to conduct precise computations that exceed the limits of brute-force classical simulation; and second, by identifying problems linked to quantum circuits that benefit from these devices' unique strengths [63]. In the work by [63], the authors concentrate on the first approach, focusing on demonstrating accurate computations without aiming to implement quantum circuits for problems with established speed-ups.

The authors employ a superconducting quantum processor featuring 127 qubits to execute quantum circuits that include up to 60 layers of 2-qubit gates, incorporating a total of 2880 CX gates. Such extensive quantum circuits are impractical for classical methods relying on brute-force approaches [63]. According to [63], these experimental outcomes are made possible by significant advancements in the coherence and calibration of superconducting processors at this scale, as well as the ability to effectively characterize and manage noise across such a large system [64].

Their benchmark circuit involves the Trotterized time evolution (TTE) of a two-dimensional transverse-field Ising model, which aligns with the topology of the qubit processor (see Fig. 2a). The Ising model is a cornerstone in various fields of physics and has inspired innovative extensions in recent simulations investigating quantum many-body phenomena, including quantum scars [78], time crystals [79, 80], and Majorana edge modes [81]. In terms of assessing the practical capabilities of quantum computation, the time evolution of the 2D transverse-field Ising model is particularly significant in

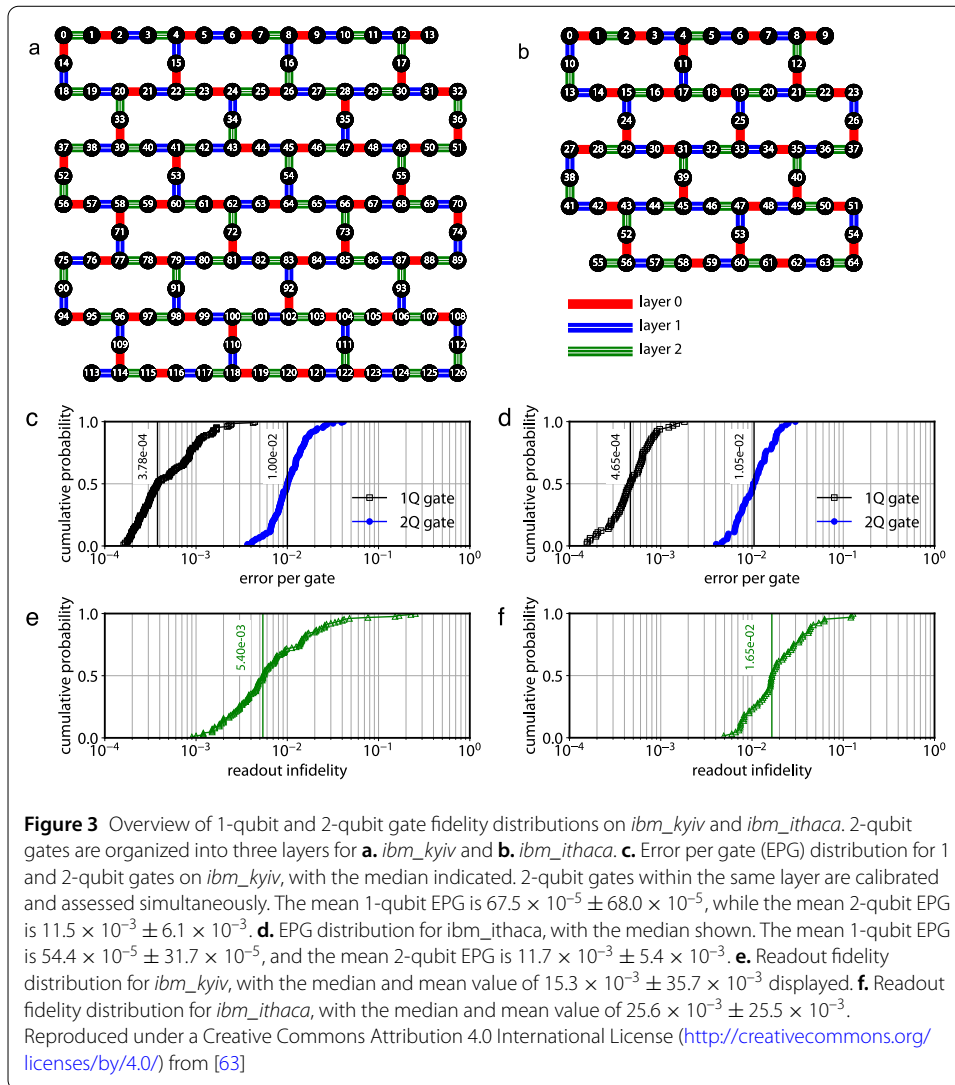


scenarios characterized by substantial entanglement growth, where traditional classical approximations face considerable challenges.

5.3.2 Device performance

For the experimental implementation, the team utilized the IBM Eagle processor, *ibm_kyiv*, which consists of 127 fixed-frequency transmon qubits [21] featuring heavy-hex connectivity. While *ibm_ithaca* was utilized to develop calibration and data-processing strategies, as well as to explore the effects of device non-idealities on error mitigation outcomes [82, 83]. The median T_1 and T_2 times are 288 μs and 127 μs , respectively—unprecedented coherence times for superconducting processors of this scale, enabling the circuit depths achieved in this study [63]. The 2-qubit CX gates between neighboring qubits are realized through the calibration of the cross-resonance interaction. Since each qubit connects to at most three neighbors, all ZZ interactions can be executed in three layers of parallelized CX gates (Fig. 2b). Within each layer, the CX gates are calibrated for optimal simultaneous operation.

One-qubit gates are calibrated to utilize a Gaussian envelope pulse with a width of 49.78 ns, incorporating DRAG correction. A summary of the device properties can be found in Table 4. Two-qubit gates are implemented using the echoed cross-resonance (ECR) pulse sequence [86, 87]. Since no qubit is connected to more than three neighbors, the 2-qubit gates can be organized into three depth-1 layers. Figures 3(a) and 3(b) depict the arrange-



ments for *ibm_kyiv* and *ibm_ithaca*, respectively. Each layer is designed to maintain a uniform 2-qubit gate length of 611.56 ns for *ibm_kyiv* and 512.00 ns for *ibm_ithaca*.

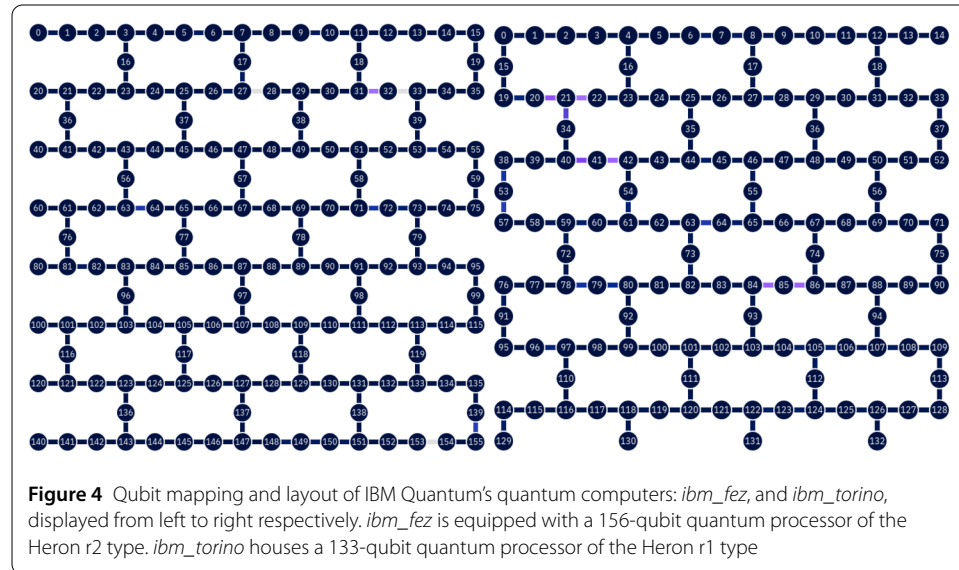
The 2-qubit gates are calibrated simultaneously, and the distribution of 2-qubit randomized benchmarking results performed per layer is summarized in Figs. 3(c) and 3(d). The readout duration for *ibm_kyiv* is set at 782 ns, with average values of $\frac{\chi}{2\pi} = 1.11 \pm 0.20$ MHz and $\frac{\kappa}{2\pi} = 3.80 \pm 1.09$ MHz. While, *ibm_ithaca* exhibits average values of $\frac{\chi}{2\pi} = 1.46 \pm 0.51$ MHz and $\frac{\kappa}{2\pi} = 0.28 \pm 0.04$ MHz, which are notably lower than those of *ibm_kyiv*. Consequently, the readout length for *ibm_ithaca* is extended to 4.55 μ s. The resulting readout fidelity is depicted in Figs. 3(e) and 3(f). As illustrated in Fig. 4, the qubit mapping and layout of IBM's quantum computers—for example: *ibm_fez* and *ibm_torino*—highlight the distinct configurations and processor types employed across these systems.

5.3.3 Quantum-centric supercomputing

IBM Quantum envisions a future centered around quantum-centric supercomputing [88, 89]. This approach aims to facilitate more advanced utility-scale applications while offering a streamlined development environment for users of IBM Quantum, potentially

Table 4 Overview of single qubit characteristics for the 127 qubits of *ibm_kyiv* and the 65 qubits of *ibm_ithaca*. The T_1 and T_2 values are calculated from the average of daily measurements taken over a period of seven days [63]

Qubit properties	<i>ibm_kyiv</i>				<i>ibm_ithaca</i>			
	min.	mean	median	max.	min.	mean	median	max.
T_1 (μs)	85.30	293.39 ± 84.78	287.87	567.55	77.92	180.52 ± 43.44	183.54	278.08
T_2 (μs)	16.18	156.73 ± 109.47	127.49	456.02	19.27	182.88 ± 91.05	183.70	407.15
f_{01} (GHz)	4.34	4.62 ± 0.11	4.61	4.96	4.54	4.74 ± 0.10	4.73	4.93
$f_{01} - f_{12}$ (MHz)	280.56	310.77 ± 11.48	311.07	356.92	316.04	333.99 ± 5.61	333.36	355.56



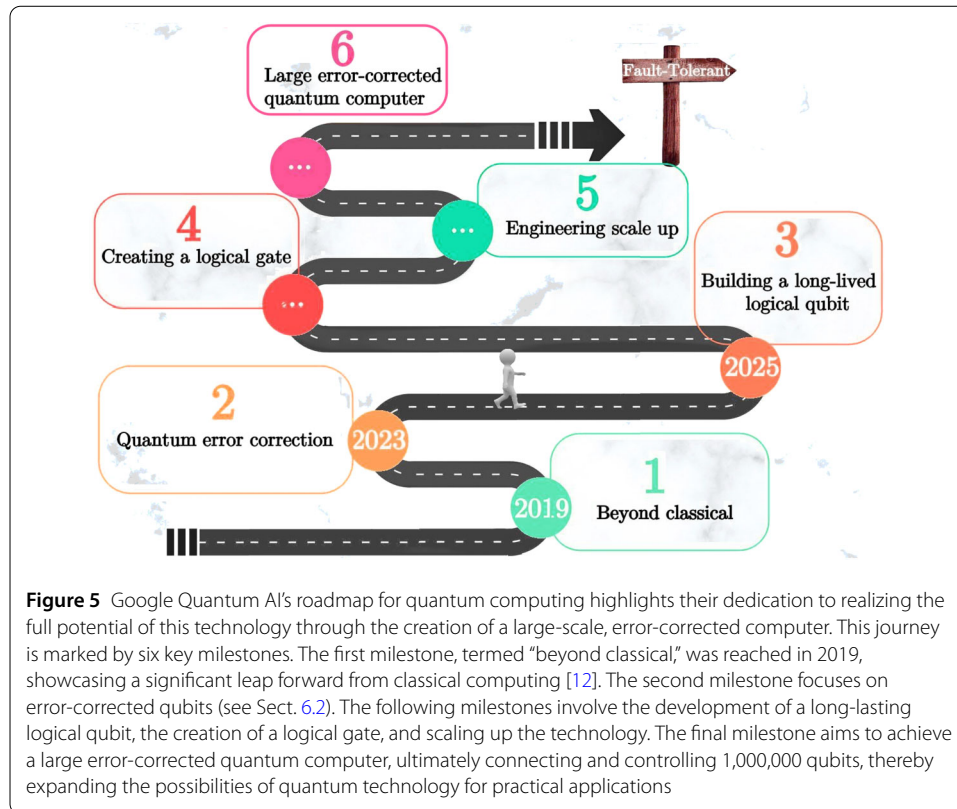
even before reaching full fault tolerance [63, 67, 76, 77, 90, 91]. Such a vision promises to enhance the accessibility and effectiveness of quantum computing, driving innovation and expanding its practical applications. For an in-depth exploration of IBM Quantum's journey in quantum computing and its notable advancements in both hardware and software, readers are encouraged to consult [28]. This review delves into the evolution of IBM Quantum's processors across different generations, highlighting performance metrics for various hardware systems and tracing their development over time.

6 Google's quantum AI

6.1 Overview

Google Quantum AI stands at the forefront of quantum computing innovation, shaping the future of the field with transformative advancements in technology and theory [12]. At the heart of Google Quantum AI's initiatives is the Quantum Artificial Intelligence Lab (QuAIL). This lab has played a crucial role in advancing quantum computing research, leading to the landmark achievement of quantum supremacy [12]. This milestone showcased the ability of quantum processors to surpass classical supercomputers in certain tasks [2, 61].

A key focus of Google Quantum AI's research is the development and optimization of superconducting transmon qubits [92, 93]. Which offer a promising pathway to scalable quantum computing due to their compatibility with current semiconductor technologies and their potential for high fidelity and coherence [94–98]. By mitigating environmental



noise and improving qubit design, Google Quantum AI has effectively prolonged coherence times [99], boosting the stability and reliability of their qubits.

Error management [100, 101] is another vital area of focus for Google Quantum AI. Google Quantum AI has achieved significant reductions in error rates by employing advanced error correction techniques and optimizing qubit designs [99, 102, 103]. These improvements lead to more precise and stable quantum computations. As illustrated in Fig. 5, Google Quantum AI's roadmap outlines six major milestones toward building a large-scale, error-corrected quantum computer, beginning with the 2019 “beyond classical” achievement and progressing toward the ultimate goal of controlling one million qubits.

Several significant advancements have been made by Google Quantum AI, particularly in the areas of quantum error correction [65, 99] and the attainment of quantum computational advantage [12]. This section highlights several of these key developments. For a thorough review of Google's advancements in quantum technology over the past decade (2013–2025), including developments in both hardware and software, readers are recommended to consult the review by AbuGhanem in [30].

6.2 Error-corrected quantum computers

Error-corrected quantum computers [27, 100, 101, 104, 105] represent a significant advancement in the quest for practical quantum computing, addressing one of the fundamental challenges posed by quantum decoherence and operational errors. These systems utilize quantum error correction codes to protect quantum information from the detrimental effects of noise, allowing for reliable computation over longer timescales [106].

By encoding logical qubits across multiple physical qubits, quantum error correction schemes enable the logical qubit to preserve its integrity and functionality, even in the presence of errors affecting some of the physical qubits [65, 99]. This approach not only has the potential to enhance the reliability of quantum computations but also paves the way toward achieving practical quantum computing [63, 67, 107–112].

6.2.1 Suppressing quantum errors

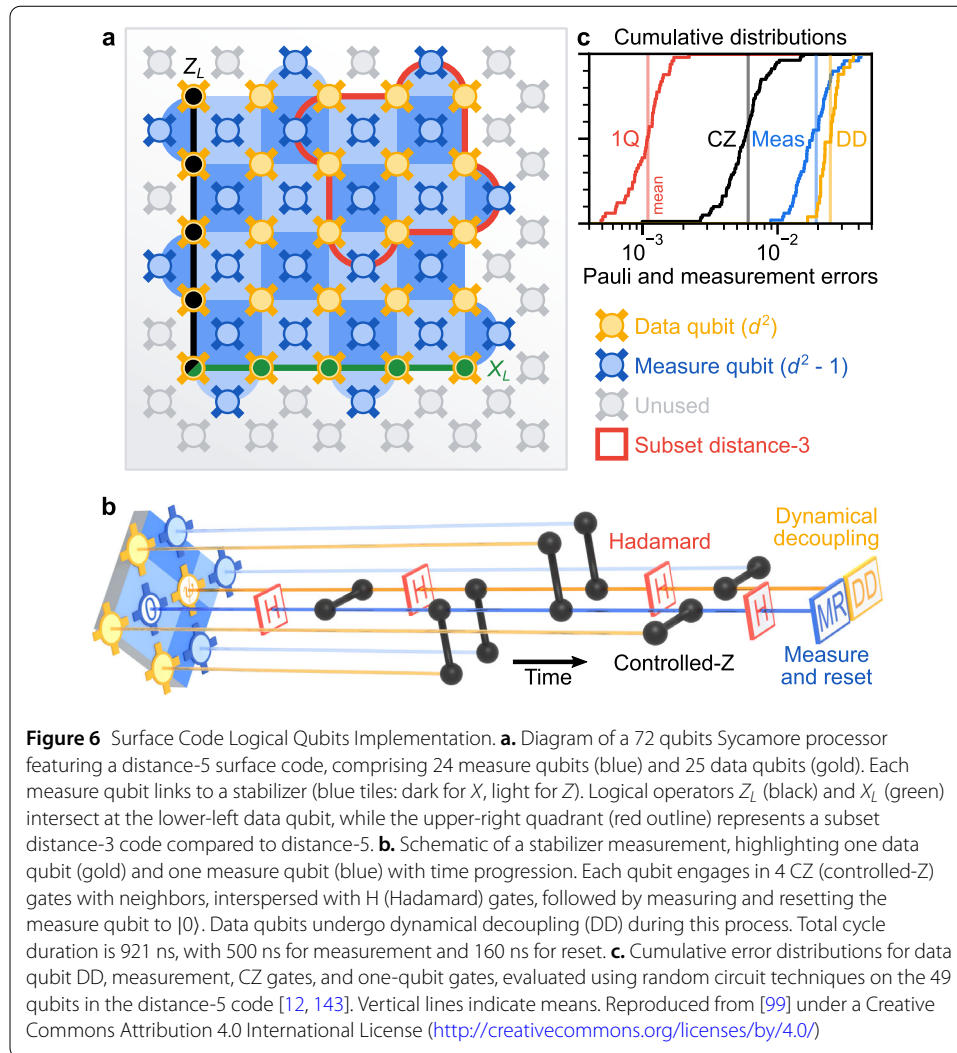
Google Quantum AI has achieved several significant advancements in the realm of quantum error correction [27, 100, 101, 104, 105]. For instance, the study [113] tackled incoherent noise in quantum processors, which disrupts long-range correlations and coherent computation. Utilizing random circuit sampling (RCS) [114–124] and cross-entropy benchmarking (XEB) [12, 19, 114, 125], researchers identified critical phase transitions related to cycle counts and error rates. Their RCS experiment with 67 qubits and 32 cycles showed that quantum processors can achieve a complex computational phase despite noise, surpassing current classical supercomputers and representing a major step forward in managing quantum noise (see Sect. 6.3.2).

A key challenge in QEC is the leakage of quantum information into non-computational states. The study in [126] employs a distance-3 Surface code and a distance-21 bit-flip, to actively eliminate leakage from qubits [127–129]. This technique achieves a tenfold reduction in steady-state leakage, keeping the average leakage population below 1×10^{-3} . By preventing leakage-induced correlated errors, this advancement enhances the practicality of scalable QEC.

Additionally, a control optimization strategy for managing the frequency trajectories of 68 frequency-tunable superconducting transmon qubits is detailed in [130]. This approach enhances gate execution and error mitigation, leading to a significant reduction in physical error rates compared to non-optimized systems. Furthermore, the strategy is anticipated to be effective for larger setups, such as a distance-23 Surface code with 1057 physical qubits, providing a scalable solution that can adapt to various quantum algorithms and architectures.

Recently, the performance of logical qubits in a superconducting qubit processors employing quantum error correction [100, 101] is detailed [99]. QEC effectively reduces operational error rates in quantum processing units, though it demands more time and qubit resources [131, 132]. Various studies have successfully implemented error correction using codes designed for single error rectification, including the color QEC code [133], distance-3 Bacon–Shor code [134], 5-qubit code [135], Surface codes [136, 137], heavy-hexagon code [138], and continuous variable QEC codes [139–142]. However, a critical question remains: will increasing the size of the error-correcting code result in lower logical error rates in practical devices? [99]. Theoretically, if physical errors are sufficiently infrequent within the quantum processor, a reduction in logical errors is expected. Yet, demonstrating this requires scaling the device to implement a code that can correct at least two errors while maintaining high performance.

Google Quantum AI addresses this question in [99], revealing that an increase in physical qubits does enhance logical qubit performance, despite introducing additional error sources. They presented a 72-qubit superconducting processor utilizing a 49-qubit distance-5 Surface code, which slightly outperforms its average subset, a 17-qubit distance-3 Surface code (see Fig. 6). The research also uncovers high-energy events that



impact performance when employing a distance-25 Repetition code, indicating both advancements and ongoing challenges in achieving low logical error rates for practical quantum computing [63, 67, 107–112].

6.2.2 Quantum error correction below the surface code threshold

The Google Quantum AI team, in collaboration with other researchers, has announced a significant advancement in QEC. Their findings, detailed in [65], highlight impressive results in reducing logical error rates relative to physical error rates across multiple Surface code implementations on superconducting quantum processors. Utilizing an upgraded 105-qubit processor, the team successfully implemented a distance-7 code with 101 qubits and a distance-5 code with around 50 qubits. The distance-7 code achieved a logical error rate of $0.143\% \pm 0.003\%$ and a suppression factor of approximately 2.14 ± 0.02 . Notably, this code also improved qubit lifetimes by a factor of 2.4 ± 0.3 compared to the best-performing physical qubit [12, 99, 139, 144, 145]. Furthermore, the data demonstrated a clear scaling of performance as the distance increased from 3 to 5 to 7, suggesting the potential to develop a distance-27 code capable of achieving a 10^{-6} error rate with 1457 physical qubits [65].

The authors also investigate the impact of correlated errors [146, 147], local noise sources, and leakage to higher energy states on logical performance [126, 148]. A comprehensive error budget indicates that correlated errors account for approximately 17% of the overall error, while local gate and measurement errors remain predominant contributors.

QEC [27, 100, 101, 104, 105] aims to achieve high-fidelity logical qubits by spreading quantum information across multiple entangled physical qubits, thereby safeguarding against errors [65]. If the physical operations remain below a critical noise threshold, the logical error rate should decrease exponentially with an increasing number of physical qubits per logical qubit [65]. For error-corrected (surface code) logical qubits [104, 105, 110], this relationship is captured by the approximate equation:

$$\varepsilon_d \propto (p/p_{(\text{thr})})^{\frac{1}{2}(d+1)} \quad (1)$$

where, d denotes the code distance, indicating that $(d\sqrt{2})^2 - 1$ physical qubits are utilized per logical qubit. ε_d and p represent the logical and physical error rates, respectively, while $p_{(\text{thr})}$ is the code's threshold error rate. Therefore, when $p \ll p_{(\text{thr})}$, the logical qubit error rate decreases exponentially with code distance. The error suppression factor

$$\Lambda = \frac{\varepsilon_d}{\varepsilon_{d+2}} \approx \frac{p_{(\text{thr})}}{p} \quad (2)$$

quantifies the reduction in logical error rate when the code distance is increased by two. Although various platforms have showcased different aspects of quantum error correction [99, 133, 136, 144, 149–151], no quantum processor has conclusively demonstrated performance below the threshold.

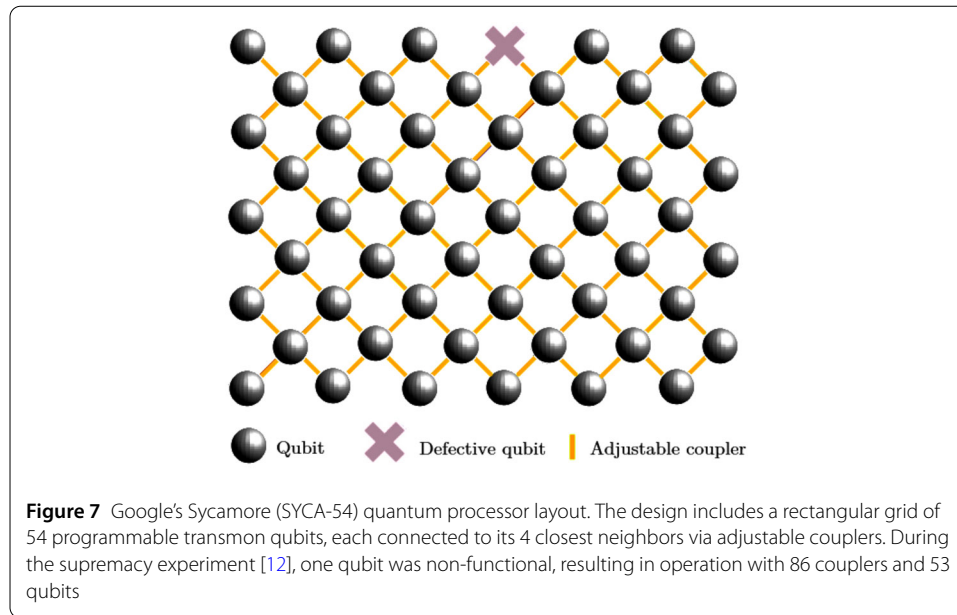
Moreover, the paper presents a real-time decoding system capable of processing error syndromes within the stringent timing constraints of superconducting processors [152–158]. This system successfully maintains below-threshold performance while achieving a decoding latency of about 63 microseconds. The comparison of real-time decoding to offline neural network decoding [159] shows only a modest reduction in accuracy, emphasizing the practicality of the approach.

These results represent significant advancements towards fault-tolerant quantum computing using superconducting qubits, specifically through the implementation of below-threshold Surface codes [65]. However, the authors acknowledge ongoing challenges, particularly concerning noise floors associated with correlated error bursts and the increasing complexity of real-time decoding as system size scales [146, 147]. Addressing these issues will be essential for realizing the full potential of fault-tolerant quantum computing and achieving practical applications in the near future [2].

Looking ahead, Google Quantum AI aims to enhance qubit performance and expand its quantum infrastructure, with the ambitious goal of building a fully error-corrected quantum computer by integrating thousands of surface-encoded logical qubits. This effort seeks to significantly lower error rates and surpass the limitations of existing technology [160].

6.3 Google quantum AI's quantum processors

Google Quantum AI's quantum processors are among the forefront of computational technology, engineered to tackle challenges beyond the reach of classical computers [12,



113]. The company has created several notable quantum processors, such as *Foxtail* [94, 161], *Bristlecone* [162], *Sycamore* [12], and, more recently, *Willow* [65]. These processors feature sophisticated control and precision, allowing for the manipulation and measurement of qubits with exceptional accuracy [12, 113].

Google Quantum AI has leveraged its quantum processors to investigate new physical phenomena, including the observation of time crystals [79] and Majorana edge modes [81], alongside experimental discoveries like robust bound states of interacting photons [163] and the noise resilience of Majorana edge modes in Floquet evolutions [81]. By consistently improving their hardware capabilities, Google Quantum AI is advancing quantum technology and enabling breakthroughs across multiple domains. This section specifically examines the performance of Google's latest quantum processors. For a comprehensive review of Google's progress in advancing quantum computing over the past decade, see the review by AbuGhanem [30].

6.3.1 Sycamore-54

Google's Sycamore (SYCA-54) quantum processor is designed with a 2-dimensional grid of 54 superconducting transmon qubits [12]. These qubits, arranged in a rectangular lattice with connections to their four nearest neighbors (see Fig. 7). Remarkably, SYCA-54 achieved a significant milestone by completing a computation in approximately 200 seconds—a task estimated to take a classical computer about 10,000 years [12].

In a landmark experiment, Google's Quantum AI team programmed SYCA-54 to perform RCS [114–124]. This computation was conducted in a complex space of 9×10^{15} , far exceeding traditional computing capabilities. While performing the same task on the Summit supercomputer was initially projected to take about 10,000 years, SYCA-54 completed it in just 3 minutes and 20 seconds, highlighting the potential of quantum computing to tackle problems that classical methods struggle with [2].

To achieve quantum supremacy [19, 61], SYCA-54 incorporated various technological innovations, allowing for the execution of Random Quantum Circuits (RQC). These cir-

Table 5 System attributes of the key features of Google's SYCA-54 quantum processor, as detailed in [12]

System features	Mean	Minimum	Median	Maximum	StDev.
Qubit frequency at readout (GHz)	5.784	5.180	5.760	6.730	0.384
Qubit idle frequency (GHz)	6.665	6.553	6.661	6.830	0.064
Qubit maximum frequency (GHz)	6.927	6.634	6.911	7.237	0.120
Qubit anharmonicity (GHz)	-0.208	-0.226	-0.208	-0.202	0.0047
T1 at Idle Frequency (μ s)	16.035	9.70	15.50	27.8	3.991
Qubit-resonator coupling (MHz)	72.10	63.90	72.30	79.10	2.794
Resonator linewidth (MHz)	0.691	0.360	0.640	1.250	0.233
Readout drive frequency (GHz)	4.588	4.470	4.618	4.690	0.076
Readout error $ 1\rangle$ isolated (%)	4.984	2.40	4.5	9.40	1.774
Readout error $ 0\rangle$ isolated (%)	1.158	0.20	1.0	3.80	0.7548
Readout error $ 1\rangle$ simultaneous (%)	5.477	2.50	5.1	15.60	2.253
Readout error $ 0\rangle$ simultaneous (%)	2.256	0.40	1.8	13.0	2.128
1-qubit RB ^a e_1	0.22	-	0.19	-	0.10
1-qubit RB e_1 ($\pi/2$ gate set)	0.16	-	0.15	-	0.06
1-qubit RB tomographic e_1 purity	0.15	-	0.14	-	0.04
1-qubit XEB ^b e_1 isolated (simultaneous)	0.15 (0.16)	-	0.13 (0.14)	-	0.05 (0.05)
2-qubit XEB e_2 isolated (simultaneous)	0.36 (0.62)	-	0.30 (0.60)	-	0.17 (0.24)
Measurement e_m isolated (simultaneous)	3.05 (3.77)	-	2.83 (3.50)	-	1.09 (1.61)

^aRandomized benchmarking.^bCross-entropy benchmarking.

circuits consist of alternating layers of single-qubit and two-qubit gates organized into cycles, with each cycle characterized by a sequence of gates culminating in a measurement.

The SYCA-54 uses transmon qubits [21], nonlinear superconducting resonators operating between 5 and 7 GHz. Each qubit utilizes its two lowest energy states and is controlled by a microwave drive for excitation and magnetic flux for frequency tuning. For state readout, qubits connect to linear resonators [92], and adjustable couplers [164, 165] enable dynamic tuning of qubit coupling from fully disengaged to 40 MHz. During the quantum supremacy experiment, one qubit malfunctioned, leaving the processor functional with 53 qubits and 86 couplers [26].

The Google Quantum AI team also tested the largest quantum circuit composed of 53 qubits and 20 cycles, collecting 30×10^6 samples over ten runs. The observed $\mathcal{F}_{\text{XEB}}$ for the truncated circuits was $(2.24 \pm 0.21) \times 10^{-3}$, with claims that the average fidelity of these circuits is at least 0.1% [12]. The key features of Google's SYCA-54 quantum processor are summarized in Table 5.

Following these results, discussions arose about the accuracy of the classical computation estimates. Huang et al. [166] presented a tensor network algorithm capable of completing the task in under 20 days. Pednault et al. [167] argued that a similar computation could be executed on the Summit supercomputer in only a few days, while Pan et al. [168] described a tensor network approach that executed the computation in 15 hours using a cluster of 512 GPUs. These findings suggest that the differences between quantum and classical computing may not be as pronounced as initially claimed [12]. Characterization of the Sycamore quantum processes through quantum tomography experiments is detailed in [169, 170].

6.3.2 Sycamore-67 and Sycamore-70

In 2023, Google Quantum AI and its partners conducted key experiments aimed at achieving quantum computational supremacy with the SYCA-67 and SYCA-70 quantum proces-

sors, featuring 67 and 70 qubits, respectively [2, 61, 113, 171]. These studies explored the interplay between quantum dynamics and noise, revealing phase boundaries that clarify how noisy quantum devices can enhance their computational capabilities. Notably, the team made significant advances in RCS [114–124], achieving a fidelity of 1.5×10^{-3} for the 67-qubit processor after 32 cycles involving 880 entanglement gates. This marked a considerable increase in circuit complexity while maintaining fidelity compared to previous research.

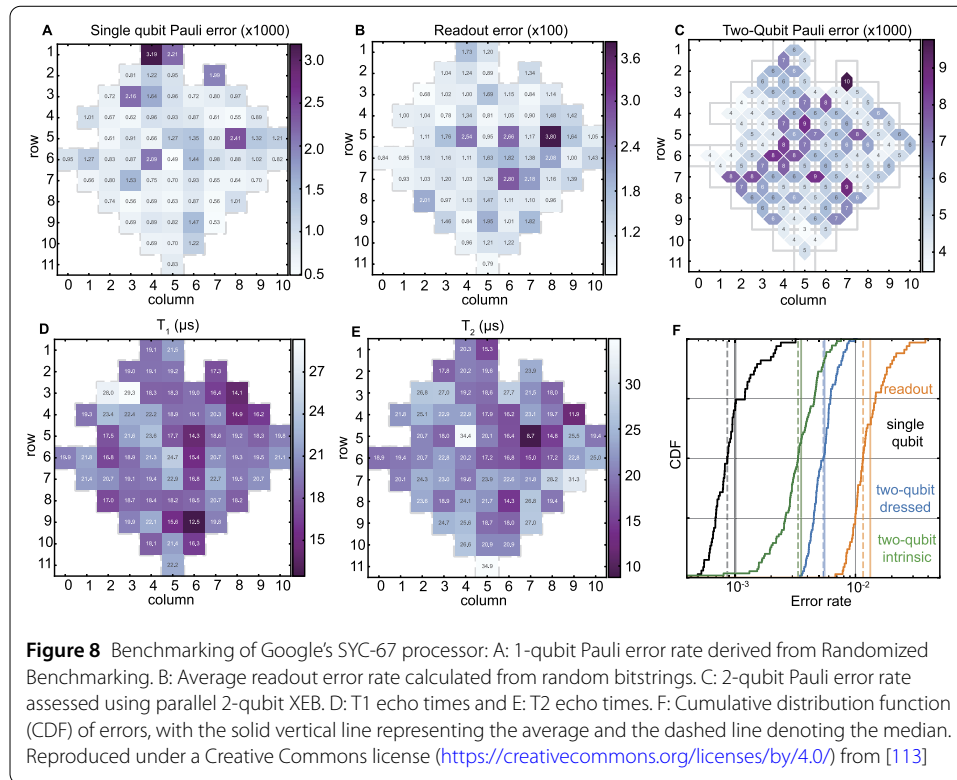
To assess the computational resources needed for simulating RCS [114–124], it's important to account for supercomputers' constraints, including floating-point operations per second (FLOPS), memory capacity, and bandwidth. The study in [113] provides estimates for simulating the largest RCS instances from previous studies [12, 32, 62] and recent Google research. These estimates involve sampling 1 million uncorrelated bit-strings with fidelity comparable to experimental results, using the leading supercomputer, Frontier, which has a theoretical peak performance of 1.7×10^{18} single-precision FLOPS and 128 GB of RAM per GPU.

The researchers demonstrated that their RCS experiment with 67 qubits over 32 cycles requires computational resources that exceed those of current classical supercomputers, even when accounting for unavoidable noise. They highlighted that global correlations [12, 19, 114, 125] in weak noise phases provide better protection against “spoofing” attacks compared to previous Boson-Sampling experiments [172]. Recent metrics in Boson-Sampling have mainly been influenced by local correlations [9, 173, 174]. For further insights into Boson-Sampling and Gaussian Boson-Sampling experiments, and their role in achieving quantum supremacy with photonic quantum computers [175], readers are referred to [1, 11].

Figures 8 and 9 provide valuable insights into the performance of the SYCA-67 and SYCA-70 quantum processors. They present key performance metrics, including one-qubit and 2-qubit Pauli error rates, readout error rates, and T1 and T2 echo times. Additionally, the figures include the cumulative distribution function (CDF) for various errors, highlighting the devices' capabilities and limitations under different conditions [113]. Figure 10 displays the upper limits on the bond dimension necessary to reach a target fidelity of $\mathcal{F} = 10^{-4}$, comparing both analytical and numerical findings as functions of the number of qubits and circuit depth [113]. It also details the memory requirements for storing complex tensors of dimensions $(2^{n/2} \times \chi)$, highlighting the different memory capacities of various computing resources, including laptops, cluster nodes, and advanced systems like Frontier.

6.4 The Willow quantum processors

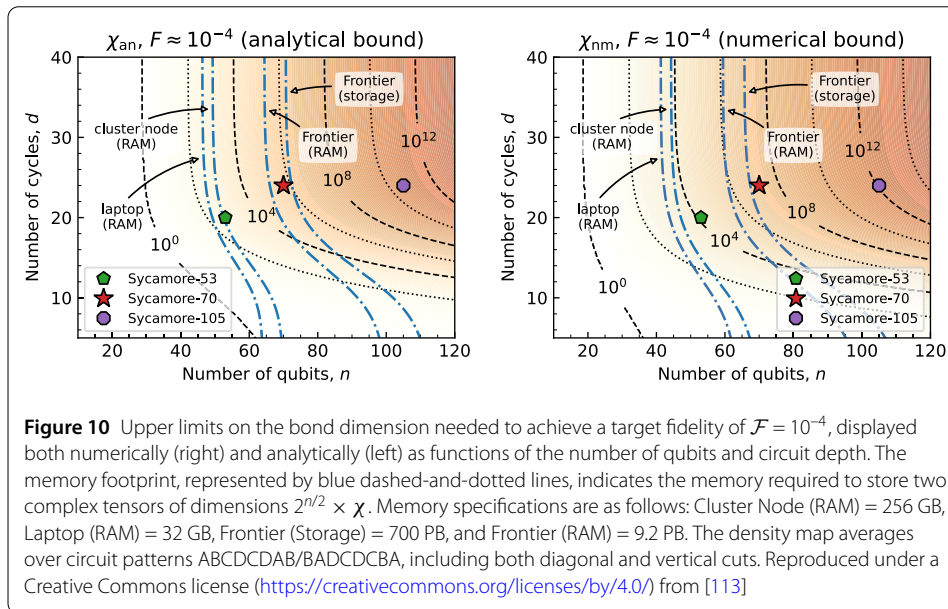
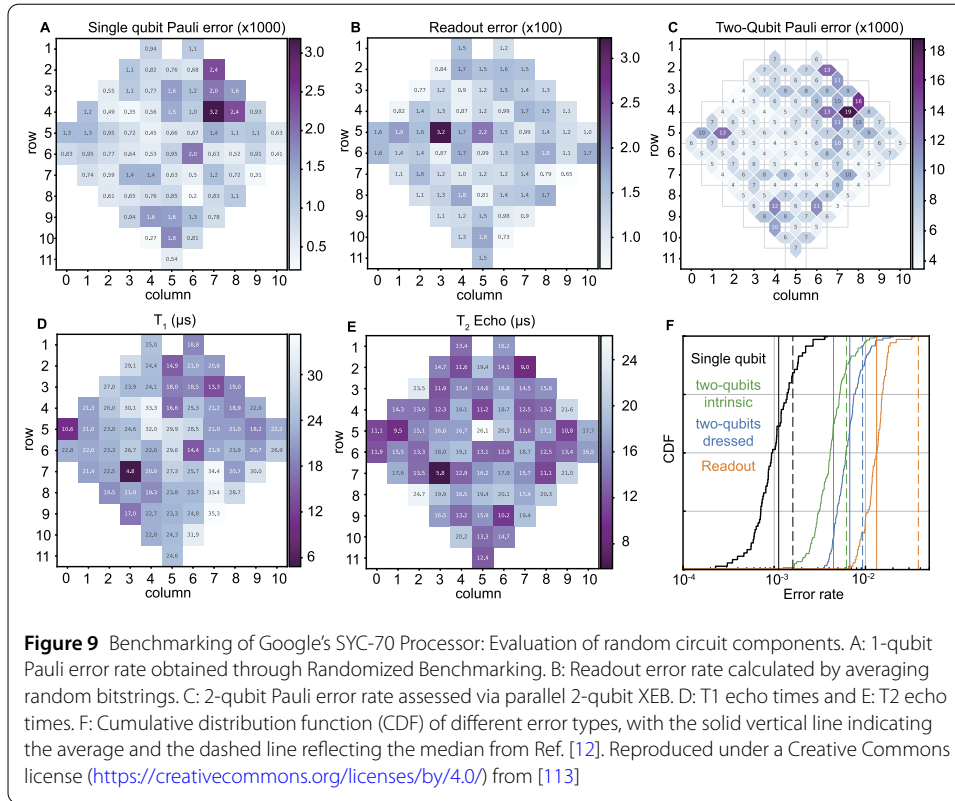
The study by Google Quantum AI and collaborators [65] demonstrates QEC operating below the surface code threshold, a critical milestone for scalable fault-tolerant quantum computing. Utilizing their latest superconducting processor, Willow, the researchers implement two surface code memories: A 72-qubit processor was used to implement a distance-5 code with a real-time decoder and a 105-qubit processor enabled the realization of a distance-7 code, which exhibits exponential suppression of logical errors. The distance-7 code, composed of 101 qubits, achieves a logical error rate of 0.143% per cycle, surpassing the lifetime of its best physical qubit by a factor of 2.4 [65]. See Fig. 11 a. Figure 11 b presents the distribution of operation error probabilities observed from this



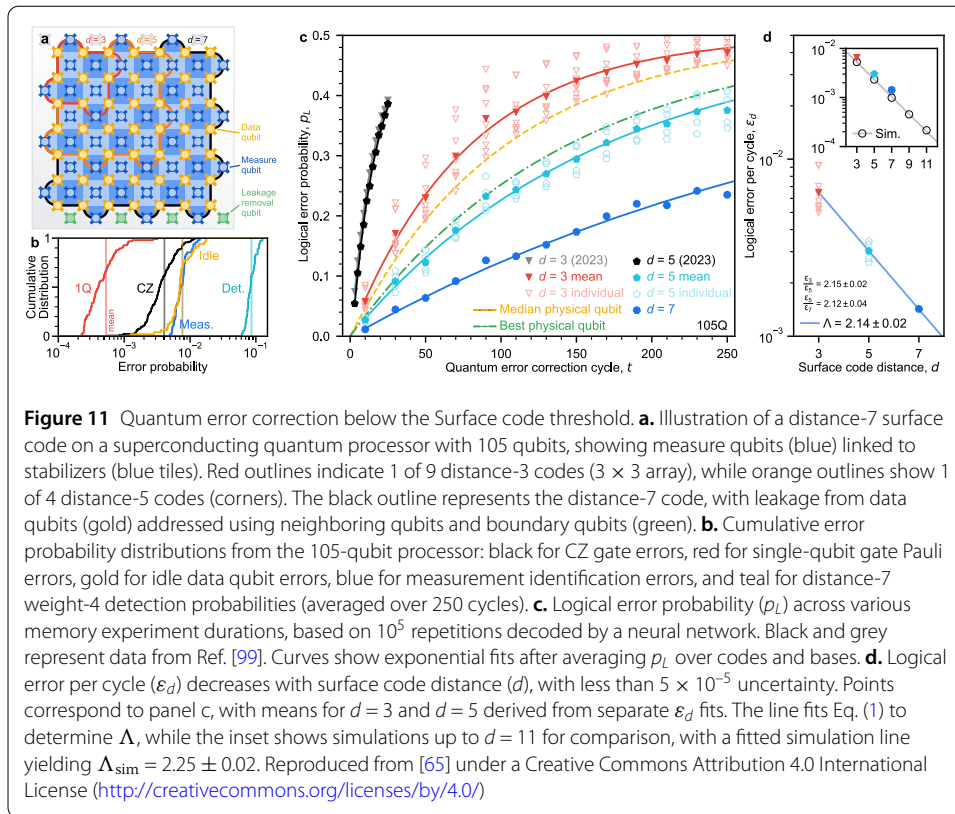
advanced processor. Additionally, the layouts of the distance-3 and distance-5 grids on the 105-qubit processor are illustrated in Fig. 12 a and b, respectively.

A key achievement of this work [65] is maintaining below-threshold performance even with real-time decoding, which achieves an average latency of 63 microseconds over a million cycles, meeting the stringent speed requirements of superconducting qubit architectures. Additionally, high-distance repetition codes (up to distance 29) were tested, revealing that logical performance is now limited by rare correlated errors occurring approximately once every hour [65]. The results indicate that with continued improvements in coherence, calibration, and error mitigation, superconducting qubit platforms may soon meet the demands of large-scale fault-tolerant quantum algorithms. Furthermore, the implementation of real-time decoding marks a significant step toward practical quantum computation, where classical processing must keep pace with quantum operations.

Additionally, Willow's ability to outperform classical supercomputers in benchmarks such as random circuit sampling highlights its potential for beyond-classical computations, pointing towards a future where quantum computers can solve practical, real-world problems that are out of reach for today's classical systems. According to [66], the Willow chip achieved a groundbreaking milestone by solving a computational problem (RCS benchmark) in less than five minutes, a task that would require the most advanced supercomputers of today an unfathomable 10^{25} years to complete. This timescale, equivalent to 10 septillion years, that is far surpassing known timescales in physics and the current age of the universe, and challenges our understanding of computational limits. These advancements, coupled with Willow's scalable architecture, pave the way for large-scale, fault-tolerant quantum algorithms, aligning closely with the goal of creating quantum computers that are not just theoretically powerful, but practically useful.



As the field advances, Google Quantum AI's ongoing contributions [30] will likely play a crucial role in shaping the future of quantum computing and fostering the next wave of technological innovation.



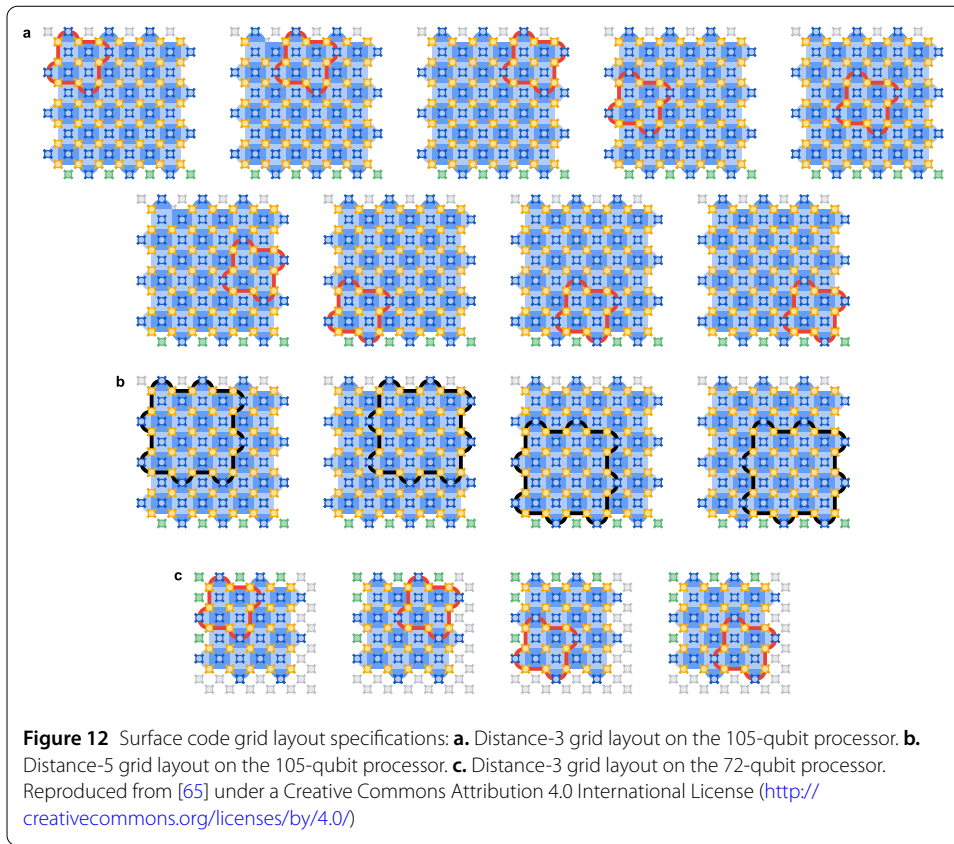
7 The Rigetti Computing

7.1 Overview

Rigetti Computing is a prominent player in the quantum computing industry, with its innovative strategies for minimizing errors in quantum computation [176–178] and the development of advanced quantum computing systems reflecting its strong commitment to advancing the field. The company's primary goal is to build the world's most powerful quantum computers using cutting-edge quantum technology. Rigetti Computing, headquartered in Berkeley, California, and was founded in 2013, offers cloud-based access to its quantum computing systems and provides users with software tools for developing quantum applications [179]. Leveraging advanced technologies and strategic collaborations, Rigetti Computing is well positioned to further advance the capabilities of quantum computing and make significant contributions to its ongoing development.

7.2 Chip design and fabrication process

Rigetti Computing's chip design and fabrication process is centered around the creation of high-quality quantum processor chips, utilizing superconducting microwave devices [178, 180–185]. The company uses advanced modeling and simulation tools to design quantum circuits and produce masks for manufacturing at their own facility, Fab-1. This captive foundry combines cutting-edge silicon semiconductor and MEMS technologies with novel techniques to create superconducting qubits and microwave device layers [178, 185]. Key materials include aluminum, indium, and niobium, which are processed through techniques like etching, lithography, and deposition to achieve ultra-low-loss superconducting devices. Rigetti also incorporates 3D integration, through-silicon vias, and



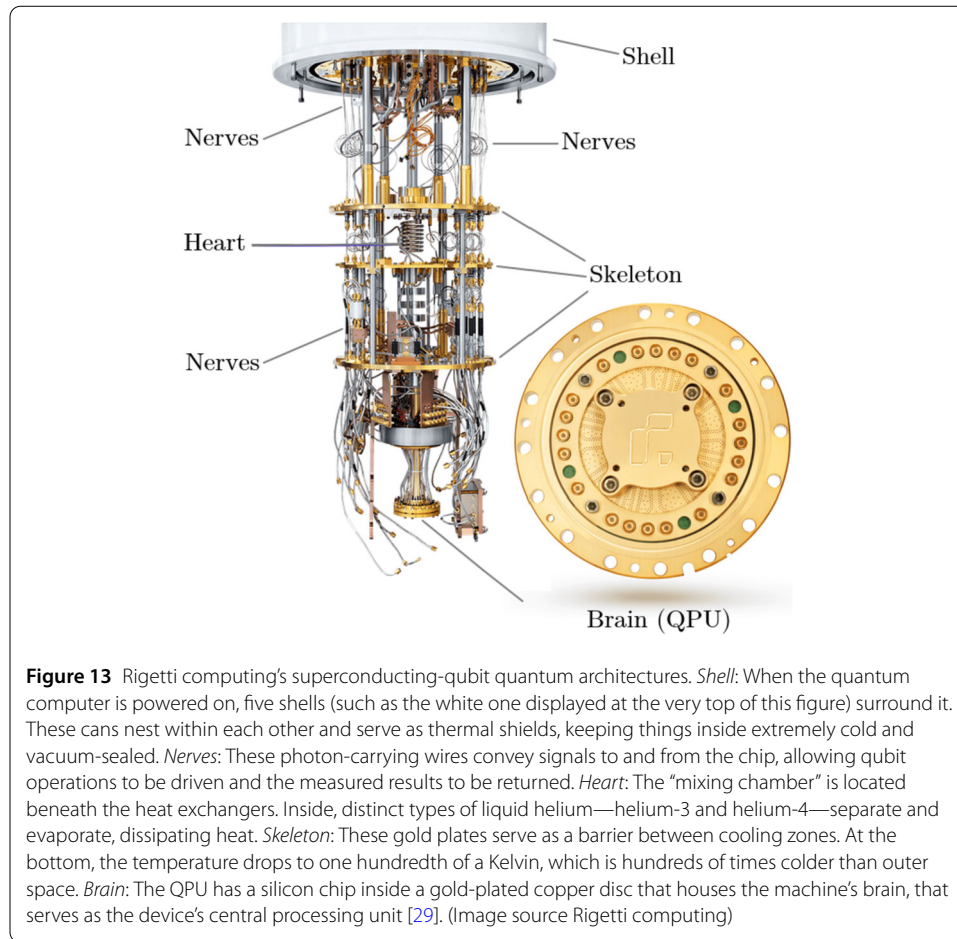
superconducting caps to enhance scalability, coherence, and minimize crosstalk. Their continuous testing and cryogenic systems enable rapid iteration and performance optimization for each design and manufacturing update.

7.3 Qubit technology

Rigetti Computing's quantum processors (as detailed in Sect. 7.5) are based on superconducting qubits, which offer scalability, fast gate times, low-latency logic, and efficient execution, making them ideal for near-term quantum applications and error correction. Each qubit is composed of a non-linear Josephson inductance and an ultra-low-loss capacitor, forming a resonant circuit for quantum operations [178, 185]. These qubits are coupled to superconducting resonators for readout and interconnected with on-chip capacitances for multi-qubit logic via microwave or DC pulses [178, 180–185]. To achieve scalability, Rigetti's Fab-1 facility employs advanced manufacturing techniques, including superconducting through-silicon vias and flip-chip bonding, which enhance qubit performance and enable better connectivity [178, 185]. Additional scalability features like multiplexed readout are integrated into the chip design to efficiently manage large arrays of qubits (see Fig. 13).

7.4 Control systems

Rigetti Computing's control systems enable the reliable and programmable manipulation of qubits in superconducting quantum computers. The system includes custom FPGA-based hardware with a powerful CPU and memory for low-latency, high-throughput



quantum-classical operations. It drives quantum processors, calibrates multi-qubit gates, and reads qubit states while ensuring phase coherence and stability [182].

Rigetti's innovative interconnect system uses high-density flexible circuits to replace traditional coaxial cables, reducing signal loss and thermal load. These circuits include passive components that improve signal quality and reduce noise. The system's 3D signal routing improves chip compactness and signal integrity, supporting scalable, high-fidelity qubit performance with minimal crosstalk. This architecture allows quantum processors to scale to hundreds of qubits, facilitating the development of advanced quantum algorithms and progress toward quantum advantage.

7.5 Rigetti Computing's superconducting quantum processors

Since 2017, Rigetti Computing has been offering cloud-based access to its quantum computers. With every new generation, the company has consistently advanced both the number of qubits and the fidelity of quantum operations. These ongoing improvements are driving Rigetti toward the crucial milestone where quantum computers are anticipated to outperform classical systems in computational capability [2].

Rigetti's quantum computing systems are built around superconducting qubit processors, which harness the principles of superconductivity to achieve extended coherence times and rapid gate operations. These processors serve as the core of Rigetti's quantum computing platform, enabling the efficient and reliable execution of sophisticated

quantum algorithms. The company has deployed a variety of quantum systems over time, including models such as AGAVE, ACORN, ASPEN-1, ASPEN-4, ASPEN-7, ASPEN-8, ASPEN-9, ASPEN-10, ASPEN-11, ASPEN-M-1, ASPEN-M-2, ASPEN-M-3, ANKAA-1, ANKAA-2, ANKAA-9Q-1 and ANKAA-9Q-3. These quantum processors are made available through the AWS Braket Service (see Sect. 20). Additionally, Rigetti has introduced its latest innovation, the NOVERA QPU, which shares the same cutting-edge technology as the ANKAA systems. Designed for high-performance applications, the NOVERA QPU marks a substantial advancement, offering superior capabilities for tackling complex computational problems.

Table 6 outlines the system characteristics of Rigetti's cutting-edge quantum processing units (QPUs), ranging from the initial 8-qubit AGAVE devices, deployed on June 4, 2017, to the advanced, large-scale ANKAA systems, including ANKAA-3, which was launched on December 20, 2024. The table includes critical performance metrics, such as qubit count, coherence time (T2), relaxation time (T1), as well as single- and two-qubit gate operation times and their corresponding fidelities, for each quantum system across this timeline.

Rigetti's quantum systems are meticulously designed to mitigate errors, a major challenge in quantum computing [176–178, 181, 186]. The company's focus on improving qubit coherence, gate fidelity, and error correction protocols plays a critical role in enhancing the overall reliability and scalability of quantum operations.

7.6 Rigetti's Novera™ QPU

The Novera™ QPU is a 9-qubit quantum processor from Rigetti, designed to offer cutting-edge performance and flexibility for research and development. Based on Rigetti's advanced ANKAA-class architecture, Novera features a square lattice of qubits with tunable couplers for high-fidelity two-qubit operations. It provides access to quantum technology, allowing users to customize and rewire the system for their specific goals. Built using the same high-quality components as Rigetti's larger systems, Novera is ideal for both experimentation and solving complex quantum challenges. Most importantly, it's available for immediate shipment, empowering researchers to take their work to the next level immediately.

7.7 QCS platform

Rigetti's Quantum Cloud Services (QCS) platform enables ultra-low latency (<1 millisecond) communication between classical hardware and its quantum processing units (QPUs), facilitating high-performance hybrid quantum-classical computation. Unlike competitors that rely on internet-based connections with long latency, Rigetti's approach tightly couples the QPU to classical resources, ensuring efficient, real-time interaction for optimal performance.

7.8 Quantum operating system and APIs

Rigetti Computing offers a suite of network APIs that integrate with various classical resources, providing access to essential quantum operating system functions like authentication, circuit submission, scheduling, and memory management. These APIs (application programming interface) are supported by Rigetti SDKs (software development kit), enabling developers to build custom software.

Table 6 System characteristics of Rigetti’s state-of-the-art quantum processing units, spanning from their 8-qubit AGAVE device, launched on June 4, 2017, to the large-scale ANKAA systems, including ANKAA-3, which was deployed on December 20, 2024. Key performance metrics are provided, such as qubit count, coherence time (T_2), relaxation time (T_1), single- and two-qubit gate operation times, and the fidelities of these gate operations for each quantum system

No.	Quantum System	System parameters ^a							
		Qubits	T1 (μ s)	T2 (μ s)	1-Q gate time	2-Q gate time	1-Q Fidelity ^b	2-Q Fidelity	Deployment Date
1	AGAVE *	8	13.38	15.05	50 (ns)	169 (ns)	96.00%	87.00%	June 4, 2017
2	ACORN *	19	20.3	10.9	50 (ns)	213 (ns)	98.63%	87.50%	December 17, 2017
3	ASPEN-1 *	16	8	12	-	-	93.23%	90.84%	November 15, 2018
4	ASPEN-4 *	13	30.47	20.13	60 (ns)	276 (ns)	99.88%	94.42%	March 10, 2019
5	ASPEN-7	28	41	35	80 (ns)	340 (ns)	99.23%	95.20%	November 15, 2019
6	ASPEN-8	31	24	12	60 (ns)	126 (ns)	99.22%	94.34%	May 5, 2020
7	ASPEN-9	32	33	16	48 (ns)	168 (ns)	99.39%	94.28%	February 6, 2021
8	ASPEN-10	32	30	13	40 (ns)	184 (ns)	99.37%	94.37%	November 4, 2021
		Qubits	T1 (μ s)	T2 (μ s)	1-Q Fidelity	CZ Fidelity ^c	XY Fidelity ^d	iSWAP ^e	
9	ASPEN-11	40	30	14	99.8%	92.7%	91.0%	-	December 15, 2021
10	ASPEN-M-1	80	29	22	99.8%	93.7%	94.6%	-	February 15, 2022
11	ASPEN-M-2	80	31	18	99.8%	91.3%	90.0%	-	August 1, 2022
12	ASPEN-M-3	80	22	24	99.9%	94.7%	95.1%	-	December 2, 2022
13	ANKAA-1	84	10.5	1.8	99.7%	93.3%	-	94.9%	March 2023
14	ANKAA-2	84	12	13	99.9%	-	-	97.5%	December 20, 2023
15	ANKAA-9Q-1	9	16.8	13	99.9%	-	-	97.8%	December 2023
16	ANKAA-9Q-3	9	21	24	99.9%	-	-	99.2%	June 27, 2024
17	ANKAA-3	84	22	19	99.9%	-	-	99.0%	December 20, 2024

^aThe parameter values for the quantum machines ranked from 1 to 4 (denoted by *) are presented as mean (average) values. In contrast, the parameter values for the quantum machines ranked from 5 to 17 are provided as median values, as reported by Rigetti Computing [180].

^bThe fidelity of the single-qubit gate.

^cThe median fidelity of the 2-qubit CZ gate per operation.

^dThe median fidelity of the 2-qubit XY gate per operation.

^eThe median fidelity of the 2-qubit iSWAP gate per operation.

7.8.1 Quil

Quil [179] (Quantum Instruction Language), developed by Rigetti, is a quantum programming language that provides an abstraction for programming quantum processors (QPUs). Resembling an assembly language, Quil combines gate-level and pulse-level descriptions of quantum circuits and incorporates classical instructions and shared memory. All Rigetti QPUs execute quantum programs described using Quil.

7.8.2 Quil-T

Quil-T is an extension of the Quil language [179] that allows for the control of continuous time dynamics in microwave signals sent to Rigetti’s quantum integrated circuits. It enables programmable gate definitions and pulse parameters, allowing users to edit pulses and adjust timing. Quil-T integrates with the standard programming workflow through pyQuil [187], the quilc compiler, and Rigetti APIs, providing access to gate calibrations on Rigetti QPUs.

7.9 Rigetti’s software tools

Rigetti offers a suite of open-source tools for quantum programming and experimentation through its Quil SDK. These tools include:

- **PyQuil:** A Python library that enables developers to write and run quantum programs using Quil [187]. It allows for easy generation, compilation, and simulation of quantum programs, either on Rigetti QPUs or using the Quantum Virtual Machine (QVM).
- **Quilc:** An optimizing compiler that converts Quil code into optimized Quil for Rigetti QPUs. It can be configured with an instruction set architecture (ISA) to target specific native gates and QPU topologies. Quilc enhances program execution fidelity by optimizing circuits based on specific QPU characteristics, with the option to disable optimizations for precise control [188].
- **QVM:** Rigetti's open-source quantum simulator that emulates the execution of Quil programs [179], with or without noise. The QVM simulates quantum operations using classical control and supports large programs with many qubits through just-in-time compilation, offering features like stochastic state evolution and noise modeling.

Together, these tools provide a comprehensive environment for quantum program development, simulation, and optimization.

8 The Zuchongzhi quantum computers

8.1 Overview

In [62], researchers explored the capabilities of Zuchongzhi 2.0, a quantum computing system with 56 superconducting qubits [62]. The system demonstrated exceptional performance in carrying out sampling tasks, completing a complex task in just 1.2 hours—an accomplishment that would take the world's most powerful supercomputer (at the time of the experiment), *Summit*, approximately 8 years to finish. Building on this success, the team introduced Zuchongzhi 2.1, an upgraded version featuring 66 qubits. This new iteration showed significant improvements in quantum advantage, surpassing the earlier version in both performance and efficiency. These findings underscore the potential of Zuchongzhi in advancing quantum information processing, marking a notable step forward in both quantum computing hardware and its ability to outperform classical computing systems.

8.2 Zuchongzhi 2.0

A cutting-edge 2-dimensional programmable superconducting quantum processor, named Zuchongzhi 2.0 [62], has been developed with a tunable coupling architecture. It incorporates 66 functional qubits and 110 couplers, making it a highly versatile system [62], see Fig. 14. To assess the system's performance, a random quantum circuits sampling experiment for benchmarking was conducted, involving a system size of 56 qubits and 20 cycles. The computational effort required to simulate this task classically is estimated to be two to three orders of magnitude greater than that of earlier work with the 53-qubit *Sycamore* processor [12]. Zuchongzhi 2.0 thus marks a significant leap in quantum computational performance, particularly in large-scale sampling tasks that remain classically intractable [62]. Beyond speed, the quantum processor excels in various operational aspects. It achieves high-fidelity 1-qubit gate operations, with an average fidelity of 99.86%, and 2-qubit gate operations with an average fidelity of 99.41%. The readout process is also highly accurate, with an average fidelity of 95.48%. These results demonstrate a quantum computational advantage, one that would be unattainable for classical systems within a practical timeframe.

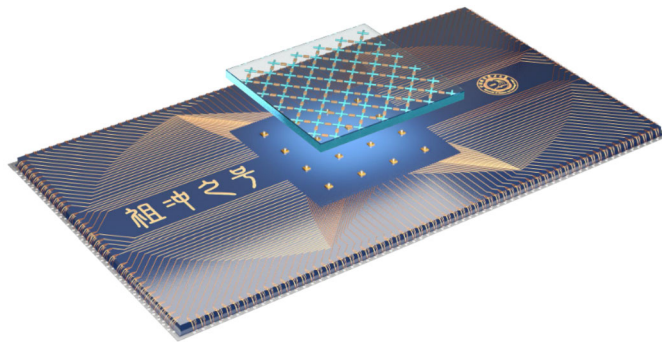


Figure 14 Schematic representation of the Zuchongzhi quantum processor's configuration. This quantum processing unit comprises a pair of sapphire chips. The first chip accommodates 66 qubits and 110 couplers, with each qubit intricately linked to its four adjacent qubits, excluding those situated at the edges. The second chip encompasses the readout elements, control lines, and interconnections. The alignment and integration of these substrates are facilitated through the utilization of indium bumps. Reproduced under Creative Commons license (<https://creativecommons.org/licenses/by/4.0/>) from [62]

8.3 Zuchongzhi 2.1

In February 2022, a quantum supremacy regime was achieved through the introduction of Zuchongzhi 2.1, a 66-qubit 2-dimensional superconducting quantum processor. Compared to Zuchongzhi 2.0 [62], Zuchongzhi 2.1 [32] exhibits substantial improvements in readout performance, with 97.74% average fidelity. Zuchongzhi 2.1 enables the execution of larger-scale RQCS, boasting a system scale of up to 24 cycles and 60 qubits. The achieved sampling task is remarkably more challenging, approximately 6 orders of magnitude, than that previously accomplished by Google's Sycamore quantum processor [12], and 3 orders of magnitude more than the task performed on Zuchongzhi 2.0 [62].

State-of-the-art classical algorithms and supercomputers would require tens of thousands of years (approximately 4.8×10^4 years) to simulate the RCS experiment, while Zuchongzhi 2.1 remarkably accomplishes it in just around 4.2 hours (10 million times faster than the fastest supercomputer at the time). This significant reduction in time consumption underscores the considerable quantum computational advantage offered by Zuchongzhi 2.1 [32].

8.4 Zuchongzhi

The Center for Excellence in Quantum Information and Quantum Physics (CEQIQP) under the Chinese Academy of Sciences (CAS) has introduced a novel quantum computing platform called Zuchongzhi, which boasts a significant improvement over its predecessor, Zuchongzhi 2.1, by featuring 176 qubits. The platform offers users the opportunity to engage in simple quantum computing programming and explore new applications through remote access [189]. By incorporating control interfaces for 110 coupled qubits, the team has empowered users to manipulate a total of 176 quantum bits, surpassing the capabilities of the previous 66-qubit chip in Zuchongzhi 2.1 [32].

The primary objective of the platform is to advance the development of quantum computing hardware and its broader ecosystem by reaching advanced global levels in key design indicators such as connectivity, fidelity, and interference time. Through remote access, industry users can leverage the platform's capabilities to explore innovative applications. Furthermore, the platform aims to maintain world-class standards by connecting

Table 7 System characteristics comparison (mean values) of Zuchongzhi 2.0 and Zuchongzhi 2.1 Superconducting quantum computers

System Specifications	Zuchongzhi Quantum Systems	
	Zuchongzhi 2.0 [62]	Zuchongzhi 2.1 [32]
Qubit Technology	Superconducting	Superconducting
Physical Qubits	56	66
Processor layout	2D array	11 × 6 2D array
Programability	✓	✓
Qubit idle frequency	4.653 GHz	4.921 GHz
Qubit maximum frequency	4.809 GHz	5.094 GHz
Qubit anharmonicity	-248.2 MHz	-246.6 MHz
Readout drive frequency	6.407 GHz	6.397 GHz
T1 at idle frequency	30.6 μ s	26.5 μ s
T2 at idle frequency	5.30 μ s	3.40 μ s
Single-qubit average gate fidelity	99.86%	99.84%
Two-qubit average gate fidelity	99.24%	99.40%
Readout average fidelity	95.23%	97.74%
The average 1-qubit state readout error e_r^a	4.52%	2.26%

^aAverage (mean) value when all gates are applied simultaneously.

Table 8 A comparison of the tensor network algorithm runtimes for different circuits on the Summit supercomputer is presented, along with the estimated classical simulation costs for the random quantum circuit sampling experiment conducted on the *Sycamore*, Zuchongzhi 2.0, and Zuchongzhi 2.1 quantum processors [32]

Quantum processor	Google's <i>Sycamore</i> vs Zuchongzhi quantum processors				
	Qubits	Circuit	Estimated runtime on <i>Summit</i>	On quantum computer	Fidelity
<i>Sycamore</i> [12]	53-qubit	20-cycle	10,000 years 15.9 days ^a	3 minutes ^b	0.224%
Zuchongzhi 2.0 [62]	56-qubit	20-cycle	8.2 years	72 minutes	0.0662%
Zuchongzhi 2.1 [32]	60-qubit	22-cycle	4800 years	60 minutes	0.0758%
Zuchongzhi 2.1 [32]	60-qubit	24-cycle	48,000 years	252 minutes	0.0366%

^aGoogle's projections suggested that the task would take a staggering 10,000 years to complete on Summit, the world's most powerful supercomputer at that time. In contrast, IBM offered a different viewpoint, claiming that Summit could finish the same task in just a few days [167].

^bAccording to a report by Google Quantum AI and its collaborators in [12], while the time required to complete the task in [32] was estimated to be 600 seconds.

multiple high-performance quantum computers to continually update and expand its capabilities. This comprehensive approach to quantum computing seeks to drive progress and establish Zuchongzhi as a cutting-edge quantum computing platform [190]. Readers are referred to [191] for more details about China's research efforts to advance the area of quantum information science from both fundamental research and industry applications.

Table 7 compares the average system characteristics of the Zuchongzhi 2.0 and Zuchongzhi 2.1 superconducting quantum processors. Meanwhile, Table 8 illustrates a comparison of runtime performance for tensor network algorithms across various circuits executed on the Summit supercomputer. It also provides an estimate of the classical simulation time required for the RQCS task on the *Sycamore* [12], Zuchongzhi 2.0 [62], and Zuchongzhi 2.1 [32] quantum processors.

9 Oxford quantum circuits

9.1 Overview

Oxford Quantum Circuits (OQC) is a leading quantum computing company based in the UK and Europe, established in 2017. It provides a comprehensive quantum computing

solution that integrates hardware, control systems, and software. Central to its technology is the *Coaxmon*, a novel three-dimensional qubit architecture that enhances scalability by relocating key components off-chip [192].

An inherent feature of all OQC systems is their capacity for low crosstalk, high coherence, and rapid gate operations. This unique architecture enables OQC to rapidly expand its quantum computing systems while upholding the highest standards of quality and performance. Notably, the parameters of low crosstalk, high coherence, and fast gate operations remain constant, even as the number of qubits may undergo changes during the scaling process. This ensures the system's robustness and stability, allowing OQC to pursue advancements in quantum computing capabilities without compromising the integrity of its quantum operations [193].

9.2 Qubit technology: the coaxmon

One of the principal obstacles in the realm of superconducting circuits for quantum computation revolves around achieving scalability in qubit numbers while upholding qubit quality and control to attain commercially viable levels of processing power [12, 194–198]. Conventional two-dimensional devices necessitate increasingly intricate engineering efforts to establish control wiring connections across the chip for each individual qubit. Unfortunately, this intricate wiring not only compromises the quality of qubits but also elevates the likelihood of costly engineering errors.

The Coaxmon architecture [192] tackles these pressing challenges by presenting a three-dimensional design that relocates vital componentry off-chip, offering several advantages such as enhanced simplicity, flexibility, ease of engineering, and, most importantly, scalability [38]. As a result, all QPUs developed by OQC exhibit long coherence times, low cross-talk, and high-fidelity superfast gate operation times. This unique Coaxmon design [38] empowers the scaling of larger two-dimensional qubit arrays, while reducing susceptibility to the limitations of qubit layout or qubit-to-qubit connectivity. By transcending the limitations of traditional two-dimensional devices [12, 195, 199, 200], the Coaxmon architecture offers a promising avenue towards overcoming the challenges of scaling quantum computing capabilities and ensuring optimal performance in quantum computation endeavors [192].

The device configuration is illustrated in Fig. 15. It comprises a superconducting charge qubit operating in the transmon regime [21, 192], featuring coaxial electrodes, referred to as the Coaxmon. This Coaxmon is coupled to a lumped element LC microwave resonator, which is meticulously fabricated on the opposite side of the chip, thereby enabling the realization of dispersive circuit quantum electrodynamics [92]. Control and measurement of the device are facilitated through coaxial ports positioned perpendicular to the plane of the chip. By adjusting the distance of these ports from the chip, the external quality factor of the circuits can be modulated accordingly. These ports offer the flexibility of independent qubit control and resonator measurement in reflection, or alternatively, they can be employed to measure the device in transmission [193].

The fabrication process involves two stages of electron beam lithography, with each side of a 0.5 mm thick sapphire chip being patterned to incorporate an aluminum LC resonator and the Coaxmon. Throughout the fabrication process, the bottom of the chip is protected by a layer of polymer resist applied through spin-coating. Moreover, chip holders are utilized to ensure the suspension of the device's bottom throughout the fabrication proce-

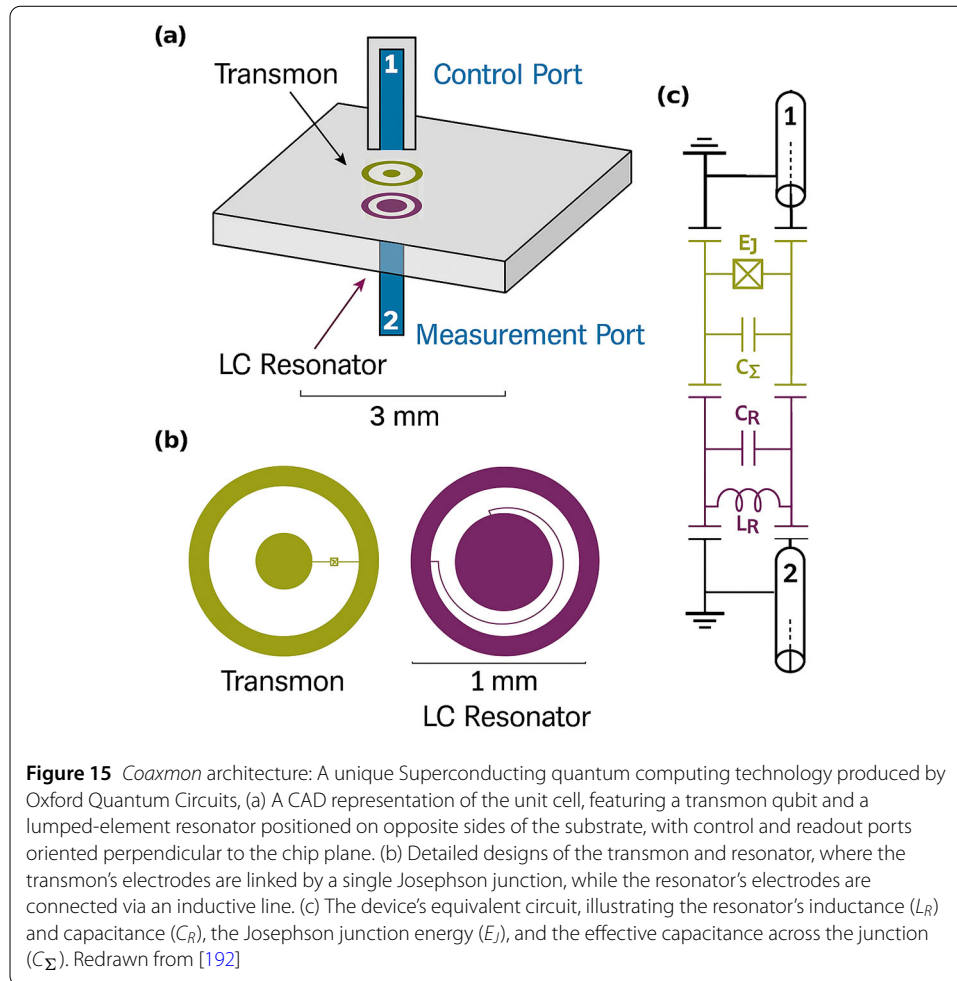


Table 9 System characteristics of Oxford Quantum Circuits' Superconducting Quantum Processor [192]

System Specifications	Experimental Values	Unit
Qubit Frequency (f_{01})	7.230	GHz
Resonator Frequency (f_{r0})	10.23	GHz
Dispersive shift ($\chi/2\pi$)	-6.340	MHz
Resonator Quality Factor	2080	-
Charging energy (E_C/h)	294.0	MHz
E_J/E_C	81.80	-
T1 Time (T_1)	4.100	μs
T2 Time (T_2)	5.650	μs
Coupling ^a ($g/2\pi$)	462.0	MHz
T2 Echo (T_{2E})	6.670	μs
Josephson energy (E_J/h)	24.10	GHz
Detuning between qubit and resonator ($\Delta_0/(2\pi)$)	-3.00	GHz

^aA simulation predicts a coupling $g/2\pi \approx 420$ MHz.

ture. For more details about Coaxmon architecture, and device characterization (see Table 9), readers are referred to [38, 192].

Table 10 Indicative Performance Metrics of OQC Toshiko Gen 1

Performance Metric	Qubits	Median T1 (μ s)	Median T2 (μ s)	1Q Median Fidelity	2Q Median Fidelity	4 Qubit Chain
Value	32	78	104	99.9%	99%	Yes

9.3 OQC Toshiko

The OQC Toshiko Gen 1 represents a pioneering enterprise-ready quantum computing platform, deployed in London and Tokyo since 2023. It incorporates superconducting qubits within the Coaxmon architecture, offering enhanced scalability. The system consists of 32 addressable qubits, arranged in a lattice structure with 2:1 and 3:1 qubit couplings. Toshiko Gen 1 supports several key quantum programming standards, including QIR, OpenPulse, and OpenQASM 2, with plans for compatibility with OpenQASM 3 in future updates. Table 10 displaying the performance metrics for the OQC Toshiko Gen 1 platform, with each metric and its corresponding value.

9.4 Through-substrate machining for superconducting quantum processors

The study in [201] introduces a novel method to integrate through-sapphire substrate machining with superconducting quantum processors to address challenges in scalability and mode mitigation for large-scale quantum computing. By leveraging sapphire's low dielectric loss as a substrate [202], the researchers employed CNC micro-milling to create 1 mm diameter apertures that enable through-substrate electrical connections (see Fig. 16). This machining process, performed with active cooling to protect sensitive components, demonstrated robustness in preserving Josephson junction (JJ) integrity, with a minor increase in resistance spread from 2.35% to 2.58% post-machining [201]. The fabricated 32-qubit QPU, referred to as "Toshiko," integrated eight sapphire apertures for inductive shunting, mitigating unwanted cavity modes and achieving median coherence times of $T_1 \approx 69.4 \mu\text{s}$ and $T_{2e} \approx 102.6 \mu\text{s}$, which are comparable to state-of-the-art silicon-based platforms [38, 203]. JJ functionality remained high, with a 99.28% success rate post-drilling, despite some mechanical chipping near drill sites. This work establishes sapphire as a viable platform for scaling superconducting QPUs and highlights its potential for future integration of materials like tantalum or titanium nitride, which offer superior qubit coherence. The study's findings open new pathways for fault-tolerant quantum computing systems.

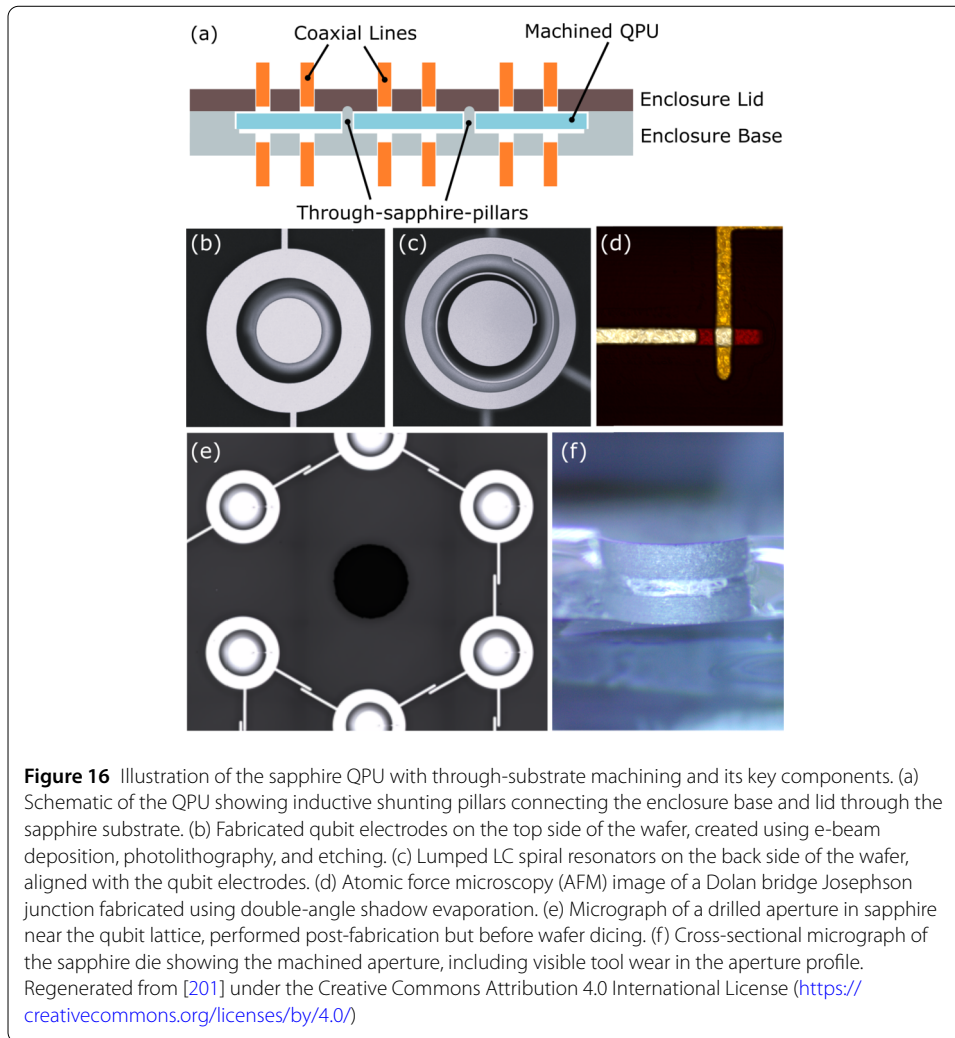
10 IQM

10.1 Overview

IQM leveraging superconducting QPUs to advance computational capabilities beyond classical limits. Founded in 2018 and based in Finland, the company focuses on developing scalable, high-performance quantum hardware tailored for various applications, including scientific research and industrial use. Its quantum processors integrate tunable couplers, advanced qubit designs, and robust cryogenic infrastructures to enhance quantum coherence and gate fidelity (see Sect. 10.5). By innovating in both hardware and software layers, IQM aims to bridge the gap between theoretical quantum advantage and practical implementations [204].

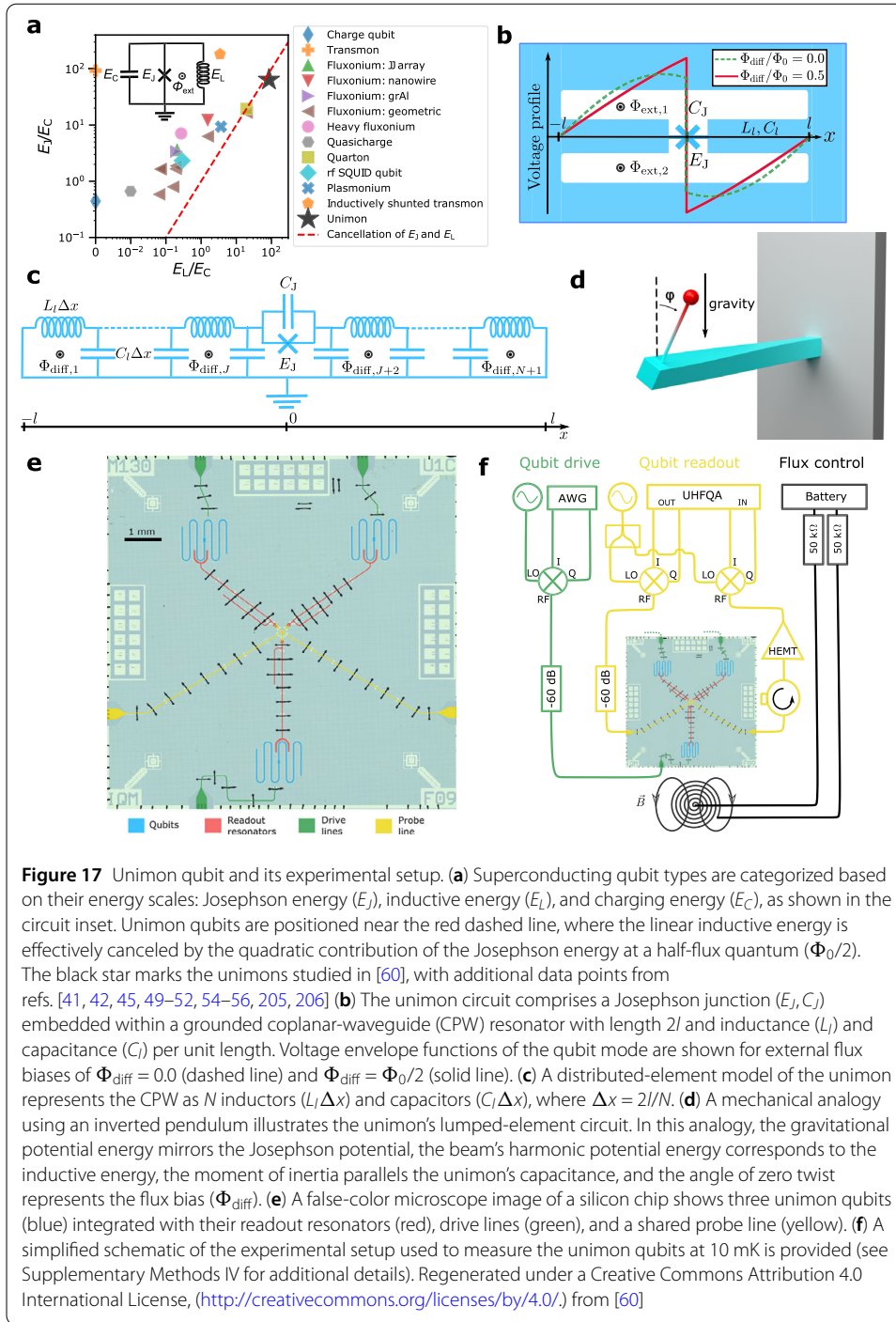
10.2 The unimon qubit

The study in [60] presents the introduction and experimental validation of a novel superconducting qubit known as the unimon. This qubit architecture features a single Joseph-



son junction shunted by both a linear inductor and a capacitor, operating within an unconventional parameter regime. In this regime, the Josephson energy nearly offsets the inductive energy, leading to a highly anharmonic qubit. Furthermore, the unimon exhibits complete immunity to low-frequency charge noise and partial resilience against flux noise (see Fig. 17). The study systematically examines the qubit's anharmonicity and transition frequencies across a broad range of flux biases, with experimental data aligning closely with theoretical first-principles models, even when tested across five distinct qubits [60]. The findings indicate that energy relaxation is primarily constrained by dielectric losses, while coherence can be preserved at a flux-insensitive operating point. A key observation from the study is that single-qubit gate fidelity improves with shorter gate durations, remaining stable at 99.9% for a 13 ns gate length, with consistent performance over several hours [60].

The unimon qubit is realized through a streamlined superconducting circuit architecture, featuring a single Josephson junction embedded within the central conductor of a superconducting CPW resonator, which is symmetrically grounded at both ends. Unlike conventional charge-based qubits, this design eliminates discrete charge islands and instead employs inductive shunting of the junction to enhance stability. Similar to the re-



cently developed fluxonium qubit [52], which leverages geometric superinductance for greater resilience against noise, the unimon belongs to a select category of superconducting qubits where a Josephson junction is inductively shunted via a geometric inductance. This configuration significantly mitigates low-frequency charge noise, improving coherence. The intrinsic nonlinearity of the Josephson junction modifies the normal resonator modes associated with current flow through the junction, transforming them into anhar-

monic oscillators conducive to quantum operations. Among these modes, the one with the lowest frequency and highest anharmonicity is chosen as the qubit state for computation.

The anharmonic mode frequencies in the unimon qubit can be adjusted by applying external magnetic fluxes, denoted as $\Phi_{\text{ext},1}$ and $\Phi_{\text{ext},2}$, to the two superconducting loops of the resonator, as illustrated in Fig. 17b. Due to its gradiometric design, the unimon demonstrates a degree of robustness against flux noise. This resilience arises because the superconducting phase difference across the Josephson junction is governed by half the difference between the externally applied fluxes, given by $\Phi_{\text{diff}} = \frac{1}{2}(\Phi_{\text{ext},2} - \Phi_{\text{ext},1})$. Significantly, the qubit's anharmonicity reaches its maximum at a special operating point, commonly referred to as the “sweet spot,” where the qubit frequency exhibits first-order insensitivity to fluctuations in the external flux difference. This optimal operating condition occurs when $\Phi_{\text{diff}} = \frac{1}{2}\Phi_0$ (plus any integer multiple of Φ_0), where $\Phi_0 = \frac{h}{2e} \approx 2.067 \times 10^{-15}$ Wb, with e representing the elementary charge and h denoting Planck's constant.

10.3 Effective Hamiltonian of the unimon qubit mode

Utilizing the distributed-element circuit model depicted in Fig. 17c, the Hamiltonian governing qubit mode m is formulated as:

$$\begin{aligned} \hat{H}_m = & 4E_{C,m}(\varphi_0)\hat{n}_m^2 + \frac{1}{2}E_{L,m}(\varphi_0)\hat{\varphi}_m^2 \\ & + E_L\hat{\varphi}_m \left(\frac{2\pi\Phi_{\text{diff}}}{\Phi_0} - \varphi_0 \right) - E_J \cos(\hat{\varphi}_m - \varphi_0) \end{aligned} \quad (3)$$

where φ_0 corresponds to the Josephson phase influenced by the direct current flowing through the junction. The terms $E_{C,m}(\varphi_0)$ and $E_{L,m}(\varphi_0)$ denote the mode-dependent capacitive and inductive energies, respectively, while E_L represents the inductive energy associated with the dc current, and E_J signifies the Josephson energy. The operators $\hat{n}_m$ and $\hat{\varphi}_m$ describe the Cooper pair number and the superconducting phase for mode m , obeying the fundamental commutation relation $[\hat{\varphi}_m, \hat{n}_m] = i$, where i is the imaginary unit. Notably, φ_0 is treated as a classical parameter that varies with the flux bias Φ_{diff} according to a transcendental equation, ensuring that $2\pi\Phi_{\text{diff}}/\Phi_0 - \varphi_0$ maintains periodicity in Φ_{diff} [60].

10.4 Experimental results on unimon qubits

The experimental study involved five unimon qubits [60], labeled Q_A through Q_E , which were distributed across two distinct chips. While all qubits shared a similar coplanar waveguide resonator structure, they exhibited differences in their Josephson energies (E_J), leading to varying degrees of cancellation in the quadratic potential energy components, proportional to $(E_{L,m}(\pi) - E_J)$. Additionally, the coupling capacitance between each qubit and its corresponding readout resonator was deliberately varied across the two chips. A summary of the key measured characteristics for all five qubits is presented in Tables 11 and 12.

To explore the factors affecting the energy relaxation time (T_1) of the unimon qubits, T_1 was measured as a function of qubit frequency. At the flux sweet spot ($\Phi_{\text{diff}} = \Phi_0/2$), the relaxation time was approximately $8.6 \mu\text{s}$, whereas at $\Phi_{\text{diff}} = 0$, it dropped to around $4.6 \mu\text{s}$. Between these two flux points, T_1 reached a minimum when the qubit frequency was close to the readout resonator frequency ($f_r = 6.198$ GHz). This trend is primarily attributed to dielectric losses, characterized by an effective quality factor of $Q_C \approx 3.5 \times 10^5$,

Table 11 The characteristic parameters for five measured unimon qubits. The data for each qubit includes the total length (2l) of the center conductor, the fitted inductance (L_l) and capacitance (C_l) per unit length of the center conductor, the Josephson energy (E_J), the inductive and capacitive energies ($E_{L,m}$ and $E_{C,m}$), the qubit frequency ($\omega_{01}/(2\pi)$), anharmonicity ($\alpha/(2\pi)$), and the location (x_g) of the unimon-resonator coupling point relative to the junction. Additionally, the table provides the measured coupling strength ($|g_{01}|/(2\pi)$) between the qubit and the readout resonator, the coupling capacitance (C_g), the dispersive shift ($\chi/(2\pi)$), the readout resonator frequency (f_r), and its linewidth ($\kappa/(2\pi)$). Values for flux-dependent parameters, such as $\omega_{01}/(2\pi)$, $E_{L,m}$ and $E_{C,m}$, $\alpha/(2\pi)$, $\chi/(2\pi)$, f_r and $g_{01}/(2\pi)$ and are reported at $\Phi_{\text{diff}} = \frac{1}{2}\phi_0$. Asterisks indicate that the inductance (L_l) and capacitance (C_l) per unit length were determined by fitting the theoretical model to the spectrum of qubit Q_2 , and the same values are applied to the other qubits, assuming identical waveguide cross sections [60]

Qubits	$\kappa/(2\pi)$ (MHz)	f_r (GHz)	$\chi/(2\pi)$ (MHz)	$ g_{01} /(2\pi)$ (MHz)	C_g (fF)	x_g (mm)	$\alpha/(2\pi)$ (MHz)	$\omega_{01}/(2\pi)$ (GHz)	$E_{C,m}/h$ (GHz)	$E_{L,m}/h$ (GHz)	E_J/h (GHz)	C_l (pF/m)	L_l ($\mu\text{H}/\text{m}$)	2l (mm)
Q_A	0.43	5.826	0.74	53.5	9.0	0.596	744	3.547	0.318	24.9	23.3	87.1*	0.821*	8.0
Q_B	1.24	6.198	1.2	70.0	10.0	0.596	434	4.488	0.297	25.2	19.0	87.1*	0.821	8.0
Q_C	9.2	5.522	9.2	79.7	12.5	0.596	343	4.781	0.290	25.3	17.4	87.1*	0.821*	8.0
Q_D	10.0	5.699	20.2	85.7	12.5	0.596	214	5.257	0.278	25.7	14.8	87.1*	0.821*	8.0
Q_E	1.8	6.156	4.1	92.3	12.5	0.596	257	5.224	0.279	25.7	15.0	87.1*	0.821*	8.0

Table 12 Results from coherence characterization and RB (randomized benchmarking) experiments for the 5 unimon qubits. It includes the measured energy relaxation time (T_1), the measured Ramsey coherence time (T_2^*), the measured echo coherence time (T_2^e), the flux noise density parameter ($A_{\Phi_{\text{diff}}}$) at 1 Hz, estimated from the echo coherence time, the average fidelity per microwave gate from standard RB experiments, and the gate duration used in these RB experiments. All measurements were performed near the sweet spot at $\Phi_{\text{diff}} = \frac{1}{2}\phi_0$ [60]

Qubits	Gate duration (ns)	RB fidelity (%)	$A_{\Phi_{\text{diff}}}(\mu\phi_0)$	T_2^e (μs)	T_2^* (μs)	T_1 (μs)
Q_A	13.3	99.81	6.1	6.8	1.9	7.2
Q_B	13.3	99.90	15.0	9.2	3.1	8.6
Q_C	40.0	99.54	11.2	9.3	2.3	5.8
Q_D	20.0	99.63	11.1	7.0	2.3	3.9
Q_E	13.3	99.86	14.3	11.4	2.5	5.6

in addition to Purcell decay via the readout resonator [60]. The data indicate that dielectric losses are the dominant source of energy relaxation at $\Phi_{\text{diff}} = \Phi_0/2$. The quality factor of this first-generation unimon qubit is nearly ten times higher than that of geometric-superinductance qubits [207], comparable to that of fluxonium qubits [47, 49], but remains an order of magnitude lower than the best-performing transmon qubits [35]. Future improvements in design, materials, and fabrication processes are anticipated to significantly reduce dielectric losses in next-generation unimon qubits.

10.5 IQM Garnet

IQM Garnet, a 20-qubit superconducting quantum computer, features a structured lattice-based QPU architecture designed for efficient qubit-qubit interactions. Its key technological advancements include flux-tunable transmon qubits, tunable couplers for reduced crosstalk, and a control system that enables precise gate operations. System-level performance evaluations [204] indicate a median two-qubit gate fidelity of 99.5%, demonstrating high reliability in executing quantum circuits. A single-qubit gate error rate of 9×10^{-4} , CZ gate error of 5×10^{-3} and a two-qubit Clifford gate error rate of 2×10^{-2} . Additionally, the system demonstrates a readout error rate of 3×10^{-2} . Benchmarking tests confirm its capability to entangle all 20 qubits into a Greenberger–Horne–Zeilinger

(GHZ) state, affirming its suitability for advanced quantum algorithms, demonstrating a GHZ state fidelity (with readout error mitigation) of 0.62, confirming genuine multi-qubit entanglement. The quantum volume is measured at 32 ($QV = 2^5$), while the system achieves a circuit layer operations per second (CLOPS) score of 2600. These metrics highlight the system's capability to perform complex quantum operations with high precision, making it suitable for quantum algorithm development and benchmarking. With a roadmap toward scaling beyond 150 qubits, IQM Garnet represents a key milestone in the company's pursuit of practical quantum computing [204].

11 Intel quantum

11.1 Overview

Intel's quantum research has focused on both superconducting and spin qubit technologies, leveraging its expertise in semiconductor manufacturing to make unique contributions to the field. Notable milestones include *Tangle Lake*, a 49-superconducting qubit chip, and full silicon wafers containing up to 26 spin qubits, which have been used for testing at Delft University of Technology. The Intel Quantum SDK serves as a simulated quantum computing environment capable of interfacing with Intel's quantum hardware, including the *Horse Ridge* II control chip and the quantum spin qubit chip.

11.2 Intel's quantum computing journey

In the year 2020, Intel, in collaboration with its partners, employed a cryogenic control chip named "Horse Ridge," designed to operate at an extremely low temperature of 3 Kelvin [208]. While 3 Kelvin is considered extremely low for conventional electronics, it is important to note that the quantum processor itself operates at significantly lower temperatures—typically in the millikelvin range—within the dilution refrigerator. These ultra-low temperatures are essential for preserving quantum coherence and enabling superconducting behavior. This cryo-controller was produced using Intel's 22-nanometer FinFET low-power CMOS technology [209]. The purpose of this chip was to enable the precise and coherent manipulation of two electron spin qubits within a silicon-based double quantum dot structure.

In 2021, Intel introduced a *cryoprobe*, a device that dramatically accelerates the testing of qubits. The *cryoprobe* allows the team to gather detailed information on quantum noise and the quality of quantum dots within minutes. This capability reduces the testing timeline for quantum devices from weeks or days to mere hours, enabling faster iteration and process optimization [210]. In 2022, Intel achieved a significant milestone with the fabrication of the industry's first silicon spin qubit devices on 300-mm silicon wafers using its fabrication equipment. This development validated the feasibility of using Intel's established transistor fabrication lines for producing high-quality quantum chips. It also marked a major step toward scaling quantum systems to the thousands or even millions of qubits required for commercial quantum computing [210].

Intel launched the "Intel" Quantum SDK, a full-stack software development kit that allows developers to program, simulate, and eventually execute quantum algorithms on real quantum hardware [211]. This SDK enables the broader developer community to interface with Intel's quantum stack, fostering innovation in quantum algorithm design and applications [212]. Intel commenced the mass production of qubits on 300-mm silicon wafers using extreme ultraviolet (EUV) lithography in a high-volume manufacturing environment. This achievement enables the integration of 10,000 quantum dot arrays on

a single wafer, paving the way for large-scale quantum systems and underscoring Intel's commitment to leveraging its semiconductor manufacturing expertise for quantum computing [210].

In 2023, Intel introduced *Tunnel Falls*, a 12-qubit silicon spin device fabricated using advanced 300-mm wafer processes. *Tunnel Falls*, made available to the research community, highlights Intel's use of cutting-edge fabrication techniques like extreme ultraviolet lithography (EUV) and CMOS (complementary metal-oxide-semiconductor) logic processing. These qubits encode information in the spin states of individual electrons, allowing for scalable and dense qubit integration [210]. In 2024, Intel demonstrated high qubit control fidelity (99.9%) and uniformity across 300-mm silicon spin qubit wafers, a first in the industry. Using cryogenic probing and high-throughput testing [213], Intel's advancements enabled significant improvements in device yield and performance analysis. This research solidifies the potential for scalable, silicon-based quantum processors manufactured using industry-standard CMOS techniques. Spin qubits, measuring approximately 100 nanometers, offer higher density than superconducting qubits, paving the way for more complex quantum systems on a single chip [210].

Intel's latest research [213] advances scalable quantum computing with silicon-based spin qubits. Using a 300-mm cryogenic wafer probing [214] system, they tested over 3700 quantum dot sites at 1.6 K, achieving a quantum dot yield of 99.8% and a full device yield of 96% for 12-dot arrays. Their optimized Si/SiGe fabrication process, aligned with CMOS methods, ensures low process variation and high uniformity, with random threshold voltage variation reduced to 59 mV. These innovations enhance device performance, enabling automated measurements and consistent single-electron operation across wafers. By integrating semiconductor industry techniques, Intel sets a foundation for scaling fault-tolerant quantum processors with large, reliable qubit arrays. Readers are referred to [210, 215] for more details about Intel's journey to building a practical quantum computer.

11.3 Tangle Lake

Intel Labs introduced a third-generation quantum processor named *Tangle Lake* [216]. Intel's *Tangle Lake* represents a significant milestone in the race toward practical quantum computing. This superconducting quantum chip integrates 49 qubits, a critical threshold where the quantum behavior of the chip becomes difficult to simulate using classical computers. This achievement positions Intel alongside industry leaders such as Google Quantum AI and IBM Quantum, who are similarly striving for quantum supremacy—the point where quantum systems outperform classical counterparts.

The 49-qubit *Tangle Lake* chip builds upon Intel's earlier 17-qubit arrays, which demonstrated the feasibility of surface-code error correction to manage disruptions in fragile quantum states. In addition to increasing qubit counts, Intel has focused on refining the overall quantum system. This includes engineering advanced packaging to minimize radio-frequency interference, utilizing flip-chip technology for compact and dense connections, and optimizing the electronics layout to control qubits and process quantum results. The chip operates at temperatures of approximately 20 millikelvin ($-273.13\text{ }^{\circ}\text{C}$), with future goals to raise operational temperatures for improved system robustness.

Intel's quantum strategy balances investments across two leading architectures: superconducting qubits and silicon-based spin qubits. While superconducting qubits currently

lead in development, spin qubits offer scalability advantages due to their smaller size and compatibility with existing semiconductor manufacturing processes. Intel has already demonstrated the capability to fabricate spin qubits using its 300-mm silicon wafer production methods.

Beyond quantum computing, Intel continues to explore complementary technologies, such as the “Loihi neuromorphic chip”, designed to mimic biological neural networks for AI applications. Together, these efforts underline Intel’s commitment to building a versatile technological foundation for next-generation computing, with Tangle Lake serving as a cornerstone in the pursuit of scalable, fault-tolerant quantum systems.

11.4 Tunnel Falls

Intel is involved in the development of both superconducting and spin qubits. The spin qubit technology, notably, offers the potential of reduced thermal constraints, boasts dimensions approximately a million times smaller than those of a superconducting qubit, and provides greater stability when compared to superconducting qubits [108, 217, 218]. Consequently, Intel aims to significantly enhance the scalability of this technology in comparison to its competitors. Furthermore, Intel’s strategy, encompassing silicon spin qubits, quantum control chips, and innovative testing methodologies, exemplifies the company’s dedication to propelling the evolution of quantum computing technologies [208].

Tunnel Falls represents Intel’s latest silicon [219, 220] spin qubit chip. This 12-qubit device reflects advancements in fabrication and manufacturing while providing a platform for further experimentation, exploration, and understanding in quantum computing [219]. As seen in Fig. 18, an illustrative depiction captured by Intel’s cryogenic probing system during the process of automation reveals the arrangement of quantum qubit mechanisms at an ultra-low temperature of 1.6 kelvins. This setup allows for the creation of quantum dots within all 16 designated sites, comprising four sensors and twelve qubit locations. Remarkably, these quantum dots can be precisely adjusted to contain just one electron, and this adjustment is achieved autonomously without the need for manual intervention from engineers. These notable outcomes owe their existence to the consistent and replicable fabrication of devices by Intel, maintaining uniformity across their design. The comprehensive data acquisition was conducted across the entirety of a semiconductor wafer.

11.5 Intel’s Roadmap Toward scalable silicon-based quantum processors

Intel’s recent work in quantum computing showcases significant strides toward the realization of scalable, fault-tolerant quantum processors using silicon-based spin qubits. Spin qubits [221–223], leveraging the quantum properties of single electrons in silicon, offer a pathway to scalable quantum systems due to their compatibility with established semiconductor manufacturing techniques [221, 224]. However, achieving scalability demands both a high-yield fabrication process and efficient, large-scale testing methods to ensure consistent performance across thousands or even millions of qubits [225–227].

In [213], Intel introduced a 300-mm cryogenic wafer probing [214] system capable of testing spin qubit devices at ultra-low temperatures of 1.6 K, achieved within approximately two hours. This system enables high-throughput data collection by aligning $100 \times 100 \mu\text{m}^2$ device pads (with a $150 \mu\text{m}$ spacing) to probe pins using machine vision algorithms, ensuring precise electrical connections for automated measurements. Once aligned, the system performs various electrical tests across the wafer, allowing statistical

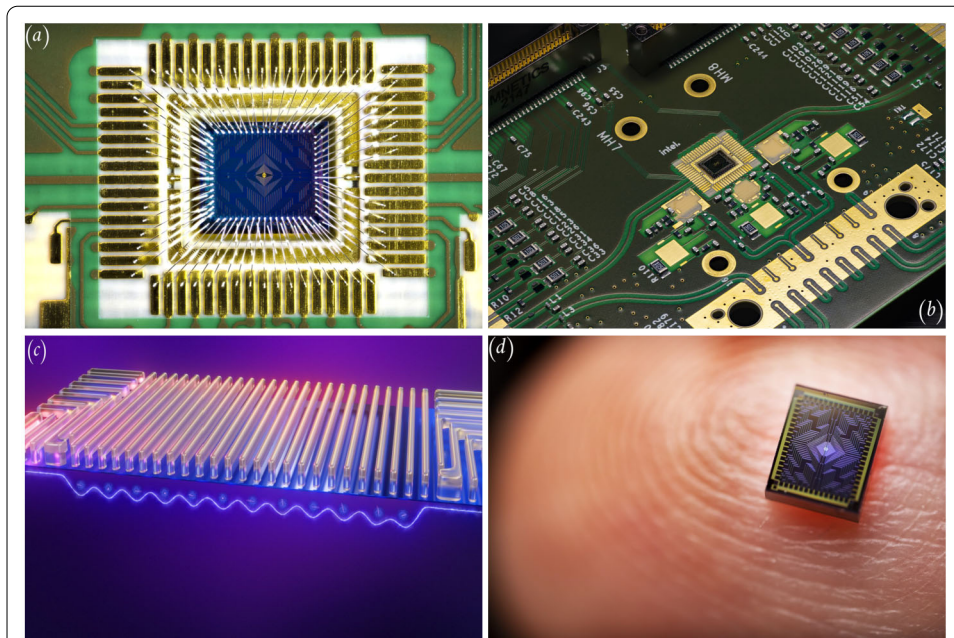
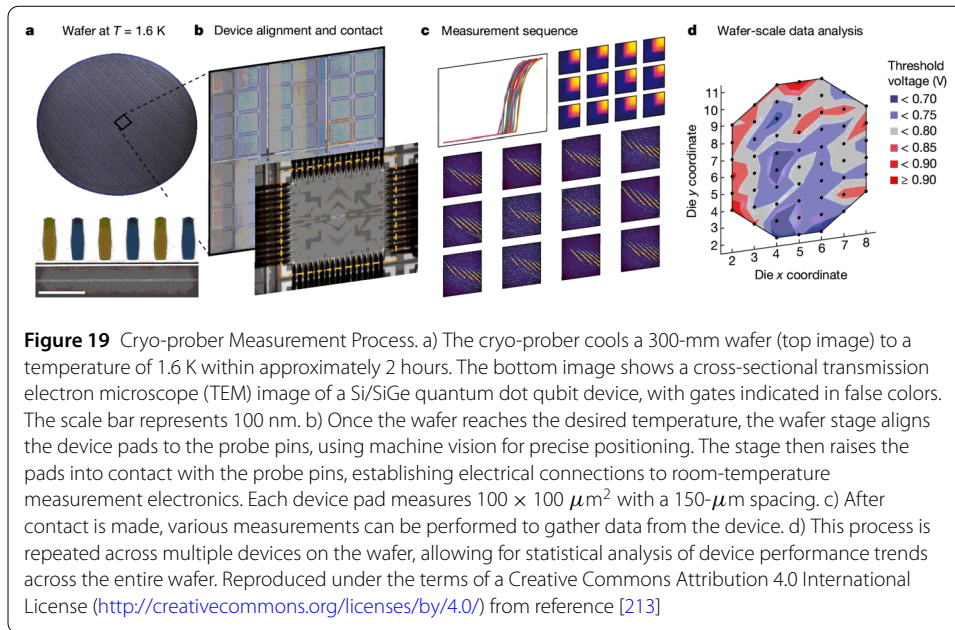


Figure 18 (a) An image depicts the Intel Tunnel Falls chip in packaging. Tunnel Falls stands as the inaugural silicon spin qubit device accessible to research institutions and universities. Its construction takes place on 300-millimeter wafers, harnessing Intel's forefront capabilities in transistor fabrication techniques. (b) A magnified view of Intel's Tunnel Falls chip. Tunnel Falls demonstrates a commendable 95% yield rate across the entire wafer, paralleling the voltage uniformity seen in CMOS logic processes. Each wafer boasts an impressive 24,000 quantum dot test chips, with a remarkable 99.8% yield meticulously tuned to the single electron level. These 12-dot configurations have the capacity to compose between four and twelve qubits, allowing for their isolation and simultaneous utilization in various operations, contingent upon the operational preferences of the respective university or laboratory systems. (c) A visual depiction of an electron confined within quantum dot gates in a 12-qubit configuration. Produced on 300-millimeter wafers, over 24,000 quantum dots per wafer. Tunnel Falls harnesses Intel's capabilities in transistor fabrication, employing techniques like EUV and advanced materials processing. This enables the chip to function as a single electron transistor and empowers Intel to manufacture Tunnel Falls with minimal modifications to a conventional CMOS logic processing line. (d) An Intel Tunnel Falls chip is juxtaposed against a human finger to emphasize its scale. The silicon spin qubits within the Tunnel Falls chip exhibit a remarkable size reduction of up to a million times when compared to other qubit variants. The compactness of the Tunnel Falls chip, measuring around 50 nanometers square, holds the potential for accelerated scalability. Reproduced with permission from Intel Corporation

analysis of performance trends across thousands of devices. Notably, this process facilitates the evaluation of over 3700 quantum dot sites per wafer, achieving a quantum dot yield of 99.8% and an overall device yield of 96% for 12-dot arrays. These results represent a significant advancement in device uniformity and scalability, overcoming challenges in variability and process control that have historically limited the adoption of spin qubits. The process is illustrated in Fig. 19, which shows the cryo-prober system setup, wafer cooling, and measurement workflow [213].

The fabrication process, optimized for Si/SiGe heterostructures, is fully compatible with CMOS industry standards. The use of extreme ultraviolet lithography and precise deposition techniques enabled high-density qubit arrays with gate pitches as small as 50 nm. Random threshold voltage variation was reduced to 59 mV, and single-electron transitions were reliably observed in 91% of the devices tested. This degree of consistency significantly enhances the predictability of qubit operation, simplifying automation and reducing tuning overhead in large arrays [213].



Intel's advancements are mutually reinforcing: the cryogenic wafer probing [214] system enables rapid feedback for fabrication optimization, while the improved manufacturing process ensures that devices are reliable enough to undergo thorough automated testing. This combination reduces disorder in the material system and improves electrostatic stability, both critical for scaling spin qubit arrays beyond current capabilities. Moreover, the study demonstrated the successful integration of a 12-quantum-dot unit cell, highlighting a modular approach that can be extended to larger arrays.

By addressing both yield and variability challenges [228], Intel's approach paves the way for scaling spin qubit arrays to meet the demands of fault-tolerant quantum computing. The high yield, combined with efficient automated testing, minimizes fabrication bottlenecks, making it feasible to integrate thousands or millions of qubits on a single chip [213]. Furthermore, the use of industry-standard CMOS infrastructure underscores the potential for quantum processors to leverage the mature supply chains and manufacturing capabilities of the semiconductor industry.

11.6 Intel Quantum SDK

Intel's Quantum SDK 1.0 [211, 212] provides a comprehensive quantum computing stack in simulation, enabling developers to program quantum algorithms and applications while preparing for the future availability of Intel's quantum hardware, such as the *Horse Ridge* II control chip and quantum spin qubit chip. The SDK offers a programming interface in C++, integrating seamlessly with Python, and includes a quantum runtime environment optimized for hybrid quantum-classical algorithms like Quantum Approximate Optimization Algorithm (QAOA) and Variational quantum eigensolver (VQE) [212].

It supports two simulation backends: the Intel Quantum Simulator (IQS), capable of simulating up to 32 qubits on a single node and more than 40 across multiple nodes, and a backend simulating Intel's quantum dot qubits. The platform is customizable, expandable, and allows for detailed compiler analysis, providing insights into algorithm optimization and system architecture requirements. Intel is also fostering a quantum ecosystem by

funding quantum curriculum development at multiple universities and encouraging exploration of real-world use cases, such as fluid dynamics, quantum machine learning, and astrophysics. The SDK is available on the OneAPI Intel Dev Cloud, with future updates planned to integrate with Intel's quantum hardware [212].

12 QuTech

12.1 Overview

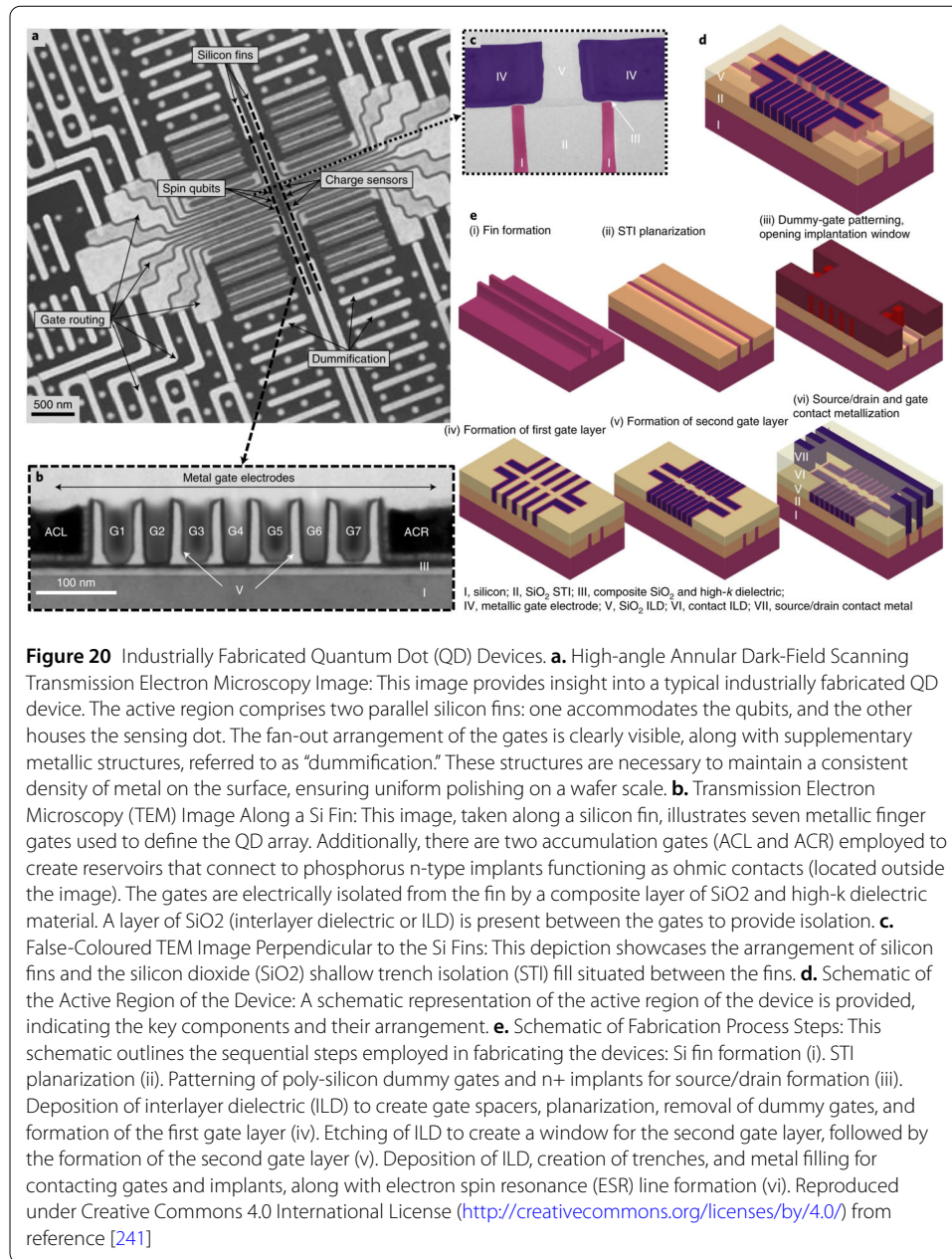
The objective of QuTech [31] lies in the translation of foundational scientific investigations into applied research, with the aim of propelling quantum technology into both industrial applications and broader societal contexts. In this pursuit, QuTech is actively engaged in the development of Quantum Inspire (see Sect. 13), a pioneering full-stack quantum computing prototype. Quantum Inspire [229] is designed to foster collaborative research and co-development within the quantum computing realm, and it stands as a platform that grants cloud-based accessibility to an array of QuTech technologies [230]. Notably, this includes a programmable quantum computer simulators boasting the capability to simulate up to 31 qubits, as well as access to a Silicon spin [231] qubit-based quantum processor that aligns with CMOS compatibility [220]. The inaugural iteration of Quantum Inspire's quantum processors encompasses a dual quantum dot system situated within an internally cultivated SiGe/²⁸Si/SiGe heterostructure. The system's delineation is accomplished through a singular layer of Al gates, signifying the initial step toward the realization of quantum technological advancements [232, 233].

It is worth noting that both Quantum Inspire (Sect. 13) and QuantWare (Sect. 14) originated within the QuTech ecosystem. Quantum Inspire is an educational and research platform developed by QuTech, while QuantWare is a commercial spin-off that emerged from its superconducting quantum processor research. Although institutionally related, we treat each entity in a dedicated section to reflect their distinct missions and independent contributions to the advancement and commercialization of superconducting quantum technologies.

12.2 QuTech's endeavors in spin qubits

QuTech has made remarkable advancements in the development of next-generation spin-qubit processors using sophisticated ²⁸Si/SiGe and Ge/SiGe heterostructures. In the realm of silicon, QuTech achieved a significant milestone by establishing a two-dimensional electron gas with notably enhanced and uniformly consistent transport characteristics across a substantial 100 mm wafer, as reported in [234]. Additionally, QuTech harnessed a comprehensive approach that amalgamates 3D atomic-level attributes, theoretical insights, and transport considerations to shed light on and statistically predict valley splitting, as documented in [235]. In the domain of germanium, QuTech accomplished impressive mobilities surpassing one million cm²/Vs, as highlighted in [236]. The team also pioneered the utilization of germanium bilayers for the development of three-dimensional quantum devices [237]. Also, QuTech achieved a noteworthy feat by designing the initial rudimentary error correction circuit, effectively implementing the phase-flip code on a four-qubit germanium quantum processor, as outlined in [238].

QuTech's achievements extended to the domain of silicon as well, with a reported two-qubit gate fidelity of 99.65% for electron spins within Si/SiGe quantum dots, an accomplishment published in [239]. This achievement marks a highly anticipated advancement



that positions the field of spin qubits closer to achieving fault-tolerant quantum computation. Further extending their prowess, QuTech successfully constructed a linear array housing six electron spin qubits within a Si/SiGe quantum dot arrangement [240]. This achievement not only surpasses the previous threshold for the number of qubits in silicon but also simultaneously achieves exceptional initialization fidelities, readout fidelities, single-qubit gate fidelities, and two-qubit state fidelities.

In close collaboration with Intel, QuTech measured the first qubits produced within the very same industrial manufacturing facilities responsible for the mass production of conventional computer chips (see Fig. 20), as highlighted in [241]. To delve into comprehensive insights regarding quantum transport and spin qubits within quantum dot devices, we recommend consulting the following review articles: [217, 221, 231, 242, 243].

Table 13 Performance metrics of the Quantum Inspire’s Spin-2 [232, 244] quantum processor

Key Performance Parameters	Spin-2 Quantum processor
Material	²⁸ Si QW in SiGe ^a
Single-qubit gate control	EDSR ^b microwave pulsing
Fidelity of single-qubit gates (RB) ^c	~ 99.0% (average)
1-qubit gate duration	250 ns (average)
Two-qubit gate control	Exchange interaction
Fidelity of two-qubit gates	> 90% (estimate - under validation)
Two-qubit gate duration	150 ns
Energy relaxation time (T1)	> 20 ms (similar device)
Phase coherence time (T2*)	> 6 μs
Fidelity of initialization and readout	~ 85% (average value)
Readout Mechanism	Spin-selective tunneling
Readout duration	2 × 300 μs = 600 μs (sequential readout)

^aSilicon-28 Quantum Well in Silicon-Germanium Matrix
^bElectrically Detected Spin Resonance (EDSR) using Microwave Pulsing
^cRandomized Benchmarking

13 Quantum Inspire

13.1 Overview

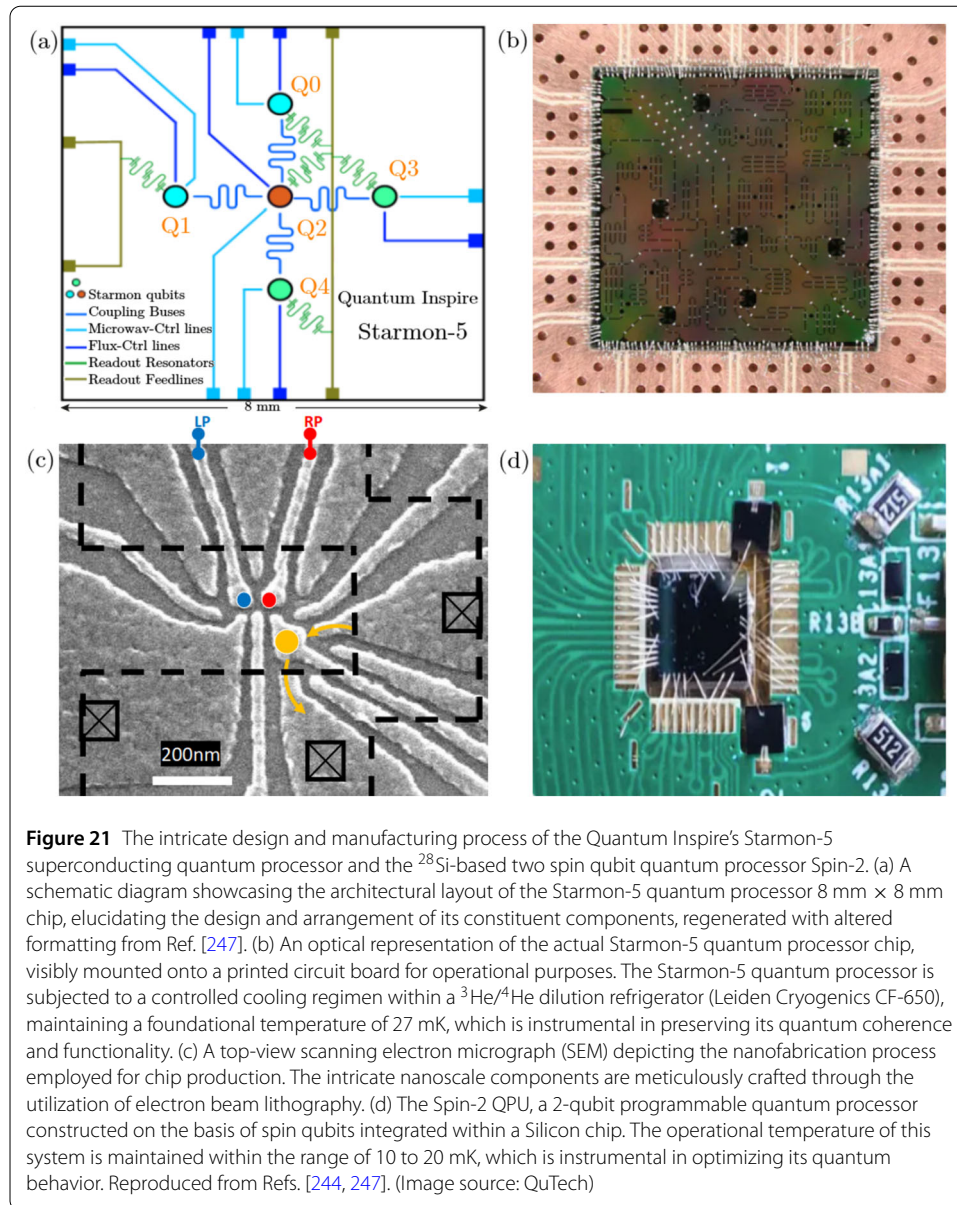
Launched in May 2020, Quantum Inspire [232] emerges as Europe’s inaugural public quantum computing platform [229]. Centralizing its efforts on education, training, and application refinement. Quantum Inspire integrates a processor founded on the promising semiconductor (spin qubits), produced via techniques akin to classic transistors but on a comparable diminutive scale [232, 244]. This characteristic lends itself to potential mass production. Additionally, the platform provides access to a processor based on superconducting (transmon) qubits, creating a distinctive amalgamation. Users are afforded the opportunity to experiment with quantum algorithms and undertake processor comparisons [229].

Distinguished by its modularity, Quantum Inspire offers cloud-based access to various QuTech quantum technologies [233], including programmable quantum processors such as: the two-qubit SPIN-2 [232, 244] and the five-qubit STARMON-5 [92, 245], founded on spin qubits and superconducting transmon qubits [231, 246] respectively. The platform also encompasses programmable quantum computer simulators, namely QX-26, QX-31 and QX-34-L, accommodating up to 34 qubits [230]. Table 13 offers performance metrics of the Spin-2 [232, 244] quantum processor.

13.2 The Starmon-5 quantum processor

The Starmon-5 [247], developed by QuTech, is a superconducting quantum processor designed using circuit quantum electrodynamics principles [92, 245]. It comprises five transmon qubits [248] arranged in an X-shaped coupling configuration on an 8 mm × 8 mm chip (see Fig. 21). The central transmon (Q2) is connected to four corner transmons (Q0, Q1, Q3, Q4) via dedicated bus resonators, enabling two-qubit operations. Each qubit is equipped with microwave and flux control lines for single- and two-qubit gates, respectively, and features dispersively-coupled resonators for high-fidelity readout [247]. The resonators include Purcell filters [249], which enhance signal isolation and reduce decay rates. The processor’s unique seven-port connectivity, forming a star-shaped layout, earned it the nickname “Starmon” [250].

The Starmon-5 operates within a ³He/⁴He dilution refrigerator, achieving a base temperature of 19 mK to maintain superconducting conditions. Its control system integrates



room-temperature electronics, including a QuTech Central Controller, Zurich Instruments' high-density arbitrary waveform generators, and quantum analyzers. These instruments provide high precision for gate implementation, ensuring timing accuracy and flexibility.

The native gate set of Starmon-5 includes several single-qubit gates, such as rotations around the X and Y axes and virtual Z gates, all implemented with optimized DRAG pulses for minimal errors [251, 252]. The processor's primary two-qubit gate is the CZ gate, realized through flux-tuned interactions. Gate durations are highly optimized, with single-qubit gates completing in 20 ns and CZ gates in 40 ns. Qubit readout is performed via frequency-multiplexed measurements of the readout resonators, with a total readout time of approximately 1 μs . The system achieves high fidelity for state preparation and measurement, supported by advanced calibration techniques [247].

14 Quantware

14.1 Overview

QuantWare, founded in July 2021, is dedicated to making quantum hardware more accessible and accelerating the development of practical quantum computing. The company focuses on scaling superconducting QPUs to drive meaningful computational advancements. Its goal is to offer quantum processors that enhance computational capabilities while ensuring economic accessibility for global enterprises. This enables organizations to build quantum computing infrastructures at a fraction of the cost of competing solutions.

With a firm commitment to an open architecture philosophy, QuantWare is actively involved in developing cutting-edge technologies aimed at exponentially increasing the number of qubits within a single processor (see Sect. 14.2). The company also collaborates with leading experts and organizations to advance progress in the quantum technology field. Ultimately, these efforts are geared toward creating processors capable of performing meaningful quantum computations in the near future.

14.2 Quantware's quantum processors

QuantWare develops state-of-the-art quantum processors, including the 21-qubit fixed coupler QPU *Contralto*, the 5-qubit QPU *Soprano*, and the 64-qubit QPU *Tenor*. These processors feature fully interconnected qubits, high T1 coherence times, and exceptional gate fidelities, enabling significant advancements in NISQ algorithms, error correction, and quantum research. Additionally, QuantWare's *Crescendo* traveling wave parametric amplifier sets new standards in quantum efficiency and multiplexed readout. Figure 22 provides an overview of QuantWare's superconducting quantum processors, showcasing devices from the 5-qubit *Soprano* to the 64-qubit *Tenor*, along with their associated performance metrics and hardware components. Together, these innovations highlight QuantWare's commitment to making quantum computing more powerful, practical, and accessible for researchers and enterprises worldwide. In this section, we briefly explore the quantum processors developed by QuantWare.

14.2.1 The Soprano 5-Qubit QPU

The *Soprano* quantum processor, specifically the *Soprano-D* model, offers a high-performance 5-qubit fixed coupler quantum processing unit designed to streamline the development of quantum computers. It features up to five interconnected, fully-controllable superconducting transmon qubits with impressive coherence times and low error rates (60 μ s average T1 coherence). The processor includes dedicated drive and flux lines for each qubit and offers the option of adding two additional isolated qubits for benchmarking purposes. Additionally, *Soprano-D* supports seamless integration with QuantWare's *Crescendo-S* amplifier to enhance readout fidelities, reaching up to 97.5%. The processor is fully packaged, cryogenically characterized, and comes with expert support, ensuring ease of implementation and operation. Available with rapid procurement times, *Soprano-D* provides a reliable and efficient solution for building quantum computing systems at an accessible cost.

14.2.2 The Contralto 21-Qubit QPU

The *Contralto-D* quantum processor by QuantWare is designed to support the development of high-performance, industry-scale quantum computers. It features up to 21

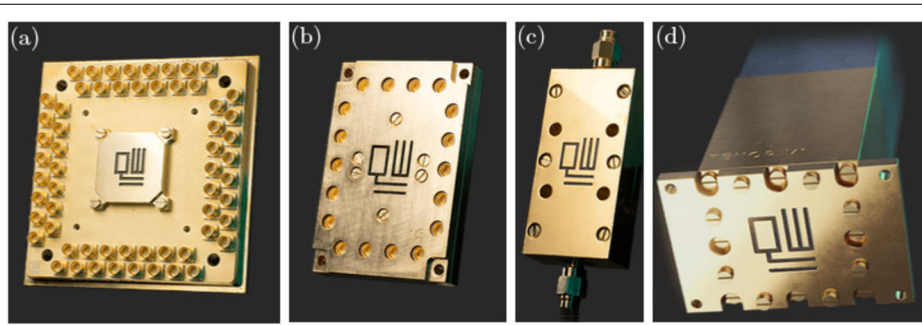


Figure 22 QuantWare Superconducting quantum processors: (a) The 21-qubit fixed coupler QPU Contralto. A cutting-edge quantum processor, offers 21 fully interconnected qubits and 4 isolated qubits, empowering research in NISQ algorithms and advanced error correction. With dedicated control lines, Contralto achieves 10 to 60 μs T1 coherence and 99.9% single-qubit gate fidelity. (b) The 5-Qubit QPU Soprano. The first commercially available superconducting 5-qubit QPU, accelerates quantum research with a T1 coherence of 10 to 60 μs and 99.9% single-qubit gate fidelity, delivered fully packaged with short lead times. (c) TWPA Crescendo: Crescendo, QuantWare's pioneering traveling wave parametric amplifier, sets new standards with 18 dB gain, 70% quantum efficiency, -100 dBm saturation power, and 50 Ohm impedance. Ideal for multiplexed readout, it operates flux-insensitively and serves 1 GHz readout bands from 5 to 9 GHz. (d) The 64-Qubit QPU Tenor. Tenor 64, the world's most advanced quantum processor, features 64 fully controllable qubits and QuantWare's 3D scalable architecture. Researchers explore quantum error correction and NISQ algorithms on their open architecture systems (currently for early access partners only). Images source QuantWare [253]

fully interconnected superconducting transmon qubits and four isolated qubits for benchmarking. With advanced performance characteristics, such as a 39 μs T1 coherence time, dedicated drive and flux lines for each qubit, and QuantWare's Hamiltonian parameter design, *Contralto-D* offers superior reliability for large-scale quantum experiments, including NISQ algorithms and error correction schemes. This processor comes with pre-delivery cryogenic characterization, high-density cabling, and expert support for seamless integration. Additionally, it can be paired with the *Crescendo-S* amplifier to achieve readout fidelities of up to 97.5%. Available in multiple configurations, *Contralto-D* is believed to provide an accessible yet cutting-edge solution for quantum computing at a fraction of the cost of competing hardware.

14.2.3 The Tenor 64-Qubit QPU

The *Tenor* quantum processor by QuantWare represents the world's largest commercially available quantum processor, featuring 64 fully controllable qubits, fortified by QuantWare's proprietary next-generation 3D scalable architecture. Powered by QuantWare's VIO platform, *Tenor* is designed to scale effectively with reduced crosstalk, making it ideal for large-scale quantum computing applications. Currently available in Early Access, *Tenor* allows researchers and companies to build advanced quantum computers with unprecedented qubit capacity. This groundbreaking processor offers the performance and scalability needed for future quantum breakthroughs. Pre-orders are now available for those looking to integrate *Tenor* into their quantum computing systems.

15 OriginQ Cloud

15.1 Overview

OriginQ Cloud, also known as Origin Quantum Cloud, is a pioneering company in the field of quantum computing, dedicated to advancing the development and practical appli-

Table 14 Performance metrics of OriginQ Wukong superconducting quantum computer

Performance metric	Value
Qubits	72
Average T1 Time (μ s)	18.72
Average T2* Time (μ s)	1.46
Single-qubit gate average fidelity	99.68%
Two-qubit gate average fidelity	95.71%
CLOPS ^a	370,000

^aCLOPS (Circuit layer operations per second) is a metric that reflects the speed at which a quantum processor can run circuits. Specifically, it measures how quickly the processor can execute layers of a parameterized model circuit, similar to those used in Quantum Volume assessments [254].

cation of quantum technologies. The company offers a comprehensive platform that provides cloud-based quantum computing services, integrating quantum, supercomputing, and intelligent computing to optimize task processing efficiency. Through its Quantum Cloud Services, Origin Quantum facilitates easy access to quantum computing resources and tools, empowering users to develop and implement quantum algorithms.

At the core of its software ecosystem is QPanda, an open-source quantum programming SDK that supports efficient quantum algorithm development, hardware integration, and high-performance simulations, while also addressing error mitigation and noise analysis. Other key offerings include VQNet for quantum machine learning, and pyqpanda-algorithm, a software package for common quantum algorithms. QRunes and Qurator further enhance the development environment with specialized programming languages and tools. Origin Quantum's solutions supports various industries, including AI, big data, finance, biochemistry, and information security, while also promoting education through quantum computing curricula, competitions, and educational frameworks. These initiatives drive innovation and expand the potential of quantum technologies.

15.2 OriginQ Wukong

The OriginQ “Wukong” superconducting quantum computer, launched in 2024, represents a significant advancement in quantum computing technology. This third-generation, self-developed quantum processor integrates fully controlled hardware, chips, operating systems, and application software, with a localization rate surpassing 80%. As shown in Table 14, the performance metrics of the OriginQ Wukong superconducting quantum computer include 72 qubits, an average T1 time of 18.72 μ s, and an average T2* time of 1.46 μ s. The system demonstrates high precision, with a single-qubit gate fidelity of 99.68% and a two-qubit gate fidelity of 95.71%. Additionally, it achieves 370,000 CLOPS (circuit layer operations per second [254]), highlighting its high computational efficiency.

OriginQ “Wukong” is complemented by a range of high-performance quantum simulators, including full-amplitude quantum simulator (35 qubits), noise-inclusive quantum simulator (34 qubits), partial-amplitude quantum simulator (68 qubits), and single-amplitude quantum simulator (200 qubits), which enable diverse quantum simulations. The platform provides a comprehensive set of tools for quantum programming and application development, such as graphical programming tools, QPanda SDK, and integrated development environments. These features facilitate both the learning and practical application of quantum algorithms, underscoring the platform's role in advancing quantum computing research and innovation.

15.3 Quantum programming frameworks and tools

Origin Quantum offers a comprehensive suite of programming tools designed to enhance quantum algorithm development and implementation. QPanda 3.0 is a high-performance, hardware-software collaborative quantum programming framework that supports Python, providing a robust environment for quantum computing tasks. For earlier versions, QPanda 2 offers both a core open-source quantum computing framework and a Python-based version, pyqpanda, enabling users to build, run, and optimize quantum algorithms. In addition, VQNet is a quantum machine learning framework tailored for constructing and optimizing quantum algorithms, accompanied by extensive API and example documentation. The pyqpanda-algorithm package consolidates essential quantum algorithms and functions, providing a foundational toolkit for developers. QRunes, a process-oriented, command-based quantum programming language, facilitates the creation of quantum algorithms with a straightforward, script-like syntax. To streamline the programming experience, Qurator integrates quantum program compilation into a one-stop development environment, simplifying installation and usage.

15.4 Quantum application software and solutions

In addition to quantum programming tools, Origin Quantum has also developed specialized software for quantum applications. ChemiQ, built on QPanda, uses the inherent advantages of quantum superposition and entanglement to compute molecular energy, structure, chemical reactions, potential energy curves, and dynamic trajectories on real or simulated quantum computers. Similarly, pyChemiQ, a Python library, supports the simulation and computation of fermionic systems on both real and virtual quantum machines. The OriginQ software platform utilizes quantum linear solvers based on the HHL and variational algorithms to replace classical solvers, accelerating hybrid quantum-classical computations with advanced preprocessing techniques. Lastly, QLayout, the first domestic quantum chip layout tool, provides a user-friendly graphical interface for chip design, enabling a broader range of users to participate in quantum chip layout creation without requiring programming expertise. These tools collectively contribute to advancing quantum computing capabilities across a range of applications, from machine learning to chemistry and hardware design.

16 Alibaba Cloud

16.1 Overview

In July 2015, Alibaba Cloud [255], a subsidiary of Alibaba Group, announced the establishment of a new quantum computing research laboratory. The primary objective of this laboratory is to confront and potentially exceed the constraints of Moore's Law by exploring next-generation computing techniques that promise exceptionally high speeds [26]. These advancements leverage Alibaba Cloud's technical expertise in areas such as artificial intelligence, classical computing algorithms, and cloud computing, establishing the company as a potential key player in quantum innovation.

16.2 Alibaba Cloud quantum development platform

A series of reports outlining Alibaba's initiatives in developing the is provided in [256, 257]. The AC-QDP is an integrated toolkit aimed at promoting the development of quantum computing algorithms and quantum processors. It utilizes a robust classical simulator hosted on the Alibaba Cloud infrastructure to enhance its capabilities.

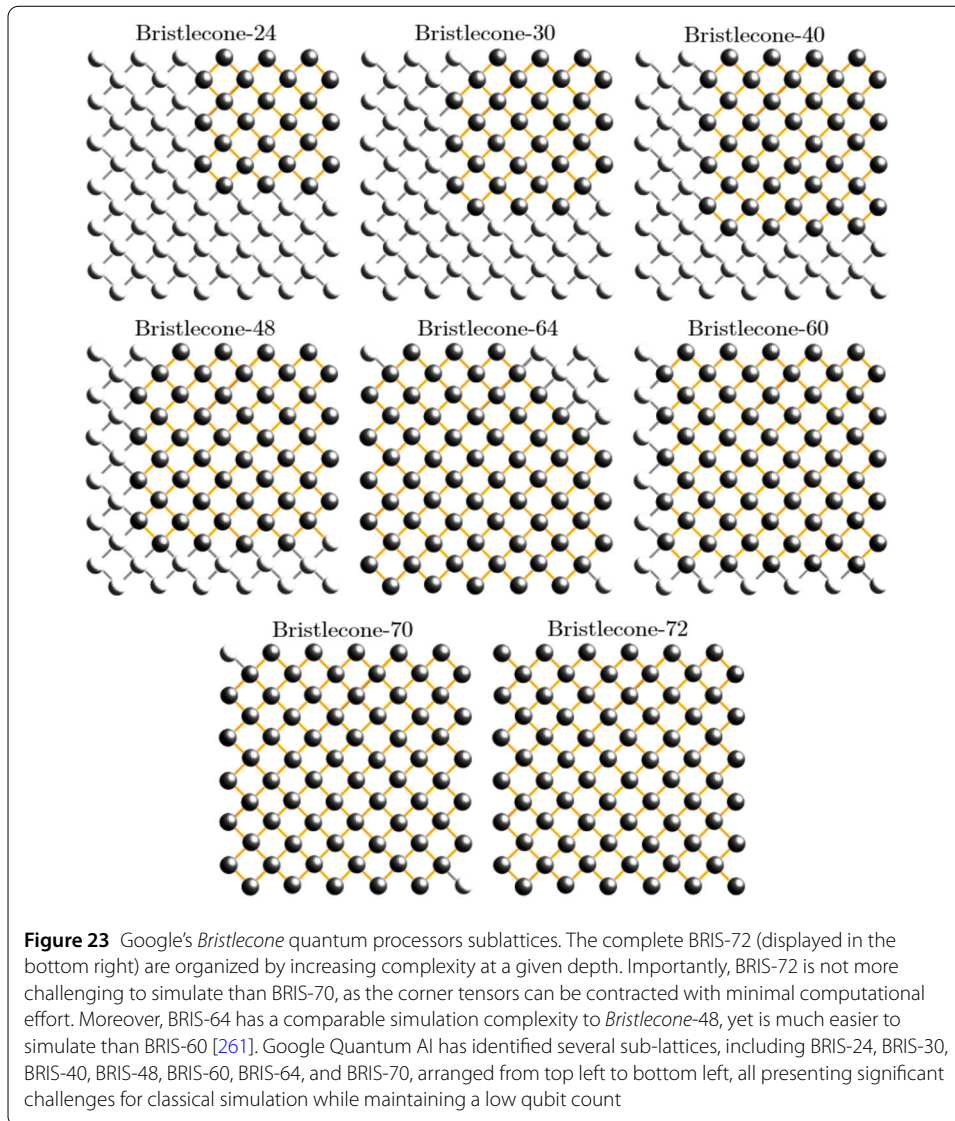
Table 15 Projected time required to simulate Google's *Bristlecone-70* circuits with 127,512 CPU cores [256]. The data indicates that a greater number of calculated amplitudes is associated with a reduction in execution time per amplitude (P.A)

Circuits	Amplitudes	Subtasks P.A.	Computational Resources	Time (s) P.A.
<i>Bristlecone-70</i> $\times$ (1+28+1)	1.00	256.0	(2/ vCPU; 4/ GB)s for $l = 1, 2, 4, 8$	9.71
	256.0	256.0	(2/ vCPU; 4/ GB)s for $l = 1, 2, 4, 8$	0.03
	1000	256.0	(2/ vCPU; 4/ GB)s for $l = 1, 2, 4, 8$	0.024
	200,000	256.0	(2/ vCPU; 4/ GB)s for $l = 1, 2, 4, 8$	0.02
	200,000	256.0	(88 vCPU; 160 GB) $\times$ 1449	0.03
<i>Bristlecone-70</i> $\times$ (1+32+1)	1.00	1024	(2/ vCPU; 4/ GB)s for $l = 1, 2, 4, 8$	23
	1000	1024	(2/ vCPU; 4/ GB)s for $l = 1, 2, 4, 8$	0.28
	10,000	1024	(2/ vCPU; 4/ GB)s for $l = 1, 2, 4, 8$	0.27
	10,000	1024	(88 vCPU; 160 GB) $\times$ 1449	0.36
<i>Bristlecone-70</i> $\times$ (1+36+1)	1.00	65,536	(2/ vCPU; 4/ GB)s for $l = 1, 2, 4, 8$	7.62
	10.00	65,536	(2/ vCPU; 4/ GB)s for $l = 1, 2, 4, 8$	3.3
	100.00	65,536	(2/ vCPU; 4/ GB)s for $l = 1, 2, 4, 8$	2.69
	100.00	65,536	(88 vCPU; 160 GB) $\times$ 1449	4.56
<i>Bristlecone-70</i> $\times$ (1+40+1)	1.00	4,194,304	(2/ vCPU; 4/ GB)s for $l = 1, 2, 4, 8$	412.25
	5.00	4,194,304	(2/ vCPU; 4/ GB)s for $l = 1, 2, 4, 8$	389.43
	1.00	4,194,304	(88 vCPU; 160 GB) $\times$ 1449	480.17

They showcased the computational capabilities of the AC-QDP through a series of experiments [256, 258, 259]. This includes the role of AC-QDP in evaluating large-scale quantum algorithms, specifically the QAOA [260]. The AC-QDP framework for QAOA enables the simulation of thousands of qubits across four layers, providing a robust platform for testing and optimizing quantum solutions [258]. To assess its classical simulation performance, they adopted random quantum circuits originally designed for Google's *Bristlecone* (BRIS) QPU as a benchmark [114, 162].

Remarkably, they managed to simulate BRIS-70 circuits with a depth of $1 + 32 + 1$ in a mere 0.43 seconds per amplitude. This achievement was made possible by employing 1449 Alibaba Cloud Elastic Computing Service (ECS) instances, each equipped with 88 Intel Xeon (*Skylake*) Platinum 8163 vCPU cores running at 2.5 GHz and boasting 160 gigabytes of memory (see Table 15). In comparison, the previously best-reported results for the same computational tasks were 104 seconds and 135 seconds, using NASA's HPC *Pleiades* and *Electra* systems, respectively [261]. Furthermore, they went on to report simulations of *Bristlecone-70* circuits with depths of $1 + 36 + 1$ and $1 + 40 + 1$, accomplishing these simulations in 5.6 seconds and 580.7 seconds per amplitude, respectively. According to [256], these simulations represent the first successful instances at such depths. An architectural overview of Google's BRIS-72 quantum processor and series of sub-lattices (BRIS-24, BRIS-30, BRIS-40, BRIS-48, BRIS-60, BRIS-64, and BRIS-70) are shown in Fig. 23.

Additionally, a simulation involving a distance-3 logical qubit that was encoded within a 17-qubit surface code was conducted in [257]. The authors used experimental noise parameters specific to transmon qubits in a planar circuit Quantum Electrodynamics (QED) architecture [262]. Notably, their simulation included the consideration of crosstalk induced by ZZ-interactions [165, 263–267]. Their findings revealed that, under the prevailing noise conditions, crosstalk played a substantial role in causing dephasing of the logical qubit. This resulted in a total phase-flip probability of approximately 0.6%, which was approximately 60% higher than what would be anticipated without accounting for crosstalk. These results provide quantifiable evidence of the considerable impact of simple, correctable crosstalk [268–270]. This underscores the imperative need to address even



the secondary effects of crosstalk in order to effectively implement error-correction techniques and realize the benefits they offer.

16.3 Classical simulation of Google's *Sycamore*-54 quantum supremacy circuits

Google quantum AI and collaborators demonstrated their achievement of quantum computational advantage using their superconducting quantum processor, *Sycamore*-54 [12]. This marked a significant milestone as the task for which quantum supremacy was claimed had initially been estimated to take approximately 10,000 years for *Summit*, the world's most powerful supercomputer at that time, to complete. Remarkably, *Sycamore*-54 accomplished the same task in just 200 seconds.

However, in a subsequent study [166], Alibaba Quantum Laboratory, introduced a classical simulation algorithm based on tensor networks [12, 114, 256–259, 261]. They conducted simulations using a cluster comparable in performance to *Summit* and found that their simulator could complete the same quantum supremacy task proposed by Google in less than 20 days. On smaller instances of the problem, they managed to reduce the

Table 16 Extrapolated benchmarks for simulating RCS (random circuit sampling) on Google's *Sycamore-54* with AC-QDP version 2.0 on an Alibaba Cloud cluster, synonymous with the *Summit* supercomputer. For m -cycles, $m = 12, 14$, one sample was explicitly run on a single V100 [166]

Benchmarking results					
No. Cycles	12	14	16	18	20
No. Amplitudes in batch	64	64	64	64	64
$\mathcal{F}_{\text{XEB}}^a$	1.3%	1.0%	0.7%	0.4%	0.2%
FLOPS efficiency and Contraction cost for 1 sample					
FLOPS efficiency	14.56%	14.97%	14.05%	16.67%	14.74%
Contraction cost	1.09×10^{13}	7.16×10^{13}	1.15×10^{17}	2.59×10^{17}	6.66×10^{18}
No. Subtasks	1×2^7	1×2^{10}	1×2^{20}	1×2^{21}	1×2^{25}
Time for generating 1 perfect sample					
<i>Summit</i>	0.0014 s	0.0088 s	15.06 s	28.69 s	833.75 s
One V100	38.02 s	243.71 s	4.82 d	9.18 d	266.80 d
Time for generating 1,000,000 samples with $\mathcal{F}_{\text{XEB}}$					
No. Perfect samples	13,000	10,000	7000	4000	2000
<i>Summit</i>	18 s	88 s	1.2 d	1.3 d	19.3d

^a $\mathcal{F}_{\text{XEB}}$: XEB fidelity.

runtime from years to mere minutes, achieving a significantly faster computational speed than *Sycamore-54* itself (see Table 16). These estimates were derived from explicit simulations of parallel subtasks and accounted for all potential computational costs, leaving no room for hidden expenditures.

16.4 Alibaba's fluxonium qubits quantum processor

The Alibaba Quantum Laboratory, has developed a quantum processor based on fluxonium qubits [45, 47, 49], offering a promising alternative to traditional superconducting qubit platforms for achieving high-fidelity quantum operations [46]. This fluxonium-based quantum processor combines several key features, including exceptional qubit coherence, rapid frequency tunability, and the ability to individually address qubits for operations such as resetting, readout, and gate operations [46].

Utilizing straightforward and efficient gate schemes, the researchers at Alibaba Quantum Laboratory have achieved remarkable results in terms of qubit fidelity. They have attained an impressive average single-qubit gate fidelity of 99.97% and a two-qubit gate fidelity of up to 99.72%. These fidelity levels are on par with, and in some cases even surpass, the best-reported results in the existing literature on superconducting transmon circuits [109, 193, 271–278]. The specifications of Alibaba's fluxonium qubits quantum processor are listed in Table 17.

This work represents a significant milestone in the field of superconducting qubits [46]. It introduces an innovative approach to fault-tolerant quantum computing that competes with and provides an alternative to the widely used transmon system [271], offering new possibilities for advancing the development of reliable quantum computing technologies. For an in-depth understanding of the fabrication processes and experimental setups used in Alibaba's Fluxonium quantum processor, readers are referred to [46].

Suddenly, Alibaba has terminated its quantum computing initiatives. In November 2023, Reuters reported that the company dismantled its quantum computing laboratory and team within its research division, donating the laboratory and associated experimental equipment to Zhejiang University [279].

Table 17 Device metrics of Alibaba's two-fluxonium qubits quantum processor [46]

Parameter	E_C/h^a	E_L/h^b	E_J/h^c	$(\omega_{21} - \omega_{10})/\omega_{10}^d$	$\omega_{10}/2\pi^e$	$\omega_{21}/2\pi^f$	$\omega_{R,0}/2\pi^g$	$\chi/2\pi^h$	T_1^i	$T_{2,Echo}^j$
QA	1.398	0.523	2.257	1.771	1.09	3.02	6.696	0.63	80.00	30.00
QB	1.572	0.537	2.086	1.771	1.33	3.20	6.753	0.40	57.0	17.0
units	GHz	GHz	GHz	GHz	GHz	GHz	GHz	MHz	μs	μs

^aCharging energy.^bInductive energy.^cJosephson energy.^dThe qubit anharmonicity.^eThe transition frequency between the ground and first excited states.^fThe transition frequency between the first and second excited states.^gResonance frequencies, in the dispersive coupling regime for qubit preparing in the $|0\rangle$ state.^hThe resonance shift.ⁱThe coherence times T_1 .^jThe coherence times $T_{2,Echo}$.

17 Anyon systems

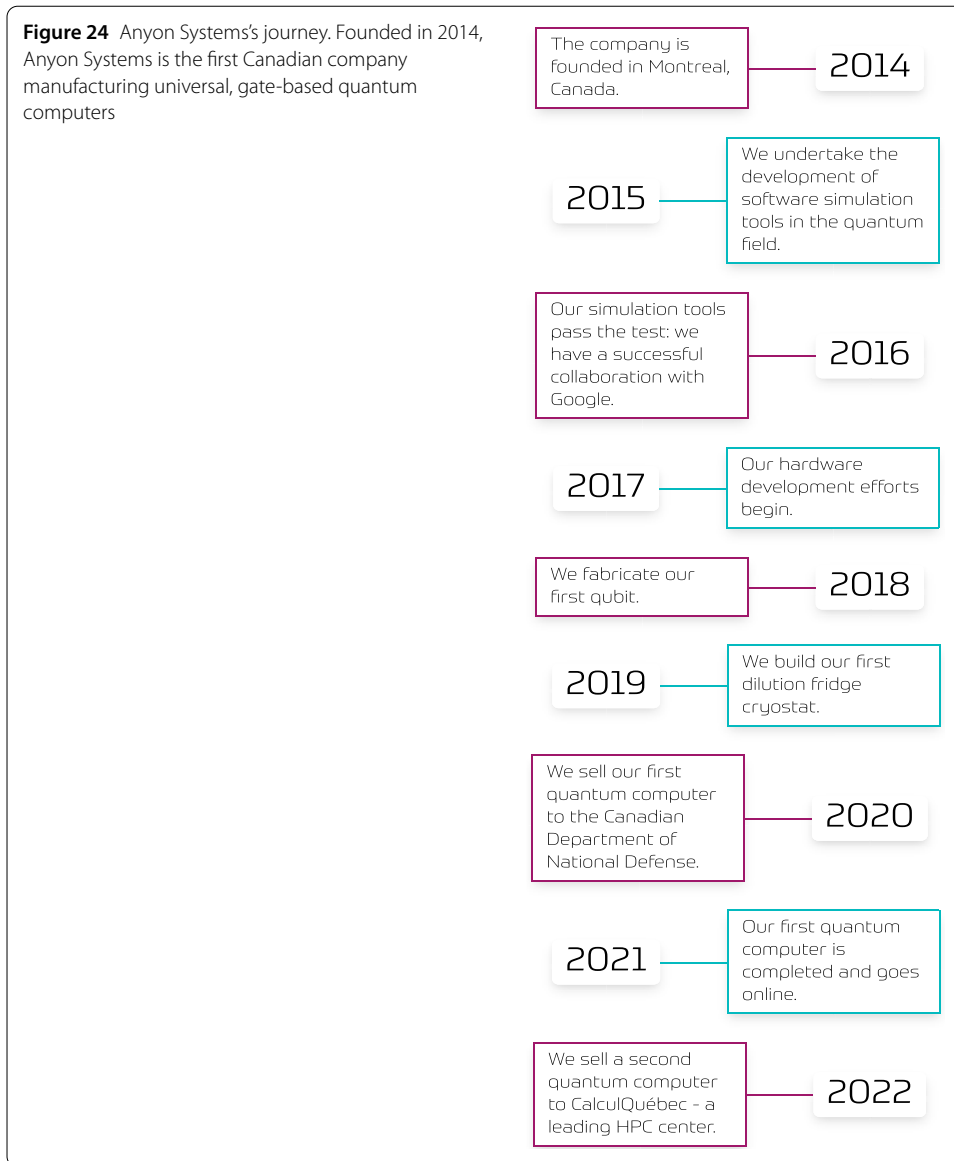
17.1 Overview

Established in 2014, Anyon Systems Inc. [280] stands as a pioneering Canadian organization dedicated to the production of universal gate-based quantum computers, primed for comprehensive quantum computation. Anyon's primary objective centers on providing high-capacity, fault-tolerant quantum computers to a broad spectrum of early adopters, encompassing government agencies, high-performance computing centers, and educational institutions, in the immediate future. Anyon holds the belief that its exclusive toolset holds the potential to expedite the design and development cycles of superconducting quantum chips, thereby offering substantial contributions to the quantum computing domain.

On December 20, 2021, Anyon Systems Inc. reached a significant milestone by successfully delivering an integrated quantum computer to the Defense Research and Development Canada (DRDC), a vital component of Canada's Department of National Defense (see Fig. 24). This quantum computing system showcases Anyon's cutting-edge Yukon generation superconducting quantum processor, complete with the company's proprietary cryogenic cooling system and in-house quantum electronics. The exceptional performance metrics of this quantum computer corroborate Anyon's vision, which prioritizes early investments in tools and vertical integration, enabling the design and deployment of integrated quantum computers and establishing a unique foundation for rapid scalability. These include the proprietary cryogenic system, quantum electronics, and software stack, collectively contributing to the realization of cutting-edge quantum computing capabilities.

17.2 Qube™

The Qube™ represents an all-in-one superconducting quantum computing system designed for seamless integration within existing High-Performance Computing (HPC) centers. This innovative system boasts a self-contained unit that combines both cryogenic and electronic components. The Qube™ also features a user-friendly human-machine interface and requires minimal maintenance, thanks to its closed-cycle cryogenic system. Moreover, its scalable design allows for the straightforward incorporation of quantum chips with an expanding number of qubits.



Additionally, Anyon's SDK is firmly rooted in the Julia programming language, providing users with a convenient interface for engaging with the quantum computing facility to execute algorithms by specifying a quantum circuit. Subsequently, Anyon's quantum compiler optimizes this circuit, ensuring the most efficient execution on the quantum processor.

17.3 Performance metrics of Qube™

Anyon Systems Inc. released the performance metrics for its Yukon generation superconducting quantum processor. The Anyon team takes immense pride in achieving the performance metrics, as presented in Table 18, which distinctly establishes Anyon as one of the preeminent superconducting quantum computing companies globally. This accomplishment imbues Anyon Systems Inc. with a high degree of optimism regarding their capacity to persevere in the execution of their strategic roadmap, aimed at enhancing qubit performance and scaling up the integrated quantum computers to be supplied to early adopters.

Table 18 Qube™ Quantum Performance Metrics. The Qube™ is an all-in-one superconducting quantum computer designed to integrate effortlessly into current high-performance computing (HPC) facilities

Gate Type	Fidelity	Time ^a
Average Fidelity Metrics		
Single Qubit Gate (Individual)	99.8%	15 ns
Single Qubit Gate (Parallel)	99.7%	15 ns
Two Qubit Gate	95.6%	35 ns
Coherence Type		
Coherence Time		
T_1 Coherence Time	10 μ s	
T_2 Coherence Time (echo)	8 μ s	
T_2^* Coherence Time (Ramsey)	4 μ s	

^aGate duration.

The Qube™ exemplifies remarkable fidelity in its quantum operations, substantiated by the following performance metrics. In the realm of single qubit gate operations, when conducted individually, it achieves an impressive fidelity rate of 99.8%, with a mere 15-nanosecond gate duration. In parallel single qubit gate executions, the system maintains a notably high fidelity of 99.7%, again with a 15-nanosecond gate duration. Additionally, when orchestrating two qubit gate operations, the Qube™ offers a fidelity level of 95.6%, with a slightly extended 35-nanosecond gate duration [281].

The Qube™ showcases impressive coherence times, a critical factor in the realm of quantum computing operations. Notably, the coherence time denoted as T_1 extends to a commendable 10 microseconds, providing ample time for quantum processes to unfold. Furthermore, the T_2 coherence time, particularly in the echo configuration, exhibits robustness with an extended duration of 8 microseconds [281]. In the Ramsey configuration, the T_2^* coherence time remains substantial at 4 microseconds. These coherence times underpin the reliability and effectiveness of the Qube™ in quantum operations, enabling extended periods for quantum processes and computations.

18 Alice and Bob

18.1 Overview

Alice & Bob is an emerging leader in the quantum computing sector, focused on overcoming key engineering challenges related to scaling and error correction in quantum systems. The company has developed a novel technology based on “cat qubits,” a form of superconducting qubits designed to address the scalability bottleneck that has historically hindered the development of fault-tolerant quantum computers. Unlike traditional quantum error correction methods, Alice & Bob’s cat qubits exhibit exceptional protection against bit-flip errors, thereby significantly reducing the resources needed for error correction and enabling the production of logical qubits with remarkably high fidelity (99.99%) [282]. The company’s five-year roadmap (see Sect. 18.3) presents a clear plan to develop a universal fault-tolerant quantum computer, aimed at addressing real-world problems, by 2030.

18.2 Cat qubits

Cat-qubit technology offers a transformative solution to the challenge of quantum error correction by providing inherent protection against bit-flip errors. Unlike traditional

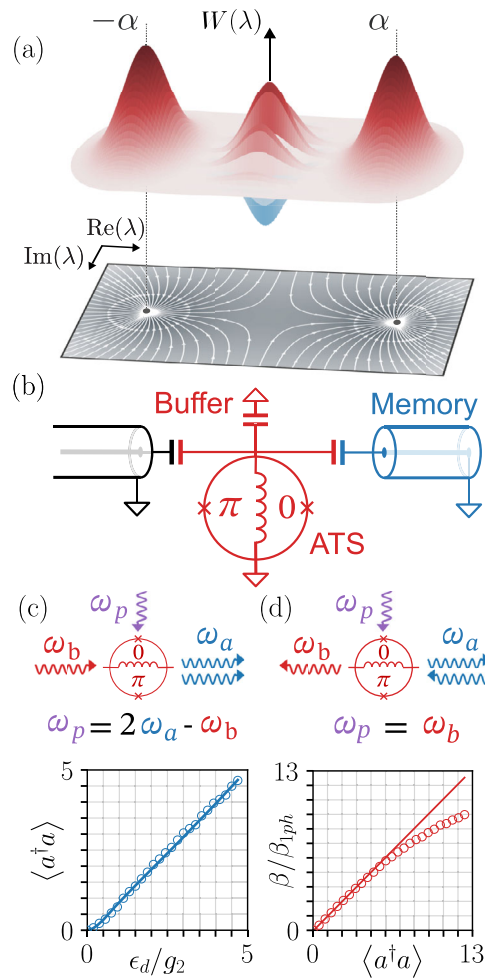
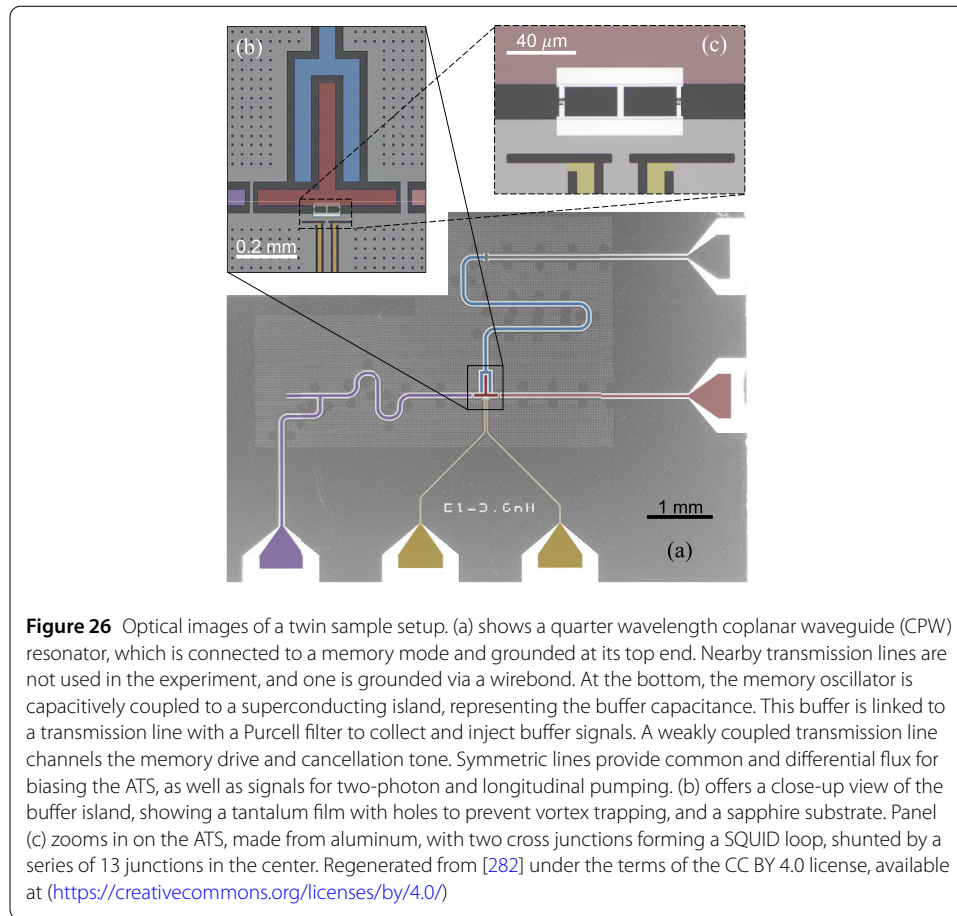


Figure 25 Quantum information encoding in a Bistable system. (a) Classical trajectories (solid lines from $H_{2ph} = g_2 a^{\dagger 2} b + g_2^* a^2 b^{\dagger} - \epsilon_d^* b - \epsilon_d b^{\dagger}$, $L_b = \sqrt{\kappa_b} b$, where a and b represent the annihilation operators for the memory and buffer modes, respectively, ϵ_d is the amplitude of the resonant drive applied to the buffer, and κ_b is the buffer's energy dissipation rate) lead to convergence in two stable states, where quantum information is stored in a space defined by these states, offering resilience to bit-flip errors. The Wigner function illustrates a superposition state. (b) The system is implemented as a circuit: A quarter-wavelength transmission line mode (blue, representing memory) is linked to its environment (black) through a buffer mode (red) that utilizes an asymmetrically threaded SQUID (ATS). (c) A combination of pump (purple) and buffer drive (red) via the ATS facilitates the injection of photon pairs into the memory mode (blue). The reverse process, which removes photon pairs, is not shown. Experimental measurements (open circles) display a steady increase in the photon count within the memory (y-axis) as the drive amplitude (x-axis) increases, consistent with theoretical predictions (solid line). (d) A pump at frequency ω_b (purple) displaces the buffer mode (red) based on the photon count in the memory mode (blue), a key feature for quantum tomography. The buffer amplitude (y-axis) versus memory photon number (x-axis) after displacement reveals small deviations from the expected linear relationship (solid line), indicating higher-order effects leading to compression. Regenerated from [282] under the terms of the CC BY 4.0 license, available at (<https://creativecommons.org/licenses/by/4.0/>)

qubits, which require substantial hardware overhead to mitigate both bit-flip and phase-flip errors, cat-qubits are encoded in the metastable states of nonlinear dynamical systems (see Fig. 25). These systems employ two-photon dissipation to stabilize quantum states, a process that suppresses bit-flips without causing decoherence, unlike conventional dissipation methods that typically degrade quantum superpositions [283, 284].



In [282], the authors implemented a superconducting circuit-based cat-qubit with bit-flip error times exceeding ten seconds—representing a four-order-of-magnitude improvement over previous implementations [285, 286] (see Fig. 26). This advancement demonstrates the viability of cat-qubits for practical quantum computation, as they retain continuous bit-flip protection even during quantum manipulations, such as phase control operations [287]. Importantly, the achieved phase-flip times, measured at 490 nanoseconds, remain consistent with single-photon loss rates and can be addressed through error correction codes with significantly reduced hardware demands [288, 289].

The quantum tomography protocol introduced in [282] eliminates the need for ancillary hardware by leveraging the dynamical properties of the system, enabling precise state preparation and measurement without breaking the bit-flip protection [290]. These developments underscore the compatibility of cat-qubits with hardware-efficient error correction architectures, offering a promising route to scalable quantum computing [291, 292].

Despite these achievements, challenges remain. Improvements in fabrication techniques, circuit engineering, and gate design are needed to enhance state preparation fidelities, reduce gate errors, and further suppress phase-flip errors [35, 293]. Nonetheless, cat-qubit technology has demonstrated unprecedented robustness and scalability potential, paving the way for its integration into fault-tolerant quantum architectures [288, 289].

18.3 Towards fault-tolerant quantum computing

Alice & Bob's quantum computing roadmap outlines their mission to develop fault-tolerant quantum computers [294], with a focus on combating decoherence. By 2030, the company aims to launch a practical quantum computer, *Graphene*, with 100 high-fidelity logical qubits, marking a significant step into the quantum era. Their progress, beginning in 2020, has already produced notable advancements like the leading cat qubits and *Boson 4* chip. The roadmap includes five key milestones to achieve a universal, error-corrected quantum computer: mastering the cat qubit, building a logical qubit, creating fault-tolerant quantum computing, implementing universal quantum computing, and eventually delivering useful quantum computing through *Graphene*.

The company's five-year roadmap outlines a structured path toward achieving a universal fault-tolerant quantum computer (designed to solve real-world problems) by 2030. The roadmap is delineated across five critical milestones:

1. *Mastering the cat qubit (2024)*: The company successfully developed a reproducible cat qubit that can store quantum information with resilience to bit-flip errors.
2. *Building the logical qubit (ongoing)*: Currently under development with the *Helium* chip series, this milestone aims to produce the first error-corrected logical qubit, operating below the error-correction threshold.
3. *Achieving fault-tolerant quantum computing*: Through the *Lithium* chip series, Alice & Bob intends to scale multi-logical-qubit systems and demonstrate the first error-corrected logical gate.
4. *Enabling universal quantum computing*: The *Beryllium* chip series will facilitate the implementation of a universal gate set, supported by magic state factories and live error correction, to enable any quantum algorithm.
5. *Delivering useful quantum computing (2030)*: The *Graphene* chip series, comprising 100 high-fidelity logical qubits, will be capable of demonstrating quantum advantage in industrial applications, integrated with existing high-performance computing infrastructure.

The comprehensive research plan for the next five years is available in [294]. Alice & Bob's approach offers a scalable solution to quantum error correction by leveraging cat qubits that inherently shield against certain external errors, reducing the complexity of error correction from a 2D to a 1D problem. This innovation significantly reduces the number of qubits required for effective error correction, paving the way for the development of practical, large-scale quantum computing systems. With a focus on industrial applications [2], including finance, healthcare, and cybersecurity [28], Alice & Bob's roadmap positions the company to achieve transformative impacts in the coming decade.

19 Microsoft Quantum

19.1 Microsoft's topological quantum computation

Microsoft Quantum is progressing towards the development of a quantum machine as an integral component of the Azure Quantum service [295, 296]. In 2022, Microsoft achieved a significant scientific milestone, overcoming a crucial obstacle in the pursuit of constructing a quantum machine equipped with "topological qubits". Although there are ongoing engineering challenges to overcome, this recent breakthrough signifies a fundamental building block in Microsoft's strategy for advancing scaled quantum computing, as detailed in [297]. Microsoft Quantum has successfully demonstrated the underlying physics

necessary for the creation of a distinct type of qubit known as “topological qubits”. The Microsoft Quantum team has documented [297] the observation of a 30 μeV topological gap within indium arsenide-aluminum heterostructures [298–304], as illustrated in Fig. 27.

Topological quantum computation [299, 305–331] represents a pathway towards achieving hardware-level fault tolerance [105, 332], potentially affording a quantum computing system possessing qubits characterized by high fidelity, rapid gate operations, and a unified module architecture [224, 318, 333–335]. The quality, speed, and physical dimensions of a topological qubit are intricately linked to a characteristic energy parameter termed “the topological gap” [336]. However, this trajectory necessitates the reliable production of a topological phase of matter and the empirical confirmation that the constituent components of a qubit exist within a topological phase, primed for quantum information processing [298, 337–339]. This endeavor is far from trivial due to the intrinsic nature of topological phases, characterized by the long-range entanglement of their ground states, a feature not readily accessible through conventional experimental methods [336].

Having successfully demonstrated the foundational physics, the subsequent phase involves the development of a topological qubit. Microsoft Quantum postulates that the topological qubit will exhibit an advantageous amalgamation of speed, size, and stability when contrasted with other qubit variants. The ultimate aspiration is for this topological qubit to serve as the keystone for a fully scalable quantum machine in the future [340, 341].

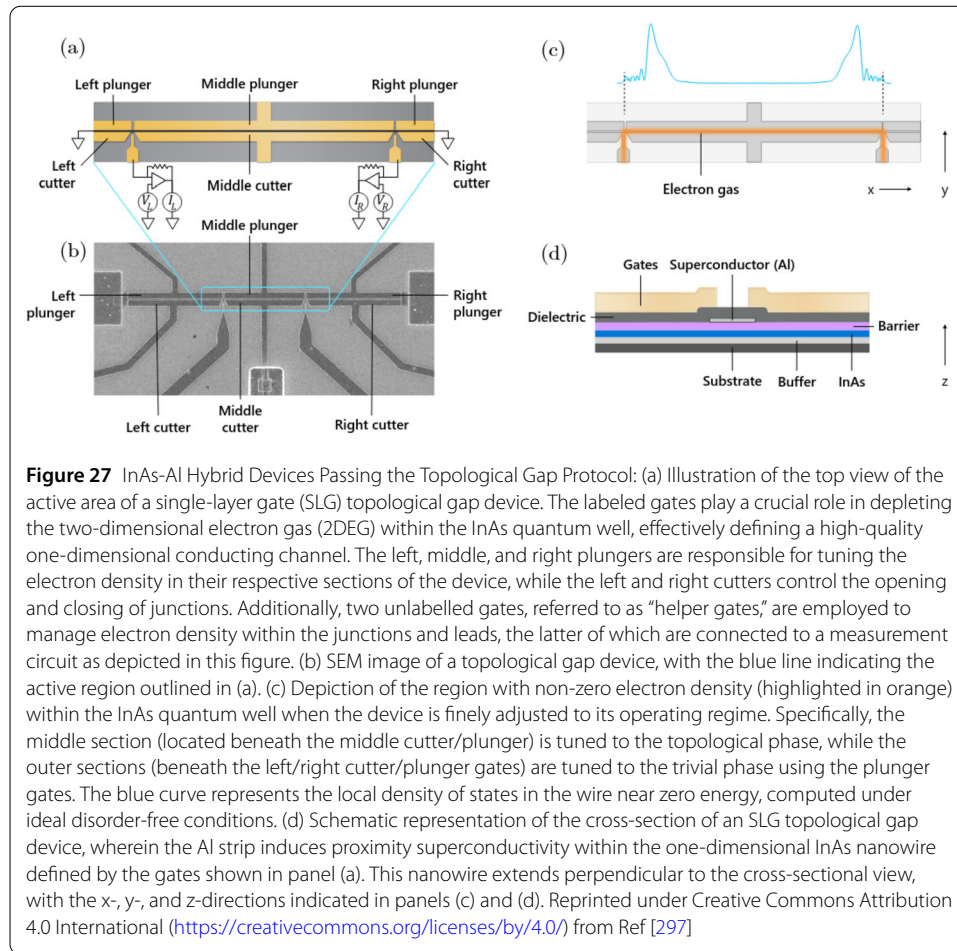
19.2 Azure quantum cloud service

The Azure Quantum Cloud Service (AQCS) [342] offers a comprehensive and robust platform for quantum computing, in collaboration with leading industry partners. As part of the Azure Quantum ecosystem, a diverse range of quantum solution providers are integrated, each contributing unique quantum hardware technologies and simulators. These resources enable users to execute quantum computing algorithms on actual quantum hardware or assess their performance through simulated quantum environments [343].

This flexibility affords users the ability to select the most suitable provider based on the specific characteristics of their computational tasks, such as the problem type, resource requirements, and desired performance outcomes. The quantum providers available on the Azure Quantum platform include Rigetti, Pasqal, Quantinuum, and IonQ, among others, each offering distinct hardware architectures and capabilities designed to address a variety of quantum computing challenges.

20 The Amazon Braket

The Amazon Braket quantum computing service (AWS) [344] is a comprehensively managed service in the domain of quantum computing, specifically designed to expedite scientific research and software development endeavors related to quantum computing. The platform offers accessibility to diverse types of gate-based quantum computers, namely superconducting, trapped ion, and neutral-atom quantum processors. Adhering to the gate-based quantum computing approach, users can avail themselves of superconducting quantum processors provided by Oxford Quantum Circuits (see Sect. 15) and Rigetti Computing (see Sect. 7), along with trapped-ion technology made available by IonQ [345]. Additionally, the platform facilitates simulations of equilibrium behavior and non-equilibrium



dynamics of Hamiltonians, catering to fields like condensed-matter physics and high-energy physics. Moreover, users can explore optimization problems using the most recent neutral-atom processor from QuEra [11]. This multifaceted access enables users to conduct extensive technology testing, evaluate the computational performance of various quantum machines for specific problem-solving purposes, and ultimately select the most suitable hardware system for their respective applications. For detailed information about the QPUs accessible via Amazon Braket, users are encouraged to visit the Hardware Provider's page [346].

Additionally, the Amazon Braket SDK [347] offers a hardware-agnostic developer framework, streamlining the process of designing and executing quantum algorithms. Its primary objective is to eliminate the necessity of coding against distinct quantum programming tools for various types of quantum hardware. By incorporating a unified developer framework, the SDK empowers developers to construct quantum algorithms and execute them seamlessly on any compatible quantum hardware or circuit simulator accessible via the Amazon Braket service. As the platform continues to incorporate new quantum technologies, users can have confidence that their development experience will remain consistent, and their existing designs and quantum algorithms can be readily tested on these novel systems [346].

21 From promise to reality: challenges and bottlenecks

The global race to build scalable quantum computers is fueled by breakthroughs in both hardware and software. Among the various quantum technologies, superconducting qubits have emerged as front-runners, thanks to their relative maturity, well-understood coherence properties, and compatibility with existing semiconductor fabrication methods. But turning these promising systems into practical, large-scale quantum computers is far from simple [87]. The field still faces several formidable challenges, from reducing error rates and improving fabrication precision to integrating effective quantum error correction. Tackling these issues requires coordinated efforts across academia, industry, and government, making collaboration as essential as technical innovation.

One of the most pressing issues is noise-induced errors, which currently limit the performance of superconducting quantum processors. Even though quantum supremacy has been achieved for specific tasks (see Sect. 4) today's quantum systems remain highly error-prone. This restricts their use in practical applications like machine learning [348], chemistry simulations [349], physics modeling [199], and optimization problems [350].

We're still operating in what's known as the NISQ era [2, 26], where quantum computations are limited by gate errors and decoherence. To appreciate the challenge, consider that the fidelity of a quantum computation — essentially how accurately a quantum circuit performs — depends on both the number of qubits and the depth of the circuit (the number of sequential operations) and decays exponentially with circuit depth. For a circuit with depth N , where each gate has fidelity $\mathcal{F}$, the overall fidelity is approximately $\mathcal{F}^N$ [60, 278]. That means even small imperfections compound rapidly. For example, to achieve at least 50% overall fidelity in a 100-qubit circuit with a shallow depth of five, gate fidelities must exceed 99.9%. However, such shallow circuits are insufficient for meaningful quantum error correction or executing practical quantum algorithms. Truly useful quantum computations will require much deeper circuits and even higher gate fidelities—on the order of 99.99% or beyond—highlighting the critical need for continued progress in qubit coherence and gate quality.

To mitigate these errors, researchers are pursuing two complementary strategies: error mitigation and quantum error correction. Error mitigation techniques aim to reduce the impact of noise without adding extra qubits, often by using clever calibration and post-processing methods [82, 85]. But while these approaches help in the short term, they can't fully eliminate errors in larger, more complex computations. For long-term scalability, quantum error correction is essential [101].

Among the various error correction methods, surface codes stand out as particularly promising for superconducting qubits [110, 351]. Their two-dimensional layout aligns well with the physical architecture of superconducting qubit arrays, and they offer a relatively high error tolerance threshold — around 99% gate fidelity. This benchmark was first achieved with superconducting transmon qubits in 2014 [352], and further progress has been made since, as seen in recent work [139, 141]. While significant progress has been made in applying distance-2–5 surface codes to superconducting quantum processors [33, 99, 136–138, 198, 353], there is still a need for further improvement in gate and readout fidelities, with a target of exceeding 99.99%, to ensure that quantum error correction is effective and feasible with a practical number of qubits.

Despite these challenges, the collaborative efforts of companies, academic institutions, and government agencies are pushing the field forward at an unprecedented pace. By shar-

ing benchmarks, open-sourcing tools, and engaging in joint research, these stakeholders are accelerating progress toward fault-tolerant, scalable quantum computing. Achieving that goal will require not only technical breakthroughs but also continued investment and interdisciplinary collaboration — the very foundation on which the future of superconducting quantum computing rests.

22 The path toward real-world applications of superconducting quantum processors

Superconducting quantum processors are widely seen as one of the most promising technologies in the quantum computing landscape. But turning that promise into reality—moving from theoretical models and controlled lab demonstrations to scalable, practical systems—remains a major challenge. It's no longer just a question of whether these systems will work; it's about how soon they can be deployed to solve complex, real-world problems.

In the near term, one of the most exciting opportunities lies in applying superconducting quantum processors to industry-specific challenges. Companies at the forefront of this field are investing heavily in developing systems with the potential to revolutionize areas like drug discovery through molecular simulation, optimize global supply chains, and advance financial modeling. These practical use cases mark a significant step toward realizing the full potential of quantum computing.

What's driving this progress even faster is the growing wave of industry collaborations and cross-sector partnerships. Leading quantum hardware developers are teaming up with established players in industries like pharmaceuticals, aerospace, and finance—ensuring that the technology being built is not only innovative but directly aligned with business needs. This kind of collaboration brings together technical expertise and real-world application, pushing the boundaries of what these processors can achieve.

At the same time, both private and public investments are pouring into quantum research and development, making the ecosystem around superconducting quantum processors more robust than ever. This influx of capital and talent is expected to accelerate the commercialization of quantum technologies, reducing the time it takes to bring practical quantum solutions to market. But despite the momentum, some serious technical hurdles remain. Scaling qubits, improving coherence times, and implementing effective quantum error correction are among the biggest challenges—and solving them is essential if superconducting quantum computers are to transition from experimental prototypes to commercial systems ready to tackle industry-scale problems.

Beyond the technology itself, real-world adoption also depends on building the right support systems. Policy frameworks, regulatory backing, and common technical standards will play a crucial role in ensuring interoperability and easing the widespread deployment of quantum systems. Without these structures in place, even the most advanced quantum hardware could struggle to find its footing in commercial settings.

In short, superconducting quantum processors may still be in the early stages of development, but their potential to transform industries is increasingly clear. With continued focus on scalability, error correction, and real-world applications—alongside deep industry collaboration and supportive policy efforts—superconducting quantum computing is well on its way to becoming a key player in solving some of the world's most complex challenges.

Superconducting quantum computing (SQC) is one of the most promising frontiers in quantum technology, with the potential to transform industries and scientific research. But this technology also comes with a dual-use nature—meaning it can be applied for both civilian and military purposes. As SQC systems advance from today's NISQ devices toward fault-tolerant architectures, the line between civilian innovation and non-civilian application becomes increasingly blurred. This dual potential raises important questions around ethics, security, and policy.

23 Conclusion

Quantum technologies exclusively are expected to open new hopes to solve enormous challenges in various fields such as: energy, security, health, artificial intelligence and environment. It would radically change our thinking of computation, programming, and complexity. The journey toward fully fault-tolerant superconducting quantum computers is a challenging yet inspiring endeavor that embodies the forefront of technological innovation. Superconducting qubits have proven to be one of the most viable platforms, with their scalability, compatibility with established semiconductor processes, and demonstrated performance advancements. However, as this review highlights, the realization of fault-tolerant systems requires addressing critical challenges, including quantum error correction, coherence time improvements, and scalable system integration.

The global effort to achieve these milestones is being driven by an ecosystem of diverse contributors, from industry giants like IBM Quantum, Rigetti Computing and Google Quantum AI to innovative startups. These organizations, along with academic and government collaborations such as QuTech and Quantum Inspire, are accelerating progress through a mix of competition, open science, and strategic partnerships. Their efforts are shaping the quantum computing landscape, bringing us closer to realizing practical quantum systems capable of addressing real-world challenges in areas like cryptography, optimization, and materials science.

As we transition from the NISQ era to the era of fault-tolerant quantum computing, this article bears witness to a historic moment in superconducting quantum computing—highlighting the hardware performance and distinctive features that have defined the NISQ era and offering an in-depth analysis of their capabilities. Furthermore, it explores the quest for fault-tolerant quantum computation, shedding light on the steps taken and the obstacles faced as the field seeks to overcome the inherent limitations of NISQ devices, offering profound insights into its potential impact on future quantum technologies.

While significant hurdles remain, the roadmap to fault-tolerant quantum systems is becoming clearer. Early-stage devices, such as those operating in the NISQ era, are already demonstrating valuable applications, fostering an iterative cycle of discovery and innovation. The collective advancements in hardware, algorithms, and system-level integration are steadily bridging the gap between experimental prototypes and commercially viable quantum computers.

In conclusion, superconducting quantum processors represent not just a technological breakthrough but a transformative opportunity to redefine computing itself. The combined efforts of industry leaders, researchers, and policymakers are creating a robust foundation for the future of quantum technology. By continuing to overcome technical barriers and fostering collaborative ecosystems, the path toward fully fault-tolerant superconducting quantum computers is both achievable and within reach. This progress will undoubt-

edly pave the way for unprecedented scientific, industrial, and societal advancements in the decades to come.

As superconducting quantum computing advances toward scalable, fault-tolerant systems, its potential for both civilian and non-civilian applications becomes increasingly clear. Beyond its promise in fields like pharmaceuticals and materials science, this technology also raises strategic and security considerations, underscoring the need for responsible innovation and governance.

Abbreviations

AFM, Atomic force microscopy; API, An application programming interface; AQCS, Azure Quantum Cloud Service; BRIS, Google's Bristlecone; CAS, Chinese Academy of Sciences; CDF, Cumulative distribution function; CEQIQP, The Center for Excellence in Quantum Information and Quantum Physics; CLOPS, Circuit layer operations per second; CMOS, Complementary Metal-Oxide-Semiconductor; CPW, Coplanar-waveguide; CX, The controlled-X gate; CZ, Controlled-Z gates; DD, Dynamical decoupling; ECR, Echoed cross-resonance; EPG, Error per gate; ESR, Electron spin resonance; EUV, Extreme ultraviolet lithography; f_{01} , Qubit frequency; f_{r0} , Resonator frequency; $\chi/2\pi$, Dispersive shift; $\Delta_0/(2\pi)$, Detuning between qubit and resonator; d , The code distance; $\mathcal{F}$, Fidelity; E_C/h , Charging energy; E_J/h , Josephson energy; ϵ_d , Logical error rates; FLOPS, Floating-point operations per second; GHz, Gigahertz; GHZ, Greenberger–Horne–Zeilinger state; $g/2\pi$, Coupling; H, Hadamard gates; HPC, High-Performance Computing; Heron r1, r2, Quantum processors produced by IBM Quantum; IQS, Intel Quantum Simulator; ISA, Instruction set architecture; JJ, Josephson junction; MHz, Megahertz; NISQ, Noisy intermediate-scale quantum; ns, Nanoseconds; OQC, Oxford Quantum Circuits; p , Physical error rates; p_{thr} , The code's threshold error rate; Q_i , The i th qubit; QAOA, Quantum Approximate Optimization Algorithm; QCS, Quantum Cloud Services; QED, Quantum Electrodynamics; QEC, Quantum error correction; QPUs, Quantum processing units; QVM, Quantum Virtual Machine; QV, Quantum volume; QuAIL, Quantum Artificial Intelligence Lab; QuLL, Quantum Instruction Language; QD, Quantum dot; RCS, Random circuit sampling; RB, Randomized benchmarking; RQCS, Random quantum circuits sampling; RQC, Random quantum circuits; SDK, Software Development Kit; SEM, Scanning electron micrograph; SiO₂, Silicon dioxide; STI, Shallow trench isolation; SQC, Superconducting quantum computing; SYCA, *Sycamore*; TEM, Transmission electron microscope; TTE, Trotterized time evolution; T_1 , Relaxation time; T_2 , Dephasing time; T_2^* , The coherence time (Ramsey); T_{2E} , T2 Echo; μ s, Microseconds.

Acknowledgements

We express gratitude to Intel Corporation for granting the necessary permissions to incorporate images of Intel's Tunnel Falls quantum chip into this research. The views and conclusions expressed in this work are solely those of the author and do not reflect the official policy or position of: IBM Quantum, Google's Quantum AI, Rigetti Computing, The Zuchongzhi Quantum computers, Oxford Quantum Circuits (OQC), Intel Quantum, IQM, QuTech, Quantum Inspire, Alibaba Cloud, Microsoft Azure Quantum, Quantware, OriginQ Cloud, The Amazon Braket (AWS), Pasqal, Quantinuum, IonQ, QuEra, Delft University of Technology, Zhejiang University, The Chinese Academy of Sciences, the Defense Research and Development Canada (DRDC), Canada's Department of National Defense, Reuters or any affiliated organizations. This research is dedicated to my mother, Fatemah. Though words fall short, I offer this to you as you face this difficult journey, with all my love, wishing you strength and a swift recovery.

Author contributions

M. AbuGhanem: Conceptualization, methodology, literature review, data curation, formal analysis, statistical analysis, visualization, Investigation, Validation, Writing, Reviewing and Editing. The author has approved the final manuscript.

Funding information

Open access funding provided by The Science, Technology & Innovation Funding Authority (STDF) in cooperation with The Egyptian Knowledge Bank (EKB). The author declares that no funding, grants, or other forms of support were received at any point throughout this research work.

Data availability

No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

All authors have approved the publication. This research did not involve any human, animal or other participants.

Competing interests

The authors declare no competing interests.

Received: 3 March 2025 Accepted: 1 August 2025 Published online: 19 August 2025

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